Universitat de les Illes Balears

MASTER'S THESIS

PARAMETER ESTIMATION AND SKY LOCALIZATION OF LENSED IMAGE PAIRS OF BINARY BLACK HOLE MERGERS

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Master's Degree in Advanced Physics and Applied Mathematics

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Summary

Around 1920, a journalist asked the British astrophysicist Arthur Eddington whether it was true that only three people in the world understood general relativity. Eddington paused for a moment and replied: "I am trying to think who the third person is." The anecdote may or may not be historically accurate, but Eddington's claim certainly was not. Within months of Einstein publishing his field equations in 1915, exact solutions were already being found: Schwarzschild's black hole, the de Sitter and anti-de Sitter cosmologies, Friedmann's expanding universe. The theory proved far more accessible than its reputation suggested [1].

That reputation, however, is not entirely undeserved. General relativity rests on tensor analysis and differential geometry, branches of mathematics that most physics students encounter only when they need them for this very purpose. Yet beneath the formalism lies a surprisingly simple physical idea: spacetime is a dynamical entity, shaped by matter and energy, and what we call gravity is nothing more than the curvature of that entity. As John Wheeler famously put it, "matter tells spacetime how to curve, and spacetime tells matter how to move" [2].

One hundred and ten years after the publication of the field equations, general relativity has passed every experimental test devised. Among its most striking predictions are gravitational waves (GWs), ripples in the fabric of spacetime produced by the most violent events in the universe. First detected in September 2015 by the LIGO Scientific Collaboration [3], they opened an entirely new observational window onto the cosmos, one based not on light but on the oscillation of spacetime itself. In the decade since, more than three hundred compact binary mergers have been detected, and the field is entering what can only be described as its golden age.

This thesis sits at the intersection of two of the richest areas in modern gravitational-wave astronomy: strong gravitational lensing and sky localization. When a GW passes near a massive galaxy on its way to Earth, its path is deflected, and it can arrive at the detector as multiple copies of the same signal, separated in time and amplitude. No confirmed lensed GW event has been observed to date, but forecasts for the upcoming O5 observing run predict more than one such event per year. Identifying these lensed image pairs, and analyzing them jointly rather than in isolation, opens the possibility of dramatically improving our knowledge of where on the sky the source lies, a measurement that with only two detectors operating is often poorly constrained.

That is the central question this work sets out to address: can the joint analysis of two lensed images of the same binary black hole (BBH) merger meaningfully improve sky localization, and

by how much? To answer it, a population of 500 strongly lensed BBH events detectable by the H1 and L1 detectors under projected O5a sensitivity is generated with the LeR framework, and a realistic detector duty cycle is applied to obtain a final injection set of 157 events, of which 150 were successfully carried through the full analysis. Lensed waveforms are generated with Hanabi using the `IMRPhenomXPHM` approximant, and each image pair is analyzed with Bilby, both individually and through a coherent joint fit that shares the intrinsic source parameters between images while allowing their lensing parameters (relative magnification, time delay, and image type) to vary independently. The joint analysis reduces the median 90% credible sky area from $571\,\mathrm{deg}^2$, obtained from the better of the two single-image analyses, to $99.9\,\mathrm{deg}^2$, a typical improvement by a factor of ~ 5, and brings 3.3% of events below the $10\,\mathrm{deg}^2$ threshold, a regime that no single-image analysis reaches in this two-detector configuration.

The structure of this thesis is organized as follows:

Chapter 1: Introduction to general relativity. Develops the theoretical foundations of GR, from the equivalence principle and the mathematics of curved spacetime to the Einstein field equations and the Schwarzschild solution.

Chapter 2: Gravitational wave theory. Derives gravitational waves from the linearized field equations, characterizes their polarization states and effect on matter, and introduces the detection principle and Bayesian parameter estimation framework.

Chapter 3: Gravitational lensing of gravitational waves. Develops the strong lensing formalism and characterizes the signatures lensing imprints on gravitational wave signals, motivating the joint analysis of lensed image pairs for sky localization.

Chapter 4: Simulation setup and implementation. Describes the construction of the strongly lensed BBH population with LeR, including the detectability and duty-cycle selection funnel used to obtain the final injection set.

Chapter 5: Parameter estimation pipeline. Covers the lensed waveform generation with Hanabi, the Bilby joint-analysis configuration, and the automation infrastructure deployed on the Picasso cluster.

Chapter 6: Sky localization. Presents the sky map generation and area statistics, comparing the 90% credible sky area obtained from single-image and joint analyses across the injected population.

Chapter 7: Conclusions. Summarizes the main findings of this work and discusses possible directions for future research.

Appendix A: Parameters of compact binary coalescences. Complete description of the parameter space of precessing binary black hole systems.

Appendix B: Reproducibility tutorial. Steps required to reproduce the results presented in this work.

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Abbreviations

BH	**B**lack **H**ole
CBC	**C**ompact **B**inary **C**oalescence
CW	**C**ontinuous gravitational **W**ave
EFE	**E**instein **F**ield **E**quations
EMRI	**E**xtreme **Mass** **R**atio **I**nspiral
GR	**G**eneral **R**elativity
GW	**G**ravitational **W**ave
HPC	**H**igh **P**erformance **C**omputing
IGWN	**I**nternational **G**ravitational-Wave **O**bservatory **N**etwork
IMBHs	**I**ntermediate **M**ass **B**lack **H**oles
LISA	**L**aser **I**nterferometer **S**pace **A**ntenna
NS	**N**eutron **S**tar
O1, O2, O3, O4, O5	**O**bservation run 1, 2, 3, 4, 5 of the LVK network
PE	**P**arameter **E**stimation
SBH	**S**tellar-mass **B**lack **H**ole
SGWB	**S**tochastic **G**ravitational **W**ave **B**ackground
SMBH	**S**uper**m**assive **B**lack **H**ole
SNR	**S**ignal-to-**N**oise **R**atio

Part I

Theoretical introduction

The first three chapters provide the theoretical foundation of this thesis. Chapter 1 introduces general relativity, developing the mathematical framework of curved spacetime and the Einstein field equations. Chapter 2 derives gravitational waves from the linearized field equations and describes the current network of interferometric detectors and the Bayesian parameter estimation framework. Chapter 3 introduces gravitational lensing, develops the strong lensing formalism for gravitational waves, and motivates the joint analysis of lensed image pairs for sky localization.

Chapter 1

Introduction to General Relativity

Nowadays, general relativity (GR) is fundamental to our understanding of modern astrophysics. Whether we are studying black holes (BHs), quasars, or gravitational lensing, Einstein's theory provides the essential framework. Efforts to detect gravitational waves (GWs) are a direct test of one of his key theoretical predictions, marking a remarkable chapter in the story of one of the most influential scientific achievements of the 20th century.

1.1 Motivation and Historical Background

The centenary of GR marks not only one of the greatest theoretical advances in physics, but also the foundation of modern astrophysical exploration. From its inception in 1915, Einstein's theory has evolved far beyond corrections to planetary orbits, becoming a central pillar in our understanding of the cosmos [4].

1.1.1 Key Historical Milestones

- **1905–1915:** Following his groundbreaking special relativity paper [5], Einstein seeks a relativistic framework for gravitation, which results in the formulation of GR in 1915.
- **1916:** Within months, Karl Schwarzschild finds the first exact solution to Einstein's equations [6], revealing the theoretical existence of BHs, a concept reinforced in 1917 when Einstein applies his theory to cosmology.
- **1919:** Arthur Eddington's eclipse expedition confirms the bending of starlight by the Sun's gravity [7], providing one of the first key empirical validations of the theory.
- **1927–1929:** Georges Lemaître and Edwin Hubble independently establish the expanding universe paradigm [8], linking GR to the observable evolution of the cosmos.
- **1960s–1970s:** After decades of theoretical neglect, interest in GR resurges. Joseph Weber pioneers early experimental searches for GWs [9], and the discovery of the Hulse-Taylor pulsar in 1974 offers indirect but compelling evidence that energy is indeed carried away by gravitational radiation.
- **1980s–1990s:** Interferometric efforts gain traction on both sides of the Atlantic, with prototype detectors developed in Europe and the United States; the resulting collaborations produce GEO600 and pave the way for LIGO and Virgo [10].

- **2015–2017:** The detection of GW150914 by LIGO in September 2015 [3], publicly announced in February 2016, marks the first direct observation of GWs, fulfilling a century-old prediction and inaugurating the era of GW astronomy.

This historical path underscores GR's pivotal role beyond theoretical elegance. The final formulation emerged through a series of four weekly submissions to the Prussian Academy of Sciences in November 1915 [11, 12, 13, 14], culminating in the field equations presented on November 25th. Among the key results was the correct prediction of Mercury's perihelion shift of 45" per century [13], which had previously eluded Newtonian mechanics.

1.1.2 Equivalence principle

The equivalence principle played a central role in the development of GR. In 1907, Einsteinrealized that a freely falling observer does not locally experience gravity, a insight he later described as "the happiest thought of my life" [15, 16].

1.1.2.1 The weak equivalence principle

The roots of GR reach back to the 17$^{\text{th}}$ century, when Galileo Galilei and Simon Stevin (1548–1620) began to recognize that the velocity reached by freely falling objects is independent of their mass [1], a realization that contradicted the Aristotelian view of physics. Galileo derived his conclusions on free fall by rolling spheres down inclined planes, results presented in his 1590 manuscript De Motu [17].

The universality of free fall reflects a striking feature of Newtonian mechanics: the equivalence between inertial mass m_i and gravitational mass m_g, also called the weak equivalence principle. In Newtonian theory, inertial mass appears in Newton's second law,

$$\vec{F} = m_i \, \vec{a} \ , \tag{1.1}$$

while gravitational mass governs the gravitational potential between two bodies,

$$V = \frac{G_N \, m_\mathrm{g} \, M_\mathrm{g}}{r} \ . \tag{1.2}$$

The motion of an object in free fall is then governed by

$$-m_\mathrm{g} g = m_\mathrm{i} \ddot{z} \ , \tag{1.3}$$

whose solution,

$$z(t) = -\frac{1}{2}\frac{m_\mathrm{g}}{m_\mathrm{i}} g t^2 + v_0 t + z_0 \ , \tag{1.4}$$

is independent of mass only if $m_g/m_i =$ constant, which experiments confirm to be universal and set to 1 by an appropriate choice of units.

This equivalence has been tested with remarkable precision. In 1888, Eötvös used a torsion balance to reach a precision of $|(m_g - m_i)/m_g| \sim 10^{-9}$ [18], later improved by Dicke in 1964 to

$< 10^{-11}$ [19]. Currently, the MicroSCOPE satellite mission has achieved a precision of 10^{-14} [20, 21].

1.1.2.2 The strong equivalence principle

Although Newtonian mechanics offered no fundamental reason for $m_g = m_i$, Einstein believed this could not be a mere coincidence. The breakthrough came when he realized that an observer in free fall does not feel their own weight and cannot distinguish, through any local experiment, whether they are in a gravitational field or drifting freely in empty space.

This reasoning extends to the case of uniform acceleration: an observer inside an enclosed rocket accelerating at $a = g$ cannot tell whether the apparent downward pull is due to gravity or to the rocket's acceleration. This leads to two equivalent formulations:

> **Equivalence Principle** (for uniform gravitational fields): *An observer in free fall within a constant gravitational field is equivalent to an inertial observer in the absence of gravity. It is impossible to distinguish between these two situations by means of physical experiments.*

> **Equivalence Principle** (for constant accelerations): *An observer undergoing uniform acceleration is equivalent to an inertial observer in a uniform gravitational field. It is impossible to distinguish between these two situations through physical experiments.*

However, this equivalence breaks down for general gravitational fields, where tidal forces (arising from field inhomogeneities) allow a free-falling observer to detect the presence of gravity [1]. For instance, two freely falling objects in Earth's radial field slowly converge toward the center rather than following parallel paths. Since tidal forces are proportional to the gradient of the field, they vanish at sufficiently small scales, and the equivalence principle recovers its validity locally.

> **Equivalence Principle** (general formulation): *Free-falling observers in a gravitational field are locally indistinguishable from inertial observers. There are no local experiments that can differentiate between these two scenarios.*

The local validity of the equivalence principle has a precise mathematical counterpart: at any point of spacetime it is always possible to choose coordinates in which the metric tensor (introduced formally in §1.2.4) reduces locally to the Minkowski form $\eta_{\mu\nu}$. This idea, made rigorous through the concept of normal coordinates, underlies the geometric interpretation of gravity developed in the following sections.

1.2 The description of curved spacetime

After his work on special relativity and learning about the mathematical structure introduced by Minkowski [22], Einstein realized that space-time itself possesses a geometry. Instead of thinking of space and time as separate and absolute entities, special relativity showed that they are part of a unified, four-dimensional space-time fabric.

Before introducing gravity, it is crucial to recall this idea: space-time itself has a geometry. In special relativity, this geometry is flat (Minkowskian); however, the presence of mass and energy changes this picture.

To describe this curved geometry, Euclidean tools are no longer sufficient. Instead, we require a more general mathematical framework: Riemannian geometry, capable of capturing the curvature induced by mass and energy. In this section, we will extend that concept further.

1.2.1 From euclidean to non-euclidean geometry

The geometry most familiar to us, Euclidean geometry, describes flat space: parallel lines never meet, and the angles of a triangle always sum to 180°. These properties stem from the work of the Greek mathematician Euclid, who around 300 BC formulated five postulates from which he derived hundreds of theorems [23]. Among these, the fifth, the parallel postulate, stood out for its complexity, prompting centuries of efforts to prove it from the others [24].

Eventually, mathematicians like Nikolai I. Lobachevsky [25] and János Bolyai [26] explored what would happen if the parallel postulate were replaced, discovering entirely new self-consistent geometries (see Fig. 1.1). On a sphere, parallel lines do not exist and triangle angles exceed 180°; in hyperbolic geometry, infinitely many parallel lines pass through a point and triangle angles sum to less than 180°.

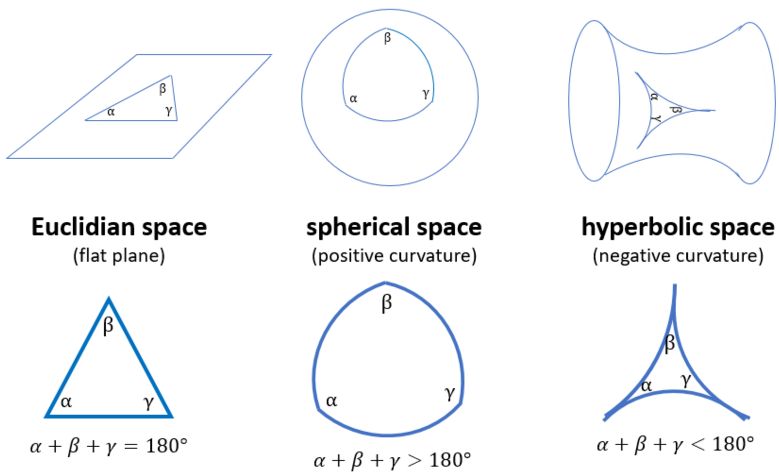


Figure 1.1: Representation of the three main types of geometry: Euclidean (flat), spherical (positive curvature), and hyperbolic (negative curvature). In each case, the sum of the angles of a triangle differs: it equals 180° in Euclidean geometry, exceeds 180° in spherical geometry, and is less than 180° in hyperbolic geometry. Source: [27].

These ideas became essential when Einstein formulated GR. In his theory, spacetime is curved by the presence of mass and energy, and the shortest paths, geodesics (see §1.2.5.4), follow the curvature of spacetime rather than straight Euclidean lines. This curved geometry, pioneered by Bernhard Riemann, provides the mathematical language to describe gravitational phenomena, replacing Newton's force with the geometry of spacetime itself.

1.2.2 Manifolds and tangent spaces

In order to describe curved spaces mathematically, we use the concept of a manifold: a space that may possess curvature or nontrivial topology, yet locally resembles Euclidean space. Formally, an N-dimensional manifold $\mathcal{M}^N$ is a space that, in a neighborhood of each point p, looks like $\mathbb{R}^N$. At every point p we can define the tangent space $T_p(\mathcal{M}^N)$, an N-dimensional vector space that provides a linear approximation to the manifold in a small region around p. Tangent spaces at different points are distinct, reflecting the fact that the manifold is not globally flat (see Fig. 1.2).

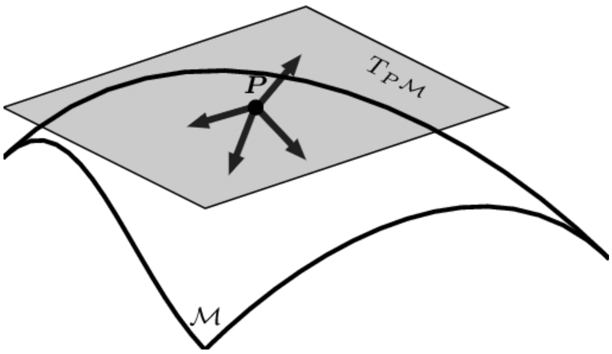


Figure 1.2: The set of all vectors defined at a point p forms the tangent space $T_p(\mathcal{M})$ at p. Assembling the tangent spaces for every point of the manifold produces the tangent bundle $T(\mathcal{M})$. Original illustration based on the source: [28].

Formally, a topological manifold satisfies three conditions: the Hausdorff property (any two distinct points can be separated by disjoint open sets), second-countability (the topology admits a countable basis), and local Euclideanness (every point has a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$). The last condition ensures that, although $\mathcal{M}$ may be globally curved, every point admits a patch that looks like flat $\mathbb{R}^n$.

Manifolds appear throughout physics: in mechanics, configuration and phase spaces are manifolds; in GR, spacetime is modeled as a Lorentzian manifold (see §1.2.4), locally isometric to Minkowski space but with nontrivial global curvature determined by the distribution of matter and energy.

1.2.2.1 Mappings, charts, and atlases

A chart on $\mathcal{M}$ is a pair (U, φ) consisting of an open subset $U \subset \mathcal{M}$ and an injective map $\varphi : U \to \mathbb{R}^n$, whose components $\varphi(p) = (x^1(p), \ldots, x^n(p))$ define local coordinates on U (see Fig. 1.3). Just as no single flat map can represent the entire Earth without distortion, a single chart rarely covers an entire manifold. For instance, covering the 2-sphere S^2 requires at least two charts.

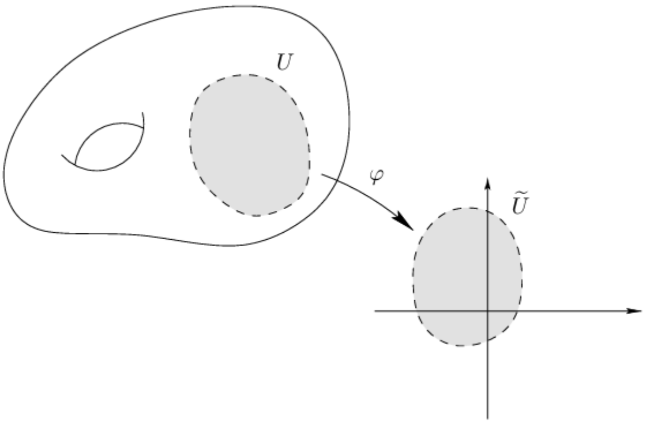


Figure 1.3: Representation of a chart (U, φ) on a manifold $\mathcal{M}$, where the subset $U \subset \mathcal{M}$ is mapped bijectively to an open set $\widetilde{U} \subset \mathbb{R}^n$ via φ. Source: [29].

An atlas is a collection of charts $\{(U_\alpha, \varphi_\alpha)\}$ whose domains cover $\mathcal{M}$, with the additional requirement that whenever $U_\alpha \cap U_\beta \neq \emptyset$, the transition map

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta) \tag{1.5}$$

is C^∞ (see Fig. 1.4). These smooth transition maps ensure that the different local coordinate patches are consistently glued together, uniquely determining the manifold's differentiable structure.

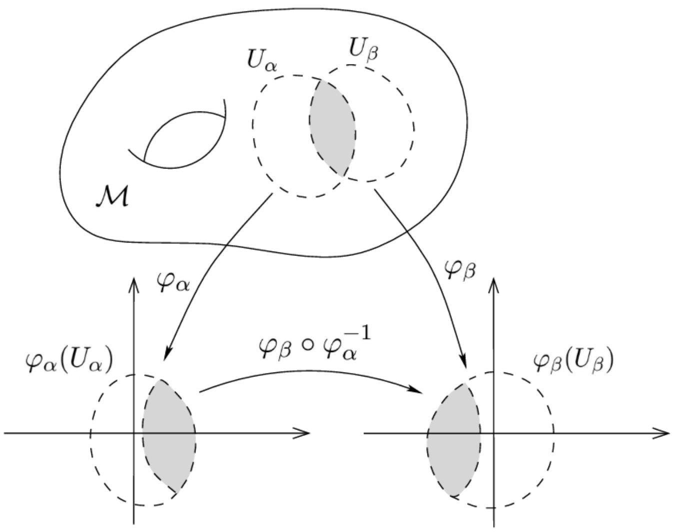


Figure 1.4: When two charts overlap, their coordinate systems are related by a smooth transition map. Original illustration based on the source: [29].

1.2.3 Mathematical interlude

1.2.3.1 Einstein summation convention and dummy indices

As we delve deeper into the mathematics of vectors and tensors, we'll notice that certain patterns appear repeatedly in equations. One such recurring structure involves expressions where an index appears more than once within the same term. For example, consider the following:

$$\sum_{\mu} V^{\mu} U_{\mu} \ . \tag{1.6}$$

In this expression, the index μ appears both as a superscript and a subscript. It is summed over, meaning it serves as a dummy index, a placeholder variable that takes on all possible values of that index. The specific letter used doesn't matter; for instance:

$$\sum_{\nu} V^{\nu} U_{\nu} \ , \tag{1.7}$$

represents exactly the same expression. The key characteristics of this type of sum are:

- The index appears exactly two times in a single term.
- It appears once as a superscript (contravariant) index and once as a subscript (covariant) index.
- It implies summation over all possible values of the repeated index.

Einstein's clever idea, known as the Einstein summation convention, was to omit the summation symbol altogether when this pattern appears. According to this convention, whenever an index is repeated once upstairs and once downstairs in a term, a summation over that index is assumed. So instead of writing:

$$\sum_{\mu} V^{\mu} U_{\mu} \ , \tag{1.8}$$

we can simply write:

$$V^{\mu} U_{\mu} \ , \tag{1.9}$$

and the summation is understood.

This notational shortcut was first introduced by Einstein in his 1916 paper The Foundation of the General Theory of Relativity [30]. At the beginning of the paper, he still wrote out summation signs explicitly. For instance:

$$dX_{\nu} = \sum_{\sigma} a_{\nu\sigma} \, dx_{\sigma} \ . \tag{1.10}$$

However, shortly after equation 7, he casually notes that whenever an index appears twice, summation is to be assumed, and from then on, he stops writing summation symbols.

The Einstein summation convention offers a powerful and elegant notational tool, widely used in physics and mathematics to simplify the representation of tensor equations [1, 31].

1.2.3.2 Tensors

Tensors are the natural language of GR: they are geometric objects whose components transform consistently under coordinate changes, ensuring that physical laws remain valid in any reference frame [32]. They generalize the familiar notions of scalars (invariant under coordinate changes), contravariant vectors (components transforming via the Jacobian of the coordinate change), and covariant vectors or one-forms (components transforming via the inverse Jacobian).

Tensors are commonly classified by their type (r, s), where r denotes the number of contravariant (upper) indices and s the number of covariant (lower) indices [33]. Some illustrative examples are:

- A scalar T carries no indices; it is a rank-0 tensor of type $(0, 0)$.
- A contravariant vector V^β is a rank-1 tensor of type $(1, 0)$, and a one-form W_α is of type $(0, 1)$.
- A mixed tensor T^α_γ is of type $(1, 1)$, and the Riemann curvature tensor $R^l{}_{ijk}$, studied in §1.2.5.5, is of type $(1, 3)$.

In general, a tensor of rank n in a d-dimensional space has d^n components, transforming under coordinate changes according to

$$T'^{i'_1 \dots i'_r}_{j'_1 \dots j'_s} = \left(\prod_{k=1}^{r} \frac{\partial x'^{i'_k}}{\partial x^{i_k}} \right) \left(\prod_{l=1}^{s} \frac{\partial x^{j_l}}{\partial x'^{j'_l}} \right) T^{i_1 \dots i_r}_{j_1 \dots j_s} \ . \tag{1.11}$$

This transformation law ensures that if a tensor equation holds in one coordinate system, it holds in all, an essential feature of any generally covariant theory such as GR. Formally, a tensor of type (r, s) can be seen as a multilinear map taking r one-forms and s vectors as inputs and returning a real number [34].

In GR, tensors are not merely mathematical tools: they encode curvature, energy, and momentum, and constitute the grammar through which the field equations are written.

1.2.4 Metric tensor

Until now, we have not introduced the most crucial element. There is a single object that, when defined on a manifold, surpasses all others in importance for understanding gravity: the metric tensor. The presence of a metric opens the door to a wide range of new concepts, which together form the framework known as Riemannian geometry [35].

A differentiable manifold equipped with a symmetric metric tensor is called a Riemannian manifold. In GR, the metric plays a fundamental role: it defines the separation between nearby points in spacetime and determines the curvature of the manifold. In flat spacetime, far from matter and energy, the metric approximates the Minkowski metric $\eta_{\mu\nu}$ of special relativity [36]. Near massive objects, such as in the Schwarzschild solution, the metric adapts to reflect spacetime curvature (see §1.4.1 for more details).

A metric tensor g on a differentiable manifold $\mathcal{M}$ is a $(0,2)$ tensor field that allows the definition of an inner product on each tangent space $T_p(\mathcal{M})$. It satisfies:

- Symmetry: $g(X,Y) = g(Y,X)$ for all vector fields X, Y.
- Non-degeneracy: If $g(X,Y)|_p = 0$ for all $Y \in T_p(M)$, then $X_p = 0$.

In a coordinate basis $\{\partial_\mu\}$, the components of the metric are given by

$$g_{\mu\nu}(x) = g\left(\frac{\partial}{\partial x^\mu}, \frac{\partial}{\partial x^\nu}\right), \tag{1.12}$$

and by definition

$$g_{\mu\nu} = e_\mu \cdot e_\nu \ , \tag{1.13}$$

where e_μ are the coordinate basis vectors.

Given coordinates x^μ, an infinitesimal displacement in the manifold is

$$ds = dx^\mu \, e_\mu \ , \tag{1.14}$$

and its squared norm (inner product with itself) is

$$(ds)^2 = (dx^\mu e_\mu) \cdot (dx^\nu e_\nu) \overset{(i)}{=} g_{\mu\nu} \, dx^\mu dx^\nu. \tag{1.15}$$

(i) We use that $e_\mu \cdot e_\nu = g_{\mu\nu}$.

This is the line element:

$$ds^2 = g_{\mu\nu}(x) \, dx^\mu dx^\nu. \tag{1.16}$$

In an n-dimensional Riemannian manifold:

$$dl^2 = g_{ij} \, dx^i dx^j \ . \tag{1.17}$$

Similarly, the inner product of two distinct vectors is

$$V \cdot W = g_{\mu\nu} V^\mu W^\nu \ . \tag{1.18}$$

The explicit form of $g_{\mu\nu}$ depends on the coordinate system [32]. In Cartesian coordinates:

$$dl^2 = dx^2 + dy^2 + dz^2, \quad g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{1.19}$$

In Minkowski spacetime:

$$ds^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2, \quad \eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{1.20}$$

In spherical coordinates:

$$ds^2 = dr^2 + r^2\,d\theta^2 + r^2 \sin^2\theta\,d\varphi^2, \quad g_{ij} = \mathrm{diag}(1, r^2, r^2\sin^2\theta)^1. \tag{1.21}$$

In general, $g_{\mu\nu}$ is a symmetric 4×4 matrix in spacetime, with 10 independent components. Because coordinate transformations can adjust 4 of them, only 6 components contain independent physical information [38].

The metric tensor is the tool that enables the computation of lengths, angles, areas, and volumes in a manifold. In physics, it encodes the gravitational field in GR. It also acts as a bridge between contravariant and covariant vectors:

$$v_\mu = g_{\mu\nu} v^\nu, \quad v^\mu = g^{\mu\nu} v_\nu\ , \tag{1.22}$$

where $g^{\mu\nu}$ is the inverse metric, satisfying

$$g_{\mu\nu} g^{\nu\rho} = \delta^\rho_\mu\ . \tag{1.23}$$

The signature of the metric distinguishes between different geometries: Riemannian metrics are positive-definite $(+, +, +, \dots)$, while Lorentzian metrics have signature $(+, -, -, \dots, -)$ or $(-, +, +, \dots, +)$ as in spacetime. Once the metric is known, the geometry and the gravitational field (in the case of GR) are completely determined.

A particularly relevant example in the context of this work is the weak-field perturbed metric, which describes spacetime in the presence of a slowly varying Newtonian gravitational potential. This metric reduces to the Minkowski form in the absence of matter and governs the deflection of GWs as they propagate through a gravitational lens, as developed in §3.2.

1.2.5 Affine connection and curvature

We've now laid the groundwork for a deeper look into the mathematics of curved space. Ultimately, our goal is to understand:

- Geodesics (see §1.2.5.4), the "straightest possible" paths between any two points on a Riemannian manifold, and thus the natural trajectories of free particles moving through curved spacetime.
- The Riemann curvature tensor (see §1.2.5.5), which encodes how space bends. A particular version of this tensor appears on the left-hand side of the fundamental object of GR, the Einstein field equations (EFE).

Parallel transport of a vector (see §1.2.5.3) plays a key role in both of these concepts. But before we can understand how to transport vectors, we must first know how to differentiate tensor fields. To make this possible, we introduce what are known as connection coefficients or Christoffel symbols (denoted by the symbol Γ). These coefficients allow us to compare vectors

[1] The notation $\mathrm{diag}(a_1, a_2, \dots, a_n)$ denotes a diagonal matrix whose entries are zero everywhere except along the diagonal, where the components are $a_1, a_2, \dots, a_n$. In this case, it means that the metric tensor g_{ij} has components $g_{11} = 1$, $g_{22} = r^2$, $g_{33} = r^2 \sin^2\theta$, and all off-diagonal components vanish [37].

located at different points on a manifold. Once the metric of the manifold is known, we can compute the connection coefficients, unlocking our ability to define derivatives and do much more. This is why the metric is central to understanding Riemannian manifolds, and therefore the geometry of spacetime itself [33].

1.2.5.1 Christoffel symbols

Our goal in GR is to express physical laws as tensor equations valid in any coordinate system. To do that, we must be able to differentiate tensors in a way that behaves well under changes of coordinates. Specifically, we want a derivative that reduces to the usual partial derivative in flat Minkowski space using Cartesian coordinates, but which still transforms as a tensor in curved spacetimes.

At first glance, differentiating a tensor might seem straightforward. For example, in Cartesian coordinates, a vector field like

$$\vec{V}(x, y, z) = (2x - y)\,\vec{e}_x + (x + 3z)\,\vec{e}_y + (y - z)\,\vec{e}_z \ , \tag{1.24}$$

can be differentiated component by component:

$$\frac{\partial \vec{V}}{\partial x} = 2\,\vec{e}_x + \vec{e}_y \ , \qquad \frac{\partial \vec{V}}{\partial y} = -\,\vec{e}_x + \vec{e}_z \ , \qquad \frac{\partial \vec{V}}{\partial z} = 3\,\vec{e}_y - \vec{e}_z \ . \tag{1.25}$$

The basis vectors $\vec{e}_x, \vec{e}_y, \vec{e}_z$ are constant in space, so we don't need to worry about them changing from point to point. We just apply the partial derivative to each component function.

However, this approach fails in curved spaces or curvilinear coordinates, where the basis vectors themselves vary from point to point. In such cases, when differentiating a tensor, we must also account for the rate of change of the basis vectors. This is where the connection coefficients, or Christoffel symbols, come into play.

Let us consider a vector expressed as $\vec{V} = V^\alpha \vec{e}_\alpha$, where $\vec{e}_\alpha$ are the coordinate basis vectors. Now let's take the derivative of this with respect to some coordinate using the product rule [33]:

$$\frac{\partial \vec{V}}{\partial x^\beta} = \frac{\partial}{\partial x^\beta}\left(V^\alpha \vec{e}_\alpha\right) = \left(\frac{\partial V^\alpha}{\partial x^\beta}\right)\vec{e}_\alpha + V^\alpha \left(\frac{\partial \vec{e}_\alpha}{\partial x^\beta}\right) \ . \tag{1.26}$$

The first term reflects the change in the vector components, while the second term captures how the basis vectors change. Since $\partial_\beta \vec{e}_\alpha$ is itself a vector, it can be expressed in terms of the basis:

$$\frac{\partial \vec{e}_\alpha}{\partial x^\beta} = \Gamma^\gamma_{\alpha\beta}\vec{e}_\gamma \ . \tag{1.27}$$

Here, the Christoffel symbols $\Gamma^\gamma_{\alpha\beta}$ quantify how the basis vector $\vec{e}_\alpha$ changes in the direction of coordinate x^β, projected onto the basis $\vec{e}_\gamma$. The lower indices α and β refer to the direction and variable of differentiation, while the upper index γ indicates the output basis direction.

We can isolate the Christoffel symbol by multiplying both sides by the dual basis vector $\vec{e}^{\,\gamma}$:

$$\vec{e}^{\,\gamma} \cdot \partial_\beta \vec{e}_\alpha = \Gamma^\gamma_{\alpha\beta}\,\vec{e}^{\,\gamma} \cdot \vec{e}_\gamma \quad \Rightarrow \quad \Gamma^\gamma_{\alpha\beta} = \vec{e}^{\,\gamma} \cdot \partial_\beta \vec{e}_\alpha \ . \tag{1.28}$$

Here, we used the property $\vec{e}^{\beta} \cdot \vec{e}_{\alpha} = \delta^{\beta}_{\alpha}$, which defines the dual basis and implies that $\vec{e}^{\gamma} \cdot \vec{e}_{\gamma} = 1$.

These coefficients do not form a tensor themselves, but they are essential for constructing tensorial derivatives.

Some key properties of Christoffel symbols are:

- They provide the rule for differentiating tensors in a curved manifold.
- They are called connection coefficients because they "connect" nearby points on the manifold, defining how vectors vary between them, crucial for defining parallel transport.
- If we know the metric $g_{\mu\nu}$, we can compute them explicitly as (see [31] for the derivation):

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\lambda} \left(\partial_{\mu} g_{\lambda\nu} + \partial_{\nu} g_{\lambda\mu} - \partial_{\lambda} g_{\mu\nu} \right) \ . \tag{1.29}$$

- In standard GR (without torsion[2]), the Christoffel symbols are symmetric in their lower indices:

$$\Gamma^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\nu\mu} \ . \tag{1.30}$$

- In an n-dimensional Riemannian manifold, there are theoretically n^3 such coefficients, though this number is reduced by the symmetry condition.

The term "Christoffel symbols" honors the mathematician Elwin Christoffel (1829–1900), who first introduced them in the context of differential geometry. Although sometimes informally dubbed "Christ-awful symbols" due to their apparent complexity, this perception tends to fade with familiarity [31]. What initially seems intricate is, in fact, a systematic way of incorporating the effects of curvature into the formulation of differential operators .

These connection coefficients are essential for extending the familiar notion of differentiation to the curved spacetimes of GR, setting the stage for the definition of the covariant derivative, which we address next.

1.2.5.2 Covariant derivative

Having introduced the Christoffel symbols (1.27) and established how to differentiate vectors in curved spacetime using equation (1.26), we are now ready to define the covariant derivative, which provides a consistent framework for differentiating tensors.

Substituting equation (1.27) into equation (1.26), we obtain:

$$\frac{\partial \vec{V}}{\partial x^{\beta}} = \frac{\partial V^{\alpha}}{\partial x^{\beta}} \vec{e}_{\alpha} + V^{\alpha} \Gamma^{\gamma}_{\alpha\beta} \vec{e}_{\gamma} \ . \tag{1.31}$$

[2] Torsion describes the amount of "asymmetry" in the way points in spacetime are connected: it indicates whether moving first in one direction and then in another ends at the same point as doing so in the reverse order. If torsion is zero, as in standard GR, both paths lead to the same point. When we study parallel transport and the Riemann curvature tensor (see §1.2.5.5), this idea will acquire a precise mathematical formulation through the torsion tensor. See [1, 39] for more details.

We relabel the dummy indices α and γ in the right-hand side term, as they are summation indices and can be freely exchanged to simplify the expression:

$$\frac{\partial \vec{V}}{\partial x^\beta} = \frac{\partial V^\alpha}{\partial x^\beta}\vec{e}_\alpha + V^\gamma \Gamma^\alpha_{\gamma\beta}\vec{e}_\alpha \ . \tag{1.32}$$

We now factor out the common basis vector $\vec{e}_\alpha$:

$$\frac{\partial \vec{V}}{\partial x^\beta} = \left(\frac{\partial V^\alpha}{\partial x^\beta} + V^\gamma \Gamma^\alpha_{\gamma\beta}\right)\vec{e}_\alpha \ . \tag{1.33}$$

We can express the components of $\dfrac{\partial \vec{V}}{\partial x^\beta}$ as:

$$\frac{\partial V^\alpha}{\partial x^\beta} + V^\gamma \Gamma^\alpha_{\gamma\beta} \ . \tag{1.34}$$

This formula represents the covariant derivative of the contravariant vector $\vec{V}$. It describes how the component V^α changes in the direction β of the coordinate system x_β. It is common to use the notation $\nabla_\beta V^\alpha$ to denote the covariant derivative, so equation (1.34) can also be written as:

$$\nabla_\beta V^\alpha = \frac{\partial V^\alpha}{\partial x^\beta} + V^\gamma \Gamma^\alpha_{\gamma\beta} \ . \tag{1.35}$$

Similarly, the covariant derivative of a one-form or covector is expressed as:

$$\nabla_\beta V_\alpha = \frac{\partial V_\alpha}{\partial x^\beta} - V_\gamma \Gamma^\gamma_{\alpha\beta} \ . \tag{1.36}$$

The covariant derivative maintains the properties of linearity and satisfies the product rule, similar to the partial derivative:

$$\begin{aligned} \nabla_\mu(\alpha V^\nu + \beta W^\nu) &= \alpha \nabla_\mu V^\nu + \beta \nabla_\mu W^\nu, \\ \nabla_\mu(V^\nu W^\rho) &= (\nabla_\mu V^\nu) W^\rho + V^\nu (\nabla_\mu W^\rho) \ . \end{aligned} \tag{1.37}$$

Moreover, it has the important property of transforming properly as a $(1,1)$ tensor:

$$\nabla_\alpha V^\gamma = \frac{\partial x^\mu}{\partial y^\alpha}\frac{\partial y^\gamma}{\partial x^\nu}\nabla_\mu V^\nu \ . \tag{1.38}$$

Furthermore, in flat space and when working with Cartesian coordinates, the covariant derivative of a vector or a one-form simplifies to the partial derivative:

$$\nabla_\alpha \phi = \partial_\alpha \phi \ . \tag{1.39}$$

Finally we can see how to construct the covariant derivative of a tensor.

We derive the covariant derivative of a tensor of type $(1,1)$, that is, a tensor with one upper (contravariant) index and one lower (covariant) index. Our goal is to construct a derivative operator that preserves the tensorial nature of the object under coordinate transformations.

Let $T^\nu{}_\mu$ be a $(1,1)$ tensor. The covariant derivative $\nabla_\rho T^\nu{}_\mu$ is given by:

$$\nabla_\rho T^\nu{}_\mu = \partial_\rho T^\nu{}_\mu - \Gamma^\lambda_{\rho\mu} T^\nu{}_\lambda + \Gamma^\nu_{\rho\lambda} T^\lambda{}_\mu \ . \tag{1.40}$$

This formula can be understood as follows:

- The first term $\partial_\rho T^\nu{}_\mu$ is the ordinary partial derivative.
- The second term $-\Gamma^\lambda_{\rho\mu} T^\nu{}_\lambda$ corrects for the transformation of the covariant index μ.
- The third term $+\Gamma^\nu_{\rho\lambda} T^\lambda{}_\mu$ corrects for the transformation of the contravariant index ν.

The general rule is: for each lower index, subtract a term involving a Christoffel symbol; for each upper index, add a term involving a Christoffel symbol. This ensures that the resulting expression transforms properly as a tensor.

In general, the covariant derivative of a tensor of type (m, n) is given by [33]:

$$\begin{aligned}\nabla_\rho T^{\mu_1\ldots\mu_m}{}_{\nu_1\ldots\nu_n} = {} & \partial_\rho T^{\mu_1\ldots\mu_m}{}_{\nu_1\ldots\nu_n} \\ & + \Gamma^{\mu_1}_{\rho\lambda} T^{\lambda\mu_2\ldots\mu_m}{}_{\nu_1\ldots\nu_n} + \cdots + \Gamma^{\mu_m}_{\rho\lambda} T^{\mu_1\ldots\mu_{m-1}\lambda}{}_{\nu_1\ldots\nu_n} \\ & - \Gamma^\lambda_{\rho\nu_1} T^{\mu_1\ldots\mu_m}{}_{\lambda\nu_2\ldots\nu_n} - \cdots - \Gamma^\lambda_{\rho\nu_n} T^{\mu_1\ldots\mu_m}{}_{\nu_1\ldots\nu_{n-1}\lambda} \ .\end{aligned} \tag{1.41}$$

To ensure the internal consistency of GR, it is postulated that the covariant derivative of the metric tensor vanishes:

$$\nabla_\beta g_{\mu\nu} = \nabla_\beta g^{\mu\nu} = 0 \ . \tag{1.42}$$

A covariant derivative satisfying this condition is referred to as metric compatible. This requirement ensures that the scalar product between vectors remains constant during parallel transport (see §1.2.5.3) across a curved spacetime manifold. While this condition holds trivially in flat spacetime (as in special relativity), it becomes a non-trivial and essential constraint in the presence of curvature.

Because the equation above is tensorial, it holds in all coordinate systems. More precisely, metric compatibility ensures that the inner product structure imposed by the metric remains invariant along curves. This allows meaningful operations such as raising and lowering indices and defining lengths and angles between vectors in curved geometry.

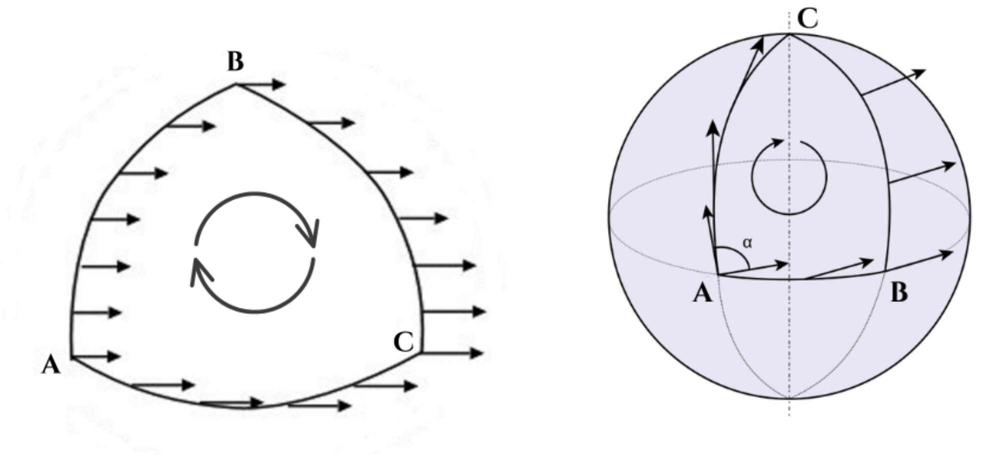
Although one can mathematically define connections for which $\nabla_\beta g_{\mu\nu} \neq 0$, such connections are not compatible with the metric and are generally not useful in the context of GR [33].

1.2.5.3 Parallel transport

To understand the geometry of curved spaces, one must move beyond the familiar tools of Euclidean geometry. In flat space, vectors at different points can be naturally compared, added, subtracted, or evaluated using inner products because transporting a vector from one point to another without changing its direction is unambiguous. However, this intuition breaks down in curved manifolds, where the concept of direction is local and the comparison of vectors at different points becomes nontrivial.

This leads to a fundamental question: how can we define the idea of keeping a vector "unchanged" as we move it along a path in a curved space? The answer lies in the notion of parallel transport. Parallel transport provides a rule for moving vectors along curves in a manifold in such a way that they remain as "constant" as possible according to the geometry of the space.

In flat space, transporting a vector around a closed loop brings it back unchanged. But in curved spaces, the result of parallel transport around a loop generally depends on the path taken, reflecting the intrinsic curvature of the manifold (see Fig. 1.5).



(a) Parallel transport in flat space. (b) Parallel transport on a curved space (sphere).

Figure 1.5: Comparison between parallel transport in flat and curved spaces. In flat space, the vector returns unchanged after a closed loop; in curved space, the final result depends on the path taken, revealing the manifold's intrinsic geometry. Original illustrations based on the sources: [33, 40].

The condition for the parallel transport of the vector $\vec{V}$ along a parametrized curve is that its covariant derivative with respect to the curve's parameter must be zero, namely,

$$\frac{d\vec{V}}{d\lambda} = 0 \; . \tag{1.43}$$

To obtain the derivative, we use equation (1.26) and replace the parameter x^β with λ, obtaining:

$$\frac{d\vec{V}}{d\lambda} = \frac{dV^\alpha}{d\lambda} e_\alpha + V^\alpha \frac{de_\alpha}{d\lambda} \; . \tag{1.44}$$

The change of the basis can be expressed as

$$\frac{de_\alpha}{d\lambda} = \frac{\partial e_\alpha}{\partial x^\beta}\frac{dx^\beta}{d\lambda} = \Gamma^\gamma_{\alpha\beta}\frac{dx^\beta}{d\lambda} e_\gamma \; , \tag{1.45}$$

were we have used the definition of the Christoffel symbols (1.27).

Substituting this result into the previous expression and replacing the index α with γ in the first term yields

$$\frac{d\vec{V}}{d\lambda} = \left(\frac{dV^\gamma}{d\lambda} + V^\alpha \, \Gamma^\gamma_{\alpha\beta} \frac{dx^\beta}{d\lambda} \right) e_\gamma \; . \tag{1.46}$$

We define the absolute derivative[3] along the curve as

$$\frac{DV^\gamma}{d\lambda} \equiv \frac{dV^\gamma}{d\lambda} + \Gamma^\gamma_{\alpha\beta} V^\alpha \frac{dx^\beta}{d\lambda} \ , \tag{1.47}$$

or equivalently,

$$\frac{D}{d\lambda}\vec{V} = \frac{dx^\mu}{d\lambda} \nabla_\mu \vec{V}. \tag{1.48}$$

Parallel transport of $\vec{V}$ along the curve is then defined by the condition

$$\frac{D\vec{V}}{d\lambda} = 0 \quad \Longleftrightarrow \quad \frac{dV^\gamma}{d\lambda} + \Gamma^\gamma_{\alpha\beta} V^\alpha \frac{dx^\beta}{d\lambda} = 0 \ . \tag{1.49}$$

This is a linear first order system of ordinary differential equations for the components $V^\gamma(\lambda)$, which, given initial data $V^\gamma(\lambda_0)$, has a unique solution along the curve.

For a tensor $T^{\mu_1 \cdots \mu_p}{}_{\nu_1 \ldots \nu_q}$ the parallel transport along a trajectory $x^\mu(\lambda)$ is defined by requiring that, for every point on the curve,

$$\left(\frac{D}{d\lambda}T\right)^{\mu_1\mu_2\ldots\mu_p}{}_{\nu_1\nu_2\ldots\nu_q} \ \equiv \ \frac{dx^\sigma}{d\lambda} \nabla_\sigma T^{\mu_1\mu_2\ldots\mu_p}{}_{\nu_1\nu_2\ldots\nu_q} \ \equiv \ 0 \ . \tag{1.50}$$

This expression constitutes a valid tensor equation because both the tangent vector $dx^\mu/d\lambda$ and the covariant derivative ∇_μ transform as tensors.

This equation plays a key role, as we will see in the next section that it serves as the starting point for deriving the geodesic equations. We will also find that freely moving or freely falling objects in spacetime follow the straightest possible paths, curves known as geodesics.

1.2.5.4 Geodesics

In Euclidean space, the shortest path between two points is a straight line. In a curved space, the analogous concept is a geodesic, a curve that, given the curvature of the space, is as "straight" as possible. On a sphere, for example, great circles play this role.

The idea of a geodesic can be formalized in several equivalent ways when considering two points A and B on a surface [31]:

1. The geodesic is the curve that minimizes the distance between A and B.
2. A geodesic is a curve whose length remains stationary under small variations of its shape.
3. A more local definition states that a geodesic is a curve that, at each point, is as straight as possible.

[3] An absolute derivative refers to the covariant derivative taken along a curve, in this case the curve parametrized by λ.

1.2.5.4.1 Geometric derivation

From a geometric perspective, a geodesic is defined as a curve whose tangent vector is parallel transported along itself. Formally, this condition requires that the covariant derivative of the tangent vector vanishes at every point of the curve [1].

Let $x^\mu(\lambda)$ represent a parametrized curve, where λ is an affine parameter. Its tangent vector is given by

$$U^\mu = \frac{dx^\mu}{d\lambda} \ . \tag{1.51}$$

The requirement of parallel transport along the curve is expressed as

$$\frac{DU^\mu}{d\lambda} = 0 \ , \tag{1.52}$$

which, using the explicit form of the covariant derivative, becomes

$$\frac{dU^\mu}{d\lambda} + \Gamma^\mu_{\rho\sigma} U^\rho \frac{dx^\sigma}{d\lambda} = 0 \ . \tag{1.53}$$

Substituting $U^\rho = \frac{dx^\rho}{d\lambda}$, one arrives at the standard geodesic equation [41]:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma} \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0 \ . \tag{1.54}$$

Thus, the geodesic equation emerges naturally as the geometric condition ensuring that the tangent vector remains invariant under parallel transport.

The main physical significance of geodesics in GR lies in the fact that they describe the trajectories of freely falling particles, objects experiencing no proper acceleration. Within this framework, the geodesic equation serves as the relativistic analogue of Newton's second law, $\vec{F} = m\vec{a}$, specifically in the case where $\vec{F} = 0$.

1.2.5.4.2 Normalization condition

To fully characterize a geodesic, one must impose a normalization condition on the tangent vector $\dot{x}^\mu = dx^\mu/d\tau$:

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = \begin{cases} -c^2 & \text{timelike geodesics (massive particles),} \\ \ \ 0 & \text{null geodesics (light rays or massless particles),} \\ +c^2 & \text{spacelike curves (non-physical).} \end{cases} \tag{1.55}$$

This condition follows directly from the spacetime interval $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$. For timelike geodesics, τ is the proper time of the particle; for null geodesics, proper time is undefined and τ becomes an affine parameter. In many cases, particularly when the spacetime admits symmetries, this condition alone suffices to determine the geodesic trajectory without solving the full geodesic equation [41].

1.2.5.5 Riemann curvature tensor

In §1.2.5.3 we established that parallel transport around a closed loop in a curved manifold generically returns a vector in a different orientation from its initial one (see Fig. 1.5). This path-dependence is not an artifact of the coordinate system but reflects the intrinsic geometry of the manifold. The Riemann curvature tensor provides a precise, coordinate-independent quantification of this effect.

To make this concrete, consider the parallel transport condition derived in §1.2.5.3,

$$\frac{DV^\gamma}{d\lambda} = \frac{dV^\gamma}{d\lambda} + \Gamma^\gamma_{\alpha\beta} V^\alpha \frac{dx^\beta}{d\lambda} = 0, \tag{1.56}$$

which governs how the components of a vector change as it is transported along a curve. If we transport V^λ around an infinitesimal closed loop, the net change in its components depends on the order in which we traverse the two directions spanning the loop. This order-dependence can be captured algebraically by examining the commutator of two covariant derivatives: acting first with ∇_μ and then ∇_ν, versus the reverse order. In flat space, partial derivatives commute, so this commutator vanishes identically. In a curved manifold, the Christoffel symbols $\Gamma^\gamma_{\alpha\beta}$ introduce additional terms that survive, encoding the failure of parallel transport to be path-independent.

Consider transporting a vector V^λ around an infinitesimal parallelogram with sides dx^μ and dx^ν (see Fig. 1.6).

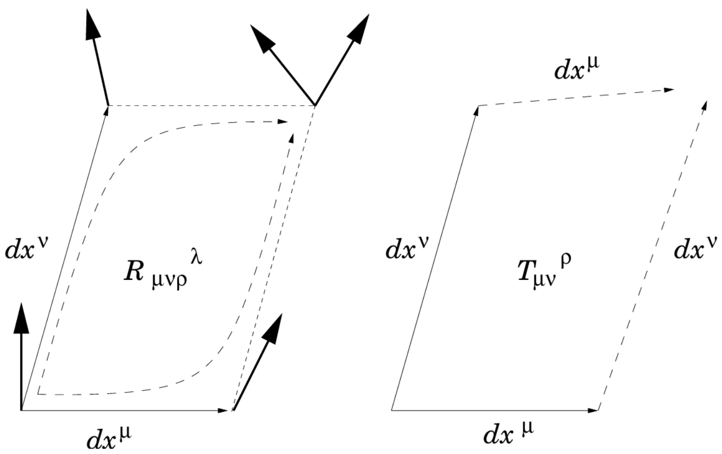


Figure 1.6: Illustration of parallel transport around an infinitesimal closed loop, showing the deviation of a vector due to curvature, characterized by the Riemann tensor $R^\lambda{}_{\mu\nu\rho}$. In the presence of torsion, described by the torsion tensor $T^\rho{}_{\mu\nu}$, the infinitesimal parallelogram fails to close, providing a geometric measure of the antisymmetric part of the connection. Source: [1].

The commutator of two covariant derivatives acting on a vector field V^ρ is defined by

$$[\nabla_\mu, \nabla_\nu]V^\rho \equiv \nabla_\mu\nabla_\nu V^\rho - \nabla_\nu\nabla_\mu V^\rho. \tag{1.57}$$

Expanding each term in terms of the Christoffel symbols $\Gamma^\rho_{\mu\nu}$, we obtain

$$\nabla_\mu\nabla_\nu V^\rho = \partial_\mu(\nabla_\nu V^\rho) + \Gamma^\rho_{\mu\sigma}\nabla_\nu V^\sigma - \Gamma^\sigma_{\mu\nu}\nabla_\sigma V^\rho. \tag{1.58}$$

Substituting $\nabla_\nu V^\rho = \partial_\nu V^\rho + \Gamma^\rho_{\nu\lambda} V^\lambda$ into the above, we get

$$\nabla_\mu \nabla_\nu V^\rho = \partial_\mu \left(\partial_\nu V^\rho + \Gamma^\rho_{\nu\lambda} V^\lambda \right) + \Gamma^\rho_{\mu\sigma} \left(\partial_\nu V^\sigma + \Gamma^\sigma_{\nu\lambda} V^\lambda \right) - \Gamma^\sigma_{\mu\nu} \left(\partial_\sigma V^\rho + \Gamma^\rho_{\sigma\lambda} V^\lambda \right). \tag{1.59}$$

Expanding the derivatives and rearranging terms, this becomes

$$\begin{aligned} \nabla_\mu \nabla_\nu V^\rho = {} & \partial_\mu \partial_\nu V^\rho + (\partial_\mu \Gamma^\rho_{\nu\lambda}) V^\lambda + \Gamma^\rho_{\nu\lambda} \partial_\mu V^\lambda \\ & + \Gamma^\rho_{\mu\sigma} \partial_\nu V^\sigma + \Gamma^\rho_{\mu\sigma} \Gamma^\sigma_{\nu\lambda} V^\lambda - \Gamma^\sigma_{\mu\nu} \partial_\sigma V^\rho - \Gamma^\sigma_{\mu\nu} \Gamma^\rho_{\sigma\lambda} V^\lambda. \end{aligned} \tag{1.60}$$

In order to compute the second term of the commutator, we do the same operation, interchanging the indices ($\mu \leftrightarrow \nu$).

Finally, we obtain:

$$[\nabla_\mu, \nabla_\nu] V^\rho = \left(\partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma} \right) V^\sigma - 2\, \Gamma^\lambda_{[\mu\nu]} \nabla_\lambda V^\rho. \tag{1.61}$$

The first group of terms corresponds to the components of the Riemann curvature tensor,

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma} , \tag{1.62}$$

while the second term involves the antisymmetric part of the connection,

$$T^\lambda{}_{\mu\nu} = \Gamma^\lambda_{\mu\nu} - \Gamma^\lambda_{\nu\mu} , \tag{1.63}$$

known as the torsion tensor. Since we have already introduced the concept of torsion earlier, we can now give a more precise interpretation: torsion quantifies the failure of an infinitesimal parallelogram to close. Specifically, if the vector dx^μ is parallel transported along dx^ν to yield dx'^μ, and dx^ν is parallel transported along dx^μ to yield dx'^ν, these two resulting vectors will not, in general, meet at the same point. In other words, the sequence $dx^\mu \to dx^\nu \to dx'^\mu \to dx'^\nu$ does not necessarily form a closed parallelogram. Only for symmetric connections (such as the Levi-Civita connection) does the torsion tensor vanish, ensuring that the paths dx^μ-dx^ν and dx^ν-dx^μ yield a closed loop [1].

When torsion vanishes, only the curvature term remains, providing a direct measure of the manifold's geometric curvature. In this case, the commutator can be expressed compactly as

$$[\nabla_\mu, \nabla_\nu] V^\rho = R^\rho{}_{\sigma\mu\nu} V^\sigma - \cancel{T^\lambda{}_{\mu\nu} \nabla_\lambda V^\rho}. \tag{1.64}$$

Thus, the Riemann curvature tensor provides a local, coordinate-independent measure of how much the geometry of a manifold deviates from flatness, encapsulating all such effects within a precise mathematical framework.

1.2.5.5.1 Symmetry properties

The Riemann tensor is antisymmetric in its last two indices:

$$R^\rho{}_{\sigma\mu\nu} = -R^\rho{}_{\sigma\nu\mu} , \tag{1.65}$$

and possesses further symmetries that significantly reduce the number of independent components. In an n-dimensional manifold, it initially has n^4 components, but these symmetries lower the count to 1 in two dimensions, 6 in three, and 20 in four (as in spacetime geometry).

It is often convenient to express all indices in lowered form using the metric tensor, as this highlights the tensor's symmetry properties in a more transparent way:

$$R_{\alpha\beta\mu\nu} = g_{\alpha\lambda} R^{\lambda}{}_{\beta\mu\nu} \ . \tag{1.66}$$

In this form, the characteristics of $R_{\alpha\beta\mu\nu}$ can be presented as follows:

- **Antisymmetry in each index pair:** The tensor changes sign when the first two indices are swapped, and likewise for the last two:

  $$R_{\alpha\beta\mu\nu} = -R_{\beta\alpha\mu\nu}, \quad R_{\alpha\beta\mu\nu} = -R_{\alpha\beta\nu\mu} \ . \tag{1.67}$$

  This means that both (α, β) and (μ, ν) behave as antisymmetric index pairs.

- **Symmetry under exchange of index pairs:** Interchanging the first and second pair of indices leaves the tensor unchanged:

  $$R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta} \ . \tag{1.68}$$

  This symmetry links curvature measured in one pair of directions to curvature in the orthogonal pair.

- **First Bianchi identity:** The cyclic sum over the last three indices vanishes:

  $$R_{\alpha\beta\mu\nu} + R_{\alpha\nu\beta\mu} + R_{\alpha\mu\nu\beta} = 0 \ . \tag{1.69}$$

  This is an algebraic identity that holds in any geometry and reflects fundamental constraints on how curvature can be arranged in a manifold.

In the special case of a flat manifold, all curvature components vanish:

$$R^{\alpha}{}_{\beta\mu\nu} = 0 \ , \tag{1.70}$$

implying that parallel transport is path-independent: any vector transported along a closed curve returns to its starting point unchanged [42].

1.2.5.6 Ricci tensor and Ricci scalar

The complete, uncontracted Riemann curvature tensor does not explicitly appear in EFE. Rather, it is contracted to yield two curvature invariants: the Ricci tensor and the Ricci scalar. These quantities are subsequently combined to construct the Einstein tensor, which forms the geometric side (left-hand side) of EFE [33].

1.2.5.6.1 Ricci tensor

By contracting the second index of the Riemann tensor with its fourth index, one obtains the Ricci tensor:

$$R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu} = g^{\rho\lambda} R_{\mu\rho\nu\lambda} \ . \tag{1.71}$$

Therefore, it is expressed in terms of the connection as follows:

$$R_{\mu\nu} = \partial_{\mu}\Gamma^{\lambda}_{\nu\lambda} - \partial_{\nu}\Gamma^{\lambda}_{\mu\lambda} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\rho} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\rho} \ . \tag{1.72}$$

1.2.5.6.2 Ricci scalar

By contracting both indices of the Ricci tensor, one obtains the Ricci scalar or curvature scalar:

$$R = R^{\mu}{}_{\mu} = g^{\mu\nu} R_{\mu\nu} \ . \tag{1.73}$$

As the name suggests, the Ricci scalar is a scalar and thus an invariant under general coordinate transformations.

1.3 Einstein field equations

The mathematical framework developed in the preceding sections, together with the physical insight of the equivalence principle, leads naturally to the central result of GR. The Einstein field equations (EFE) relate the curvature of spacetime, encoded in the Einstein tensor

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \ , \tag{1.74}$$

to the energy and momentum content of matter, described by the energy-momentum tensor $T_{\mu\nu}$ (see §1.3.1):

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \ . \tag{1.75}$$

This is a system of ten coupled, second-order nonlinear partial differential equations whose solutions determine the metric $g_{\mu\nu}$ given a distribution of matter and energy. The constant $8\pi G/c^4$ ensures consistency with Newtonian gravity in the weak-field limit. In vacuum, $T_{\mu\nu} = 0$, the equations reduce to

$$R_{\mu\nu} = \frac{1}{2} g_{\mu\nu} R \ , \tag{1.76}$$

which further simplifies, upon contraction, to $R = 0$ and therefore

$$R_{\mu\nu} = 0 \ , \tag{1.77}$$

known as the vacuum Einstein equations. Although the equations formally involve ten components, the contracted Bianchi identity

$$\nabla^{\mu} G_{\mu\nu} = 0 \ , \tag{1.78}$$

provides four constraints, leaving only six independent equations. The remaining four degrees of freedom reflect the freedom in the choice of coordinates [1].

Einstein's equations are highly nonlinear, which makes them extremely challenging to solve analytically. No general method is known, and exact solutions typically rely on assuming a high degree of symmetry. Nevertheless, they encode a remarkably rich physical content: they describe the propagation of GWs, the bending of light, the expansion of the universe, and the formation of BHs, all consequences of the same geometric framework.

1.3.1 The energy-momentum tensor

The energy-momentum tensor $T^{\mu\nu}$ appears on the right-hand side of the EFE as the source of spacetime curvature. In Newtonian physics, mass alone generates the gravitational field, but special relativity establishes that mass is just one manifestation of energy through

$$E^2 = p^2c^2 + m^2c^4 \ . \tag{1.79}$$

The source of gravity in GR must therefore encompass not only mass but all forms of energy and momentum. The tensor $T^{\mu\nu}$ encapsulates these contributions by describing the flux of four-momentum p^μ across a hypersurface of constant x^ν, and its components can be represented as a 4×4 matrix (see Fig. 1.7): T_{00} is the energy density, $T_{0i} = T_{i0}$ the momentum density or energy flux, and T_{ij} the momentum flux, with diagonal components giving the isotropic pressure and off-diagonal ones the shear stress.

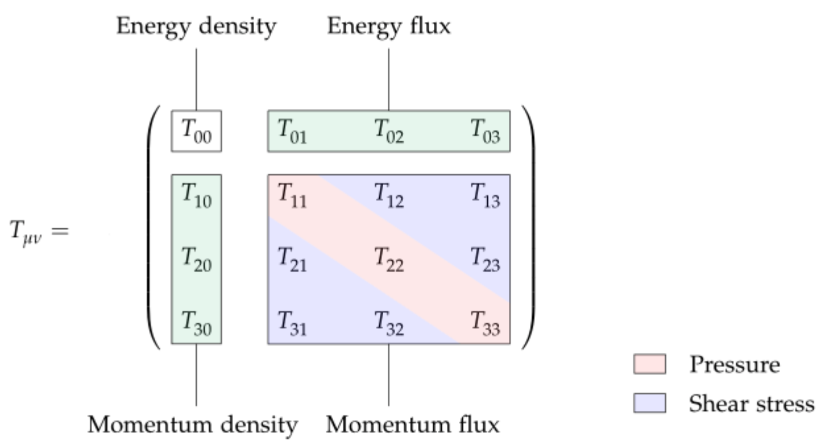


Figure 1.7: Schematic representation of the energy-momentum tensor $T_{\mu\nu}$ as a 4×4 matrix. Source: [43].

For a perfect fluid in flat spacetime, modeled as a system fully characterized by its energy density ρ and isotropic pressure P, the tensor takes the form [41]

$$T^{\mu\nu} = (\rho + P)\, U^\mu U^\nu \pm P\, \eta^{\mu\nu} \ , \tag{1.80}$$

where $U^\mu = \gamma(c, \vec{v})$ is the four-velocity of the fluid. In the comoving frame, $U^\mu = (1, 0, 0, 0)$ and the tensor reduces to $T^{\mu\nu} = \mathrm{diag}(\rho, P, P, P)$.

A key property of $T^{\mu\nu}$ is its conservation law

$$\nabla_\mu T^{\mu\nu} = 0 \ , \tag{1.81}$$

which ensures local conservation of energy and momentum in any physically meaningful system, both in flat and curved spacetime [33].

1.3.2 The cosmological constant

Among the possible modifications of the EFE (1.75), a particularly important one is the introduction of a cosmological constant Λ [41]. Including this term modifies the field equations to

$$R_{\alpha\beta} - \tfrac{1}{2} g_{\alpha\beta} R + g_{\alpha\beta}\Lambda = \frac{8\pi G}{c^4} T_{\alpha\beta} \,. \tag{1.82}$$

Historically, Einstein introduced Λ in 1917 as a way to obtain a static universe without beginning or end, effectively behaving like a repulsive pressure. However, the static solutions were later found to be unstable, and in 1929 Edwin Hubble (1889–1953) discovered that the universe is in fact expanding, a result fully consistent with the original unmodified equations. This led Einstein to famously refer to the cosmological constant as "the biggest blunder" of his life.

Ironically, observations in 1998 with the Hubble Space Telescope revealed that the expansion is accelerating. One possible explanation invokes a pervasive dark energy effectively described by Λ, which has thus regained scientific relevance despite its troubled history [33]. The modern challenge lies in the so-called cosmological constant problem: quantum field theory predicts $\Lambda \sim m_P^4 \sim 10^{76}\,\mathrm{GeV}^4$, while observations constrain it to $\sim 10^{-47}\,\mathrm{GeV}^4$, a discrepancy of roughly 10^{123} and the largest known mismatch between theory and experiment [1].

1.4 Relevant solutions to Einstein's equations

EFE are extremely challenging to solve because of their intrinsic nonlinearity. Unlike linear systems, the superposition of two independent solutions does not in general produce a new one. The physical origin of this nonlinearity is simple: spacetime curvature is generated by the presence of matter and energy, yet the curvature itself carries energy and momentum, thereby acting as an additional source of curvature. In other words, gravity is not only coupled to matter and energy but also to itself, which makes the field equations nonlinear by nature.

Concretely, to "find a solution" means to determine a metric tensor $g_{\mu\nu}$ (and, when relevant, a compatible matter configuration) that satisfies

$$R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu},$$

subject to appropriate boundary or initial conditions.

Each solution corresponds to a possible spacetime geometry associated with a particular distribution of matter and energy. Broadly speaking, they can be grouped into three categories:

- **Exact analytical solutions**: such as the Schwarzschild, Kerr, or Friedmann-Lemaître-Robertson-

Walker metrics.

- **Approximate solutions**: derived through perturbative methods, for instance weak-field expansions describing GW.
- **Numerical solutions**: obtained via large-scale simulations, such as those modeling the merger of two BHs.

The vast variety of solutions illustrates the richness of GR: each one represents a distinct possible universe governed by Einstein's equations.

Einstein himself initially believed that his equations were so complicated that no exact solution would ever be found. Remarkably, only a few months after the formulation of GR in 1915, Karl Schwarzschild (1873–1916) obtained the first exact solution, describing the spacetime of a static, spherically symmetric object. Since then, dozens, indeed hundreds, of exact solutions have been discovered [1]. In this section we will explore the Schwarzschild solution.

1.4.1 The Schwarzschild solution

In vacuum, EFE reduce to $R_{\mu\nu} = 0$. At first sight, one might be tempted to consider the trivial choice $g_{\mu\nu} = 0$. However, this is not allowed: the metric must be a non-degenerate tensor field with a well-defined inverse, a requirement already assumed in the derivation of the field equations from the variational principle[4]. This condition, $\det g_{\mu\nu} \neq 0$, immediately implies that the simplest Ricci-flat[5] solution is not the trivial one but rather Minkowski spacetime, which corresponds to flat geometry in the absence of matter or energy:

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 \ . \tag{1.83}$$

The first non-trivial exact solution of the vacuum equations was found by Karl Schwarzschild in 1916. The Schwarzschild metric describes the spacetime exterior to a static, spherically symmetric, non-rotating mass distribution. In Schwarzschild coordinates (t, r, θ, ϕ), the line element takes the form

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta \, d\phi^2 \, , \tag{1.84}$$

where M denotes the mass of the central object [33].

This solution has profound physical implications. It provides the theoretical framework to compute relativistic corrections to planetary orbits (e.g., the perihelion precession of Mercury), as well as to predict the bending of light by gravity. Most notably, it predicts the existence of BHs, one of the most striking consequences of GR that we will se in this section.

It is important to understand that there are two different Schwarzschild solutions. The first one is the exterior Schwarzschild solution (1.84), which applies to the region outside a spherical body

[4] Ultimately, this constraint likely suggests that the gravitational field is not fundamental and should be replaced by another description in scenarios where $\det g_{\mu\nu}$ becomes small [35].

[5] Metrics that satisfy the vacuum equations are referred to as Ricci-flat. Although Ricci-flat solutions are generally not flat, they still represent valid vacuum solutions (assuming no cosmological constant is present) [1].

of mass M and radius $R_0 > 2GM$. Since this exterior region is empty space, the solution is a vacuum one and describes the gravitational field produced by the object. According to Birkhoff's uniqueness theorem, the exterior Schwarzschild metric constitutes the sole spherically symmetric solution to the vacuum Einstein equations. This result implies, in particular, that no spherically symmetric vacuum solutions exist that are non-static; moreover, whenever such a solution is static, it necessarily corresponds to the Schwarzschild geometry [1].

The second is the interior Schwarzschild solution[6], also formulated by Karl Schwarzschild in 1916, which applies inside the object ($r < R_0$) and models its matter as a perfect fluid. These two solutions are fundamentally different, but when the radius of the object satisfies $R_0 > 2GM$, they match smoothly at the boundary $r = R_0$.

An important characteristic of the Schwarzschild exterior solution is that when the radial coordinate r becomes very large, the metric tends to the Minkowski form expressed in spherical coordinates, ensuring consistency with the classical limit:

$$\lim_{r\to\infty}\left(1-\frac{2GM}{r}\right)=1\ . \tag{1.85}$$

This means that, far away from the source, the gravitational field is essentially negligible and spacetime is indistinguishable from flat Minkowski space[7].

1.4.1.1 The Kretschmann scalar and the Schwarzschild radius

As $r \to 0$, the metric components diverge, signaling a singularity where spacetime curvature becomes infinite and the laws of physics break down [45]. Metrics derived from EFE can present two types of singularities. Coordinate singularities are artifacts of the chosen coordinate system with no physical content: for instance, $g_{\phi\phi} = 0$ at $\theta = 0, \pi$ in spherical coordinates, which vanishes under a simple rotation. Physical singularities correspond to genuine divergences of spacetime curvature, and Hawking and Penrose showed in 1969 that such singularities necessarily exist inside any closed trapped surface [46].

To distinguish between the two, one computes curvature invariants constructed from contractions of the Riemann tensor, which are independent of the coordinate choice. For the Schwarzschild solution, the Kretschmann scalar

$$K \equiv R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48G^2M^2}{r^6} \tag{1.86}$$

diverges only at $r = 0$, confirming a physical singularity there, while remaining finite at $r = 2GM$. The apparent pathology of the metric at $r = 2GM$ is therefore a coordinate singularity. Nevertheless, this radius has profound physical significance: it defines the Schwarzschild radius and marks the event horizon of the BH, a surface beyond which standard Schwarzschild coordinates become inadequate.

[6] The solution is given by $ds^2 = -\frac{1}{4}\left[3\sqrt{1-\frac{2GM}{r}} - \sqrt{1-\frac{2GMr^2}{R_0^3}}\right]^2 dt^2 + \left(1-\frac{2GMr^2}{R_0^3}\right)^{-1} dr^2 + r^2 d\Omega^2$, where R_0 denotes the radius of the object [44].

[7] Spacetimes that exhibit this behavior are referred to as asymptotically flat and can be regarded as representing isolated systems [33].

1.4.1.2 Causal structure and radial null geodesics

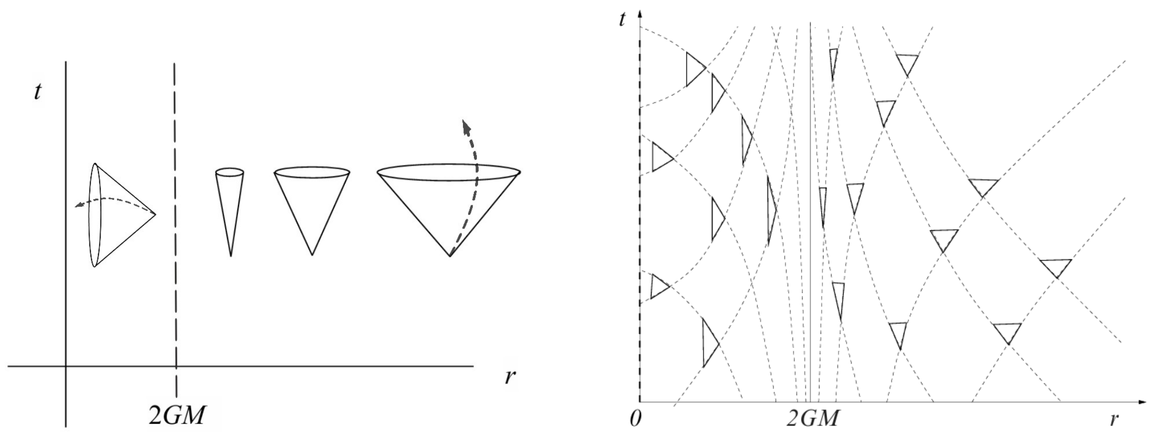
Setting $ds^2 = 0$ and $d\theta = d\phi = 0$, the slope of radial null geodesics in the (t, r) plane is

$$\frac{dt}{dr} = \pm \left(1 - \frac{2GM}{r}\right)^{-1}, \tag{1.87}$$

which recovers the Minkowski value ± 1 at large r but diverges as $r \to 2GM$, causing the light cones to close up (see Fig. 1.8). Inside the horizon, the cones flip orientation entirely, forcing all causal trajectories toward the singularity at $r = 0$. Integrating the null geodesic equation gives

$$t = \pm \left(r + 2GM \ln |r - 2GM| + C_0\right), \tag{1.88}$$

showing that an outgoing light ray requires infinite Schwarzschild coordinate time to reach $r = 2GM$. A distant observer therefore sees an infalling body freeze and fade at the horizon, while in proper time the crossing occurs in a finite interval. The issue lies in the coordinates, not the physics.



(a) Tilting of light cones in Schwarzschild coordinates.

(b) Ingoing and outgoing null geodesics in Schwarzschild coordinates.

Figure 1.8: Light cones in the t-r plane of Schwarzschild spacetime. Far from the BH, cones open symmetrically at $\pm 45°$. As $r \to 2GM$ they progressively close; inside the horizon they invert, pointing inexorably toward $r = 0$. Original illustrations based on: [1, 41].

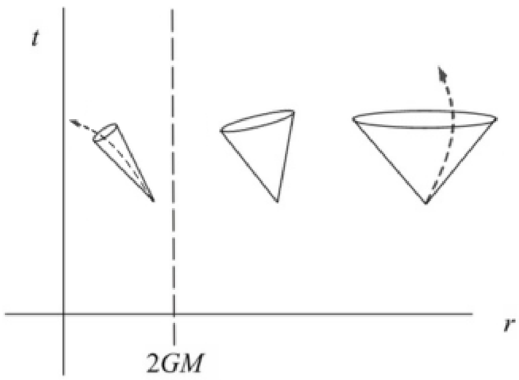
To follow geodesics continuously across the horizon, one introduces the advanced Eddington-Finkelstein coordinates [47],

$$\tilde{t} = t + 2M \log(r - 2M), \tag{1.89}$$

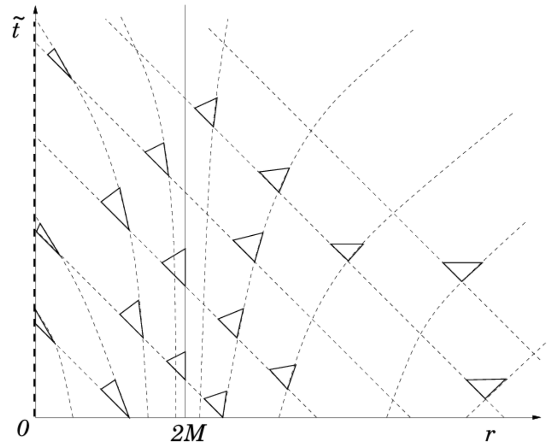
in which the metric becomes

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) d\tilde{t}^2 + \frac{4GM}{r} d\tilde{t}\, dr + \left(1 + \frac{2GM}{r}\right) dr^2 + r^2 d\Omega^2, \tag{1.90}$$

manifestly regular at $r = 2GM$ (see Fig. 1.9).



(a) Light cones in Eddington-Finkelstein coordinates.



(b) Null geodesics in Eddington-Finkelstein coordinates.

Figure 1.9: In Eddington-Finkelstein coordinates the horizon is regular. Ingoing geodesics always remain at 45°; outgoing ones tilt progressively inward until at $r = 2GM$ they are trapped, and for $r < 2GM$ all light falls toward the singularity. Original illustrations based on: [1, 33].

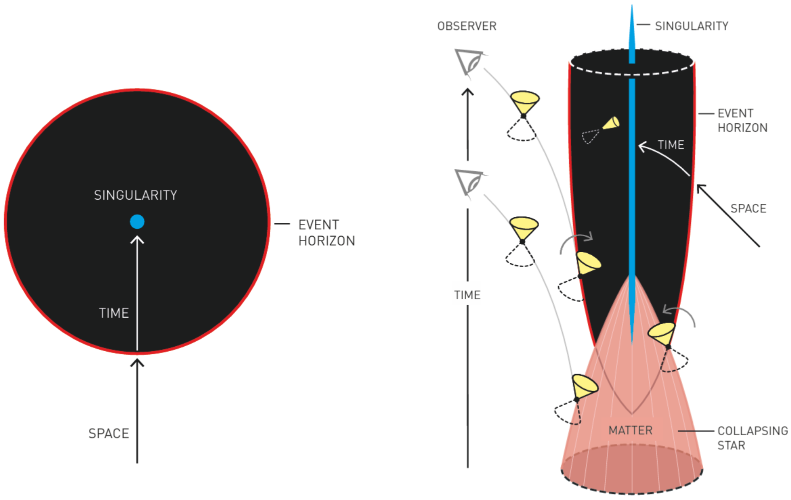
1.4.1.3 Black holes in general relativity

A discussion of GR would be incomplete without addressing BHs. In this section, we have already seen that they naturally emerge from the Schwarzschild solution. We will not delve deeply into their technical details, but we will introduce the essential concepts required to understand, in later sections, the phenomenon of GW.

BHs are regions of spacetime where gravity is so intense that nothing, not even light, can escape. According to GR, they form when matter becomes sufficiently compressed, curving spacetime to the point that a boundary, called the event horizon, appears (see Fig. 1.10a). This surface acts as a one-way limit: once matter or radiation crosses it, escape is no longer possible. Inside, all trajectories inevitably lead toward a central singularity, where curvature becomes infinite (see Fig. 1.10b). Although the term BH was popularized by John Wheeler in 1967, the idea traces back to the 18th century, when John Michell and Pierre-Simon Laplace speculated that a star of sufficient density could trap its own light. Their reasoning was based on Newtonian gravity and the notion of escape velocity. While not fully correct, their intuition coincidentally pointed to the same critical radius later derived by Schwarzschild in GR [33].

In nature, BHs most commonly arise from the evolution of massive stars. For stars like the Sun, nuclear fusion generates enough thermal pressure to counterbalance gravity, maintaining equilibrium for billions of years. Once the fuel is exhausted, smaller stars contract into white dwarfs, supported by electron degeneracy pressure. Chandrasekhar showed in 1931 that such stars cannot exceed about 1.4 $M_\odot$; beyond that threshold, gravity overwhelms electron pressure [33].

More massive stars collapse further, producing NSs, stabilized for a time by neutron degeneracy pressure. However, Oppenheimer and collaborators demonstrated that above roughly two to three solar masses, not even this force is sufficient. The star then undergoes unstoppable collapse, crossing its Schwarzschild radius and forming a BH.



(a) (b)

Figure 1.10: (a) Simplified depiction of a BH. The event horizon (red line) marks the boundary beyond which nothing can escape. Inside, all paths lead to the singularity (blue dot). (b) More detailed diagram showing the causal structure of a Schwarzschild BH. The light cones tip inward as one approaches the horizon, and inside they point entirely toward the singularity, indicating that all causal trajectories inevitably end there. Source: [48].

Astrophysical evidence confirms that stellar collapse is not the only mechanism. Observations show the existence of supermassive BHs, with millions to billions of solar masses, at the centers of galaxies, likely formed through accretion and mergers over cosmic time [1].

According to the so-called no-hair theorem, BHs are surprisingly simple objects. From the perspective of an outside observer, they can be fully described by only three physical quantities: mass M, electric charge Q, and angular momentum J. This gives rise to four main idealized solutions in GR:

- Schwarzschild BHs, defined only by mass.
- Reissner-Nordström BHs, with mass and electric charge.
- Kerr BHs, with mass and rotation.
- Kerr-Newman BHs, with mass, charge, and rotation.

In practice, astrophysical BHs are expected to be nearly neutral, as any net charge would quickly be neutralized by surrounding plasma. Therefore, the most relevant solutions are Schwarzschild (non-rotating) and Kerr (rotating) BHs. The latter are believed to describe most real BHs, as they likely inherit angular momentum from their progenitor stars or accretion disks.

BHs can also be classified by mass:

- **Stellar-mass black holes (SBHs):** Formed from the collapse of massive stars (typically with $M \gtrsim 20M_{\odot}$), they span from a few to hundreds of solar masses. They are often found in binary systems, where they can accrete matter from a companion star and emit X-rays.

- **Intermediate-mass black holes (IMBHs):** These BHs are expected to lie in the mass range 10^2–$10^5 M_\odot$. While no unambiguous detections exist, several candidates have been suggested from low-luminosity active galactic nuclei, X-ray sources and accretion disk spectra. The strongest evidence comes from GW events: GW190521, the merger of $85^{+21}_{-14} M_\odot$ and $66^{+17}_{-18} M_\odot$ BHs that produced a $142^{+28}_{-16} M_\odot$ remnant [49], and the more recent GW231123, where BHs of $137^{+22}_{-17} M_\odot$ and $103^{+20}_{-52} M_\odot$ merged into a $225^{+26}_{-43} M_\odot$ remnant [50].

- **Supermassive black holes (SMBHs):** Located at the centers of galaxies, they range from millions to billions of solar masses. Their origin is still uncertain, but they grow by accretion and mergers. The Milky Way's central BH, Sagittarius A*, has a mass of about $4 \times 10^6 M_\odot$.

In addition to these categories, theorists also propose the existence of primordial black holes (PBHs). These could have formed within the first second after the Big Bang, when extreme densities allowed matter to collapse directly into BHs. Their possible masses are thought to range from as little as 10^{-5} grams up to about $10^5 M_\odot$. Quantum effects predict that lighter PBHs would have evaporated over time, but more massive ones could still remain scattered across the universe [51].

In summary, this chapter has laid the mathematical and physical foundations of GR, progressing from the geometry of curved spacetime and the tools of differential geometry (parallel transport, geodesics, and the Riemann tensor) to the EFE and their most astrophysically relevant solutions: BHs. These concepts constitute the theoretical backbone upon which the remainder of this thesis is built. In the next chapter we turn to the central subject of this work: GWs, which emerge naturally as propagating solutions of the linearized Einstein equations and represent one of the most profound predictions of GR.

Chapter 2

Gravitational wave theory

We have now reached the central topic of this thesis and one of the most remarkable predictions of Einstein's GR: GWs. In this chapter, we introduce their theoretical foundations, beginning with their natural emergence from the linearized EFE (§2.1), and characterize their polarization states and physical action on test masses via the geodesic deviation equation. We then survey the main astrophysical source classes (§2.2) and discuss the detection principle underlying current interferometric observatories, covering antenna pattern functions, noise sources, and the global detector network (§1.10). The chapter closes with an overview of the Bayesian parameter estimation (PE) framework used to extract source properties from GW signals (§1.10), establishing the statistical foundation for the data analysis presented in subsequent chapters.

2.1 Gravitational waves in linearized gravity

In 1916, shortly after formulating the final version of his general theory of relativity, Einstein predicted the existence of GWs [52]. He showed that, within the weak-field or linearized approximation of his equations, spacetime admits wave-like solutions: transverse oscillations of the metric that propagate at the speed of light. These waves are produced by time-dependent variations of the mass quadrupole moment of the source. Einstein also realized that their amplitudes would be extremely small, which led many to doubt their physical reality for decades[1] [54].

GWs can be understood as ripples in the curvature of spacetime that carry energy and information about their astrophysical sources. They are generated by asymmetric acceleration of mass or energy, typically in compact binary systems involving BHs (BH-BH), NSs (NS-NS), or mixed BH-NS mergers. The most direct and elegant derivation of their existence arises from the linearized formulation of GR, obtained by expanding the EFE around the flat Minkowski metric $\eta_{\mu\nu}$ [55].

The first indirect evidence for GWs came from the discovery of the binary pulsar system PSR B1913+16 by Hulse and Taylor in 1974 [56]. Subsequent measurements of its orbital energy loss, provided compelling proof of their existence. This milestone, together with advances in astrophysics, established the foundation for direct GW astronomy [54].

Nearly a century after Einstein's prediction, on September 14, 2015, the LIGO Scientific Collaboration (LSC) achieved the first direct detection of GWs, GW150914 [3], a signal originating from a binary black hole merger (BBH). This event not only confirmed Einstein's theoretical insight but also inaugurated a new era of observational astrophysics, allowing us to probe relativistic systems in the dynamic, strong-field regime of gravity.

[1] Indeed, until the Chapel Hill conference in 1957 [53], the question of whether GWs were genuine physical phenomena remained under debate .

2.1.1 Linearized gravity

GR describes gravity as the manifestation of spacetime curvature. The dynamics of the gravitational field are governed by EFE [55],

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G \, T_{\mu\nu}, \tag{2.1}$$

where $R_{\mu\nu}$ is the Ricci tensor, R the Ricci scalar, and $T_{\mu\nu}$ the energy-momentum tensor.

The curvature of spacetime is encoded in the Riemann tensor, which is constructed from the Christoffel symbols. For a torsion free connection, the Christoffel symbols are defined as [1]

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\lambda} \left(\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu} \right). \tag{2.2}$$

In terms of these quantities, the components of the Riemann curvature tensor are given by

$$R^{\lambda}{}_{\rho\mu\nu} = \partial_\mu \Gamma^{\lambda}_{\rho\nu} - \partial_\nu \Gamma^{\lambda}_{\rho\mu} + \Gamma^{\lambda}_{\sigma\mu} \Gamma^{\sigma}_{\rho\nu} - \Gamma^{\lambda}_{\sigma\nu} \Gamma^{\sigma}_{\rho\mu}. \tag{2.3}$$

The Riemann tensor fully characterizes the intrinsic curvature of spacetime and vanishes identically in flat manifolds. By contracting its indices, one obtains the Ricci tensor [33],

$$R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}, \tag{2.4}$$

and the Ricci scalar,

$$R = g^{\mu\nu} R_{\mu\nu}, \tag{2.5}$$

which together determine the Einstein tensor appearing in the field equations. These geometric quantities provide the starting point for the linearized description of gravity developed in the following sections.

2.1.2 The weak field limit

As a first step toward the understanding of GWs, we wish to study the expansion of the Einstein equations around the flat space metric in the so called weak field limit. Mathematically, we can write this as

$$g_{\mu\nu} \approx n_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1, \tag{2.6}$$

where $\eta_{\mu\nu}$ is the flat space Minkowski metric $\eta_{\mu\nu} = (-,+,+,+)$ and $h_{\mu\nu}$ represents a small perturbation [55].

The assumption that $h_{\mu\nu}$ is small allows us to neglect all terms beyond first order in this quantity. The resulting approximation is known as the linearized theory of gravity, and it leads to

$$g^{\mu\nu} \approx \eta^{\mu\nu} - h^{\mu\nu} + \mathcal{O}(h^2). \tag{2.7}$$

All index operations from this point onward will be performed using the Minkowski metric $\eta_{\mu\nu}$ instead of the full metric $g_{\mu\nu}$ [41]. For instance,

$$h^{\mu\nu} = \eta^{\mu\rho} \eta^{\nu\sigma} h_{\rho\sigma}. \tag{2.8}$$

Our goal is to determine the equation of motion satisfied by the perturbations $h_{\mu\nu}$, which can be obtained by expanding Einstein's equations to first order. We begin by considering the Christoffel symbols (2.2), given by

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\lambda} \left(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu} \right) \approx \frac{1}{2} \eta^{\rho\lambda} \left(\partial_\mu h_{\nu\lambda} + \partial_\nu h_{\mu\lambda} - \partial_\lambda h_{\mu\nu} \right). \tag{2.9}$$

Here, $g^{\rho\lambda}$ has been replaced by $\eta^{\rho\lambda}$, since all terms of order higher than $h_{\mu\nu}$ are neglected. Using these connection coefficients, we can compute the Riemann tensor (2.3) as

$$\begin{aligned} R^{\rho}{}_{\mu\nu\sigma} &= \partial_\nu \Gamma^{\rho}_{\mu\sigma} - \partial_\sigma \Gamma^{\rho}_{\mu\nu} + \Gamma^{\delta}_{\mu\sigma} \Gamma^{\rho}_{\delta\nu} - \Gamma^{\delta}_{\mu\nu} \Gamma^{\rho}_{\delta\sigma} \\ &\approx \partial_\nu \Gamma^{\rho}_{\mu\sigma} - \partial_\sigma \Gamma^{\rho}_{\mu\nu} + \mathcal{O}(h^2) \\ &\approx \frac{1}{2} \eta^{\rho\lambda} \left(\partial_\mu \partial_\nu h_{\lambda\sigma} - \partial_\nu \partial_\lambda h_{\mu\sigma} - \partial_\mu \partial_\sigma h_{\lambda\nu} + \partial_\lambda \partial_\sigma h_{\mu\nu} \right) + \mathcal{O}(h^2). \end{aligned} \tag{2.10}$$

Since the connection coefficients $\Gamma^{\rho}_{\mu\nu}$ are already first order quantities, the only contributions to the Riemann tensor in the linear approximation come from their derivatives, while the quadratic terms in Γ can be safely discarded [41].

From the Riemann tensor derived above, we can obtain the linearized Ricci tensor by contracting the indices ρ and ν:

$$\begin{aligned} R_{\mu\sigma} = R^{\rho}{}_{\mu\rho\sigma} &\approx \frac{1}{2} \eta^{\rho\lambda} (\partial_\mu \partial_\rho h_{\lambda\sigma} - \partial_\rho \partial_\lambda h_{\mu\sigma} - \partial_\mu \partial_\sigma h_{\lambda\rho} + \partial_\lambda \partial_\sigma h_{\mu\rho}) + \mathcal{O}(h^2) \\ &= \frac{1}{2} \Big(\partial_\mu \partial^\lambda h_{\lambda\sigma} - \partial_\rho \partial^\rho h_{\mu\sigma} - \partial_\mu \partial_\sigma h^{\rho}{}_{\rho} + \partial^\rho \partial_\sigma h_{\mu\rho} \Big) + \mathcal{O}(h^2) \\ &= \frac{1}{2} \Big(\partial_\mu \partial^\lambda h_{\lambda\sigma} + \partial^\rho \partial_\sigma h_{\mu\rho} - \Box h_{\mu\sigma} - \partial_\mu \partial_\sigma h \Big) + \mathcal{O}(h^2), \end{aligned} \tag{2.11}$$

where $h \equiv h^{\rho}{}_{\rho} \equiv \eta^{\rho\sigma} h_{\rho\sigma} + \mathcal{O}(h^2)$ is the trace of $h_{\mu\nu}$ and $\Box \equiv \partial_\rho \partial^\rho \equiv \eta^{\rho\sigma} \partial_\rho \partial_\sigma$ is the D'Alembertian operator [33].

The Ricci scalar is obtained by contracting the Ricci tensor with $\eta^{\mu\sigma}$:

$$\begin{aligned} R \approx \eta^{\mu\sigma} R_{\mu\sigma} &\approx \frac{1}{2} \Big(\partial_\mu \partial^\lambda h^{\mu}_{\lambda} + \partial^\rho \partial_\sigma h^{\sigma}_{\rho} - \Box h - \partial_\mu \partial^\mu h \Big) \\ &= \partial_\sigma \partial^\rho h^{\sigma}_{\rho} - \Box h = \partial^\nu \partial^\rho h_{\nu\rho} - \Box h, \end{aligned} \tag{2.12}$$

where we have interchanged the dummy indices $(\lambda \leftrightarrow \rho)$ and $(\mu \leftrightarrow \sigma)$ in the first term, and used $h^{\sigma}_{\rho} = \eta^{\sigma\nu} h_{\rho\nu}$ in the final equality.

Once we have these, using the expressions (2.11) and (2.12) we can compute the linearization of the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - (1/2)\eta_{\mu\nu} R$ for a given source $T_{\mu\sigma}$ and obtain the linearized form of the EFE (2.1)

$$\frac{1}{2} \Big(\partial_\mu \partial^\lambda h_{\lambda\sigma} + \partial^\rho \partial_\sigma h_{\mu\rho} - \Box h_{\mu\sigma} - \partial_\mu \partial_\sigma h \Big) - \frac{1}{2} \eta_{\mu\sigma} \left(\partial^\nu \partial^\rho h_{\nu\rho} - \Box h \right) = 8\pi G T_{\mu\sigma}. \tag{2.13}$$

Since a weak gravitational field corresponds to small spacetime curvature, we can safely assume the absence of matter sources, setting $T_{\mu\sigma} = 0$ we obtain

$$G_{\mu\sigma} \equiv \partial_\mu \partial^\lambda h_{\lambda\sigma} + \partial^\rho \partial_\sigma h_{\mu\rho} - \Box h_{\mu\sigma} - \partial_\mu \partial_\sigma h - \eta_{\mu\sigma} \left(\partial^\nu \partial^\rho h_{\nu\rho} - \Box h \right) = 0. \tag{2.14}$$

To simplify this expression, we introduce the trace reversed perturbation [55], defined as

$$\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h. \tag{2.15}$$

Observe that the trace of the perturbation satisfies $\bar{h} \equiv \eta^{\mu\nu}\bar{h}_{\mu\nu} = h - 2h = -h$, which allows us to invert the definition in (2.15) to obtain

$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h}. \tag{2.16}$$

Using this relation, we can now express the Einstein tensor entirely in terms of the trace reversed perturbation:

$$\begin{aligned} G_{\mu\sigma} \equiv\, & \partial_\mu\partial^\lambda\left(\bar{h}_{\lambda\sigma} - \tfrac{1}{2}\eta_{\lambda\sigma}\bar{h}\right) + \partial^\rho\partial_\sigma\left(\bar{h}_{\mu\rho} - \tfrac{1}{2}\eta_{\mu\rho}\bar{h}\right) - \Box\left(\bar{h}_{\mu\sigma} - \tfrac{1}{2}\eta_{\mu\sigma}\bar{h}\right) - \partial_\mu\partial_\sigma(-\bar{h}) \\ & - \eta_{\mu\sigma}\left[\partial^\nu\partial^\rho\left(\bar{h}_{\nu\rho} - \tfrac{1}{2}\eta_{\nu\rho}\bar{h}\right) - \Box(-\bar{h})\right] = \partial_\mu\partial^\lambda\bar{h}_{\lambda\sigma} - \tfrac{1}{2}\partial_\mu\partial_\sigma\bar{h} + \partial^\rho\partial_\sigma\bar{h}_{\mu\rho} \\ & - \tfrac{1}{2}\partial_\sigma\partial_\mu\bar{h} - \Box\bar{h}_{\mu\sigma} + \tfrac{1}{2}\eta_{\mu\sigma}\Box\bar{h} + \partial_\mu\partial_\sigma\bar{h} - \eta_{\mu\sigma}\left(\partial^\nu\partial^\rho\bar{h}_{\nu\rho} - \tfrac{1}{2}\Box\bar{h} + \Box\bar{h}\right). \end{aligned} \tag{2.17}$$

Finally, by collecting and simplifying all terms, we obtain the compact linearized form of the EFE in vacuum, written in terms of the trace reversed perturbation $\bar{h}_{\mu\nu}$:

$$G_{\mu\sigma} \equiv \partial_\mu\partial^\lambda\bar{h}_{\lambda\sigma} + \partial^\rho\partial_\sigma\bar{h}_{\mu\rho} - \Box\bar{h}_{\mu\sigma} - \eta_{\mu\sigma}\partial^\nu\partial^\rho\bar{h}_{\nu\rho} = 0. \tag{2.18}$$

2.1.2.1 Gauge freedom and the Lorenz condition

With the linearized field equations at hand, the next step is to address the issue of gauge freedom. In GR, the theory remains covariant under arbitrary coordinate transformations. This means that even when we decompose the metric as a small perturbation around flat spacetime $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, the choice of coordinates is not unique. There can exist other coordinate systems in which the metric can still be expressed as the Minkowski background plus a small perturbation, but with a different $h_{\mu\nu}$. Hence, this decomposition is not unique [41].

In the linear approximation, an infinitesimal coordinate transformation

$$x^\mu \to x'^\mu = x^\mu + \xi^\mu(x), \tag{2.19}$$

induces a transformation of the perturbation tensor given by

$$h'_{\mu\nu} = h_{\mu\nu} - \partial_\mu\xi_\nu - \partial_\nu\xi_\mu, \tag{2.20}$$

where the derivatives $|\partial_\mu\xi_\nu|$ are taken to be of the same perturbative order as the metric perturbations $h_{\mu\nu}$.

To simplify the equations, we use the trace reversed perturbation (2.15) [55]:

$$\bar{h}'_{\mu\nu} = \bar{h}_{\mu\nu} - \partial_\mu\xi_\nu - \partial_\nu\xi_\mu + \eta_{\mu\nu}\partial_\rho\xi^\rho. \tag{2.21}$$

Taking the divergence of this expression leads to

$$\partial^\nu \bar{h}'_{\mu\nu} = \partial^\nu \bar{h}_{\mu\nu} - \Box \xi_\mu. \tag{2.22}$$

We can now exploit the coordinate freedom by choosing the vector ξ^μ to satisfy

$$\Box \xi^\mu = \partial^\nu \bar{h}_{\mu\nu}. \tag{2.23}$$

With this choice, the transformed perturbation fulfills

$$\partial^\nu \bar{h}'_{\mu\nu} = 0, \tag{2.24}$$

which defines the Lorenz gauge condition [55] (also called the harmonic gauge or the De Donder gauge):

$$\partial^\nu \bar{h}_{\mu\nu} = 0. \tag{2.25}$$

Imposing the Lorenz condition simplifies the linearized Einstein equations (2.18) to

$$\Box \bar{h}_{\mu\nu} = 0, \tag{2.26}$$

which is the standard wave equation in flat spacetime.

2.1.2.2 Solving the wave equation and the Transverse-Traceless gauge

We are now interested in solving the linearized equation (2.26) to find the form of GWs. The general solution of the linearized wave equation can be written as a superposition of plane waves:

$$\bar{h}_{\mu\nu}(x) = \mathrm{Re}\left\{ A_{\mu\nu} e^{i k_\alpha x^\alpha} \right\}, \tag{2.27}$$

where $A_{\mu\nu}$ is a complex, symmetric polarization tensor and k_α is the wave four-vector [57].

Substituting this ansatz into the wave equation (2.26), we obtain

$$k^\sigma k_\sigma = 0. \tag{2.28}$$

This implies that, for non-trivial solutions $\bar{h}_{\mu\nu} \neq 0$, the wave four-vector must be null. That is, the perturbations propagate at the speed of light[2] (GWs in vacuum) [41].

Furthermore, applying the Lorenz gauge condition (2.25) to the wave equation (2.27), we can prove that:

$$A_{\mu\nu} k^\mu = 0, \tag{2.29}$$

which shows that GWs are transverse with respect to their direction of propagation.

After applying the Lorenz gauge condition (2.24), there still remains a residual gauge freedom that allows further simplification. This residual freedom is a consequence of the fact that the Lorenz gauge does not completely eliminate coordinate freedom. As follows from (2.22), under

[2] A null wave four-vector satisfies $k^\mu k_\mu = 0$, which in Minkowski spacetime reads $\omega^2 - |\mathbf{k}|^2 = 0$. This relation implies $\omega = |\mathbf{k}|$, so the phase velocity of the perturbation is $v = \omega/|\mathbf{k}| = c$. Therefore, GWs correspond to massless excitations of the gravitational field that propagate at the speed of light in vacuum [35].

the transformation (2.19) the quantity $\partial^\nu \bar{h}_{\mu\nu}$ transforms homogeneously. As a result, the gauge condition $\partial^\nu \bar{h}_{\mu\nu} = 0$ remains invariant under additional coordinate transformations of the form $x^\mu \to x^\mu + \xi^\mu$, provided that the vector field ξ^μ satisfies

$$\Box \xi^\mu = 0. \tag{2.30}$$

If $\Box \xi^\mu = 0$, then the symmetric tensor

$$\xi_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu - \eta_{\mu\nu} \partial_\rho \xi^\rho, \tag{2.31}$$

also fulfills $\Box \xi_{\mu\nu} = 0$, since the flat d'Alembertian operator commutes with partial derivatives. Consequently, solutions of the wave equation for $\bar{h}_{\mu\nu}$ remain invariant under the subtraction of such terms.

This residual gauge freedom allows one to impose four additional conditions on $h_{\mu\nu}$. In particular, one may choose the gauge such that the trace vanishes, $h = 0$, implying $\bar{h}_{\mu\nu} = h_{\mu\nu}$. Furthermore, the remaining gauge freedom can be used to set $h_{0\mu} = 0$. The Lorenz condition then reduces to

$$\partial^j h_{ij} = 0, \tag{2.32}$$

while the traceless condition becomes $h^i{}_i = 0$.

In summary, we can impose the conditions

$$h_{0\mu} = 0, \qquad h^i{}_i = 0, \qquad \partial^j h_{ij} = 0, \tag{2.33}$$

which define the transverse-traceless (TT) gauge [55]. The first condition indicates that GWs possess only spatial components (since the symmetry enforces $h_{\mu 0} = h_{0\mu} = 0$). The second condition implies that the perturbation is traceless, meaning the sum of the diagonal elements of the perturbation matrix vanishes. Finally, the last relation reveals that the perturbation is transverse, such that the amplitude tensor is orthogonal to the direction of propagation.

As a result, the degrees of freedom of $A_{\mu\nu}$ are reduced from ten to only two independent components, indicating that a GW has only two physical degrees of freedom, corresponding to its two polarization states [58].

An important feature of working in the TT gauge is that the vanishing trace condition implies

$$\bar{h}_{\mu\nu} = h_{\mu\nu}. \tag{2.34}$$

Therefore, the plane wave solution can be expressed as

$$h^{\mathrm{TT}}_{\mu\nu} = A_{\mu\nu} e^{i k_\alpha x^\alpha}, \tag{2.35}$$

where $h^{\mathrm{TT}}_{\mu\nu}$ represents the metric perturbation in the TT gauge.

2.1.3 Polarization states

Considering a wave propagating in the z-direction, we can choose the wave four-vector as $k^\mu = (\omega/c, 0, 0, \omega/c)$. The Lorenz gauge (2.25) implies $A_{\mu 0} + A_{\mu 3} = 0$[3] and the additional TT gauge conditions lead to $A_{\mu 0} = 0$. Thus, the only non-zero components of the amplitude tensor are A_{11}, A_{22}, and $A_{12} = A_{21}$. The traceless condition $A^i{}_i = 0$ further implies that $A_{11} + A_{22} = 0$, allowing us to express the TT perturbation as

$$h^{TT}_{\mu\nu}(t,z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h^{TT}_{+} & h^{TT}_{\times} & 0 \\ 0 & h^{TT}_{\times} & -h^{TT}_{+} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & A_{+} & A_{\times} & 0 \\ 0 & A_{\times} & -A_{+} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}_{\mu\nu} e^{-i\omega(t-z/c)}, \tag{2.36}$$

where h^{TT}_{+} and $h^{TT}_{\times}$ represent the two independent and orthogonal polarization states of the GW, commonly referred to as the plus and cross modes [35]. Returning from the complex to the real domain, the strain can finally be expressed as

$$h^{TT}_{\mu\nu}(t,z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & A_{+} & A_{\times} & 0 \\ 0 & A_{\times} & -A_{+} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}_{\mu\nu} \cos\left[\omega\left(t - z/c\right)\right], \tag{2.37}$$

where A_{+} and $A_{\times}$ denote the amplitudes of the two polarization modes.

2.1.4 Effect of a gravitational wave on test masses

In order to describe the physical influence of a GW on a given system, we analyze the motion of test particles in its presence. A coordinate independent characterization of this effect is provided by the relative motion of nearby particles. This behavior is governed by the geodesic deviation equation [41], which captures how a GW modifies the separation between neighboring geodesics

$$a^\mu = \frac{D^2 S^\mu}{dt^2} = R^\mu{}_{\nu\rho\sigma}\, U^\nu U^\rho S^\sigma, \tag{2.38}$$

where S^μ denotes the separation between two neighboring geodesics, $U^\mu = dx^\mu/dt$ is the tangent vector to the reference geodesic, a^μ represents the relative acceleration between the two test particles and $R^\mu{}_{\nu\rho\sigma}$ is the Riemann curvature tensor,

Before analyzing the effect of a GW on an extended system of test masses, it is instructive to examine its influence on a single particle. We consider a particle initially at rest in flat spacetime, prior to the arrival of the wave. An inertial reference frame is attached to this particle, with the x-axis chosen along the direction of propagation of an incoming TT GW. The particle subsequently follows a geodesic of the perturbed spacetime generated by the wave.

[3] If the gauge condition requires $h_{\mu 0} = 0$ for all spacetime points and the metric perturbation is written as $h_{\mu\nu} = A_{\mu\nu} e^{ik_\rho x^\rho}$, the exponential factor never vanishes. Therefore, the only way for $h_{\mu 0}$ to be identically zero is that the corresponding amplitude components vanish $A_{\mu 0} = 0$ (and, by symmetry, $A_{0\mu} = 0$) [35].

The motion of the particle is governed by the geodesic equation,

$$\frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = \frac{dU^\alpha}{d\tau} + \Gamma^\alpha_{\mu\nu}U^\mu U^\nu = 0. \tag{2.39}$$

At $\tau = 0$, the particle is at rest in this frame, so that $U^\alpha = (1, 0, 0, 0) + \mathcal{O}(h)$. The acceleration induced by the GW at this instant is therefore given by

$$\left.\frac{dU^\alpha}{d\tau}\right|_{\tau=0} = -\Gamma^\alpha_{00} = -\frac{1}{2}\eta^{\alpha\beta}\left(h_{\beta 0,0} + h_{0\beta,0} - h_{00,\beta}\right). \tag{2.40}$$

However, in the TT gauge all components $h_{0\mu}$ vanish. As a consequence,

$$\left.\frac{dU^\alpha}{d\tau}\right|_{\tau=0} = 0. \tag{2.41}$$

The four-velocity thus remains constant at all times, implying that the particle experiences no coordinate acceleration due to the passage of the GW. Its coordinate position therefore remains unchanged. This result shows that the motion of a single freely falling particle is insufficient to reveal the presence of a GW. Detectable effects arise only when considering the relative motion between multiple nearby test masses [58].

Having established that the motion of a single freely falling particle does not provide a direct observable signature of a GW, we now turn to the physically relevant case of a system of test particles. Observable effects arise from the relative motion between nearby particles, which is invariant under coordinate transformations.

We consider a congruence of neighbouring timelike geodesics describing freely falling test particles. In the absence of gravitational radiation, these particles are initially at rest with respect to a common inertial frame, so that their four-velocity is

$$U^\mu = (1, 0, 0, 0) + \mathcal{O}(h). \tag{2.42}$$

At some reference time $t = 0$, a plane GW propagating along the z-direction passes through the system. We work throughout in the TT gauge, where the only non vanishing components of the metric perturbation lie in the plane orthogonal to the direction of propagation.

The relative motion of neighbouring particles is governed by the geodesic deviation equation (2.38). In the linearised regime of gravity, the Riemann tensor is already first order in the metric perturbation $h_{\mu\nu}$. Consequently, it is sufficient to evaluate the equation using the unperturbed four-velocity $U^\mu = (1, 0, 0, 0)$, and to identify the coordinate time t with the proper time along the geodesics, up to corrections of order $\mathcal{O}(h^2)$.

Using the alredy computed expression for the linearised Riemann tensor in the TT gauge,

$$R^\rho{}_{\mu\nu\sigma} = \frac{1}{2}\eta^{\rho\lambda}\left(\partial_\mu\partial_\nu h_{\lambda\sigma} - \partial_\nu\partial_\lambda h_{\mu\sigma} - \partial_\mu\partial_\sigma h_{\lambda\nu} + \partial_\lambda\partial_\sigma h_{\mu\nu}\right) + \mathcal{O}(h^2), \tag{2.43}$$

one finds that the only non vanishing components relevant for the geodesic deviation equation are

$$R^{i}{}_{00j} = \frac{1}{2}\eta^{\rho\lambda}\left(\partial_0\partial_0 h_{\lambda\sigma} - \partial_0\partial_\lambda h_{0\sigma} - \partial_0\partial_\sigma h_{\lambda 0} + \partial_\lambda\partial_\sigma h_{00}\right) + \mathcal{O}(h^2) = \frac{1}{2}\,\partial_t^2 h_{ij} + \mathcal{O}(h^2), \tag{2.44}$$

where $i, j = 1, 2, 3$ label spatial coordinates. The equation of motion for the spatial components of the separation vector then reduces to

$$\frac{d^2 S^i}{dt^2} = \frac{1}{2}\,\partial_t^2 h_{ij}^{TT}\, S^j. \tag{2.45}$$

This result shows explicitly that the GW induces tidal forces only in the plane transverse to the direction of propagation. For a wave travelling along the z-axis, the longitudinal component S^3 remains unaffected, while the transverse components (S^1, S^2) encode the physical deformation of the system.

Integrating the equation we find

$$S^i(t) = S^i(0) + \frac{1}{2}\,h_{ij}^{TT}(t)\,S^j(0). \tag{2.46}$$

For a general superposition of the two polarizations, the metric perturbation in the TT gauge can be written as

$$h_{ij}^{TT}(t,z) = \begin{pmatrix} A_+ & A_\times & 0 \\ A_\times & -A_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}_{ij} e^{-i\omega(t - z/c)}. \tag{2.47}$$

Restricting to the spatial components and inserting this expression into the solution of the geodesic deviation equation (2.46), one obtains

$$\begin{aligned} S^x(t) &= \left(1 + \frac{1}{2}A_+ e^{-i\omega(t-z/c)}\right) S^x(0) + \frac{1}{2}A_\times e^{-i\omega(t-z/c)}\, S^y(0), \\ S^y(t) &= \left(1 - \frac{1}{2}A_+ e^{-i\omega(t-z/c)}\right) S^y(0) + \frac{1}{2}A_\times e^{-i\omega(t-z/c)}\, S^x(0), \\ S^z(t) &= S^z(0), \end{aligned} \tag{2.48}$$

which are coupled equations for the transverse components of the separation vector.

Since the geodesic deviation equation is linear in the metric perturbation $h_{\mu\nu}^{TT}$, the response of the system to a GW is also linear. As a consequence, the effects produced by a general superposition of the two independent polarizations can be obtained as the sum of the responses to each polarization separately. It is therefore sufficient to analyze the cases of purely $+$ and purely $\times$ polarized waves independently, without loss of generality [35].

If the cross polarization vanishes, $A_\times = 0$, the metric perturbation reduces to

$$h_{ij}^{TT}(t,z) = \begin{pmatrix} A_+ & 0 & 0 \\ 0 & -A_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}_{ij} e^{-i\omega(t - z/c)}. \tag{2.49}$$

The corresponding evolution of the separation vector is then

$$
\begin{aligned}
S^x(t) &= \left(1 + \frac{1}{2} A_+ e^{-i\omega(t-z/c)}\right) S^x(0), \\
S^y(t) &= \left(1 - \frac{1}{2} A_+ e^{-i\omega(t-z/c)}\right) S^y(0), \\
S^z(t) &= S^z(0),
\end{aligned}
\tag{2.50}
$$

where as mentioned in (2.37), we should take the real part of the exponential. This polarization produces an oscillatory stretching and squeezing of the particle distribution along the transverse axes (see Fig. 2.1).

If the plus polarization vanishes, $A_+ = 0$, the metric perturbation becomes

$$
h_{ij}^{TT}(t,z) = \begin{pmatrix} 0 & A_\times & 0 \\ A_\times & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}_{ij} e^{-i\omega(t-z/c)}. \tag{2.51}
$$

In this case, the solution of the geodesic deviation equation reads

$$
\begin{aligned}
S^x(t) &= S^x(0) + \frac{1}{2} A_\times e^{-i\omega(t-z/c)} S^y(0), \\
S^y(t) &= S^y(0) + \frac{1}{2} A_\times e^{-i\omega(t-z/c)} S^x(0), \\
S^z(t) &= S^z(0).
\end{aligned}
\tag{2.52}
$$

This polarization induces a shear deformation in the transverse plane, with principal axes rotated by 45° with respect to the x and y directions (see Fig. 2.1).

2.1.5 Generation of gravitational waves

In the previous sections we described GWs as propagating perturbations of the space-time metric and analyzed their effect on matter. We now turn to the inverse question: how dynamical matter sources generate gravitational radiation.

A conceptual subtlety immediately appears. GW emission can take place in strongly non-linear regimes, such as the late stages of CBCs, where the full Einstein equations are required to describe the dynamics near the source. However, the radiation detected far from the source is typically very weak, so that the metric can be written as a small perturbation of flat space-time. This justifies studying GW generation in the local wave zone within the framework of linearized gravity (see Fig. 2.2).

We therefore consider the linearized Einstein equations in the presence of matter, assuming weak gravitational fields and slowly moving sources, so that the background metric can be approximated as Minkowski and internal velocities are small compared to the speed of light. Under these conditions the problem becomes analytically tractable and closely analogous to electromagnetic radiation from accelerated charges [34].

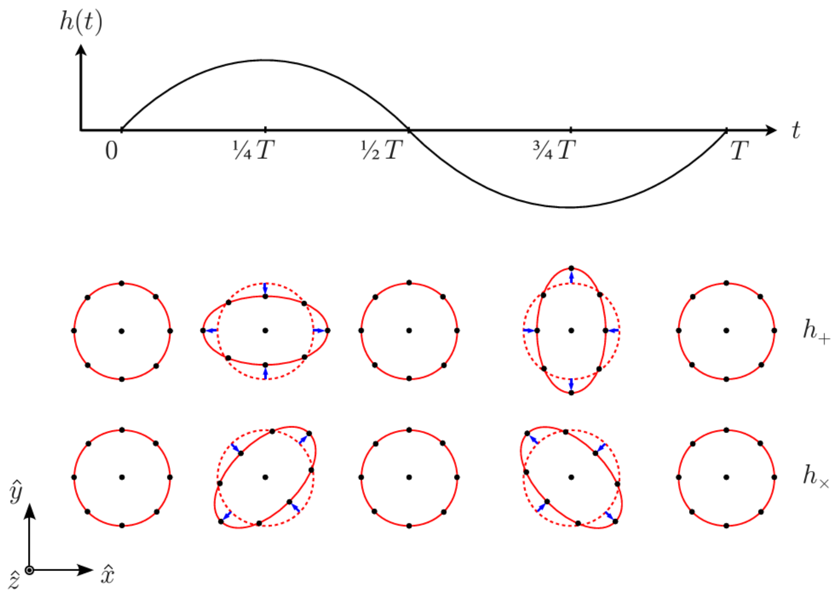


Figure 2.1: A monochromatic GW with angular frequency $\omega = 2\pi/T$ propagates in the $\hat{z}$ direction. The lower panel illustrates the deformation produced by the + and $\times$ polarization modes on a ring of freely falling test particles in a local inertial frame. Source: [34].

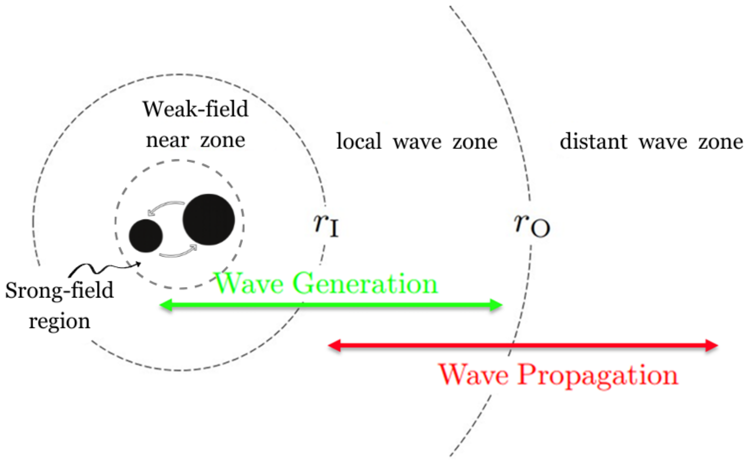


Figure 2.2: Spatial regions associated with a compact binary source of GWs. The strong-field region, weak-field near zone, and the local and distant wave zones are indicated, separated by the characteristic scales r_I and r_O. GW generation predominantly occurs in the near zone (green arrow), while wave propagation is described in the distant wave zone (red arrow). Original illustration based on the sources: [59, 60].

In contrast with electromagnetism, conservation laws strongly constrain gravitational radiation. Mass conservation excludes monopole radiation, while conservation of linear and angular momentum eliminates dipole contributions. Consequently, the leading radiative term arises from the time variation of the mass quadrupole moment of the source.

2.1.5.1 Einstein's quadrupole formula

The starting point is the linearized Einstein equation (2.26) in the presence of matter, written in harmonic gauge,

$$\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}, \tag{2.53}$$

where $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h$ denotes the trace-reversed metric perturbation and $T_{\mu\nu}$ is the energy-momentum tensor of the source (see §1.3.1). The Lorenz gauge condition $\partial^\mu \bar{h}_{\mu\nu} = 0$ is consistent provided that energy-momentum is conserved, $\partial^\mu T_{\mu\nu} = 0$.

As in electromagnetism, this set of decoupled wave equations can be solved using Green's function techniques [55]. For an isolated source localized within a finite spatial region Σ, the retarded solution is

$$\bar{h}_{\mu\nu}(t, \mathbf{x}) = \frac{4G}{c^4} \int_\Sigma \frac{T_{\mu\nu}(t_{\rm ret}, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \, d^3x', \tag{2.54}$$

where the retarded time is

$$t_{\rm ret} = t - \frac{|\mathbf{x} - \mathbf{x}'|}{c}. \tag{2.55}$$

This expression explicitly enforces causality: the metric perturbation at $(t, \mathbf{x})$ depends on the configuration of the source at the earlier time $t_{\rm ret}$ (see Fig. 2.3) [35].

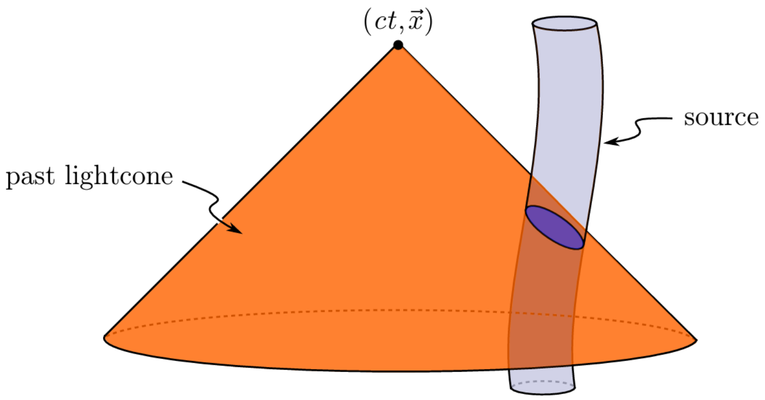


Figure 2.3: Representation of the past light cone of the observation event $(ct, \mathbf{x})$. GWs propagate at the finite speed c, so the metric perturbation measured at the detector is determined by the configuration of the source at the retarded time $t' = t - |\mathbf{x} - \mathbf{y}|/c$. Only the portion of the source worldtube intersecting the past light cone contributes to the observed gravitational field, ensuring causal propagation. Source: [34].

We are interested in the radiation observed far from the source, in the wave zone, where $r = |\mathbf{x}|$ is much larger than both the source size R and the gravitational wavelength λ. In this regime one expands

$$\mathbf{x} - \mathbf{x}'| \simeq r - \mathbf{n} \cdot \mathbf{x}', \qquad \mathbf{n} = \frac{\mathbf{x}}{r}, \tag{2.56}$$

and keeps only the leading terms in $1/r$ and in v/c.

Conservation laws severely restrict the possible radiative multipoles. By Birkhoff's theorem, any spherically symmetric mass distribution produces a static exterior solution, so a time-varying monopole term cannot radiate. Furthermore, the first time derivative of the mass dipole moment is proportional to the total linear momentum of the system, which is conserved for an isolated source. As a result, both monopole and dipole radiation are absent in GR.

The leading contribution to gravitational radiation therefore arises at quadrupole order. Keeping the lowest non-vanishing term in the multipole expansion and restricting to non-relativistic sources, one finds that the dominant spatial components of the metric perturbation are

$$\bar{h}_{ij}(t, \mathbf{x}) = \frac{2G}{c^4 r} \ddot{I}_{ij}(t - r/c), \tag{2.57}$$

where

$$I_{ij}(t) = \int_{\mathcal{S}} \rho(t, \mathbf{x})\, x_i x_j \, d^3x, \tag{2.58}$$

is the (Newtonian) mass moment of inertia tensor of the source $\mathcal{S}$ and ρ is the (Newtonian) mass density.

At this stage the solution is still written in harmonic gauge and is not yet transverse nor traceless. The radiative degrees of freedom are obtained by extracting the TT part of the spatial metric perturbation. Outside the source, where $T_{\mu\nu} = 0$, this can be done by applying the TT projection operator,

$$h_{ij}^{\mathrm{TT}} = \Lambda_{ij,kl}\, \bar{h}_{kl}, \tag{2.59}$$

with

$$\Lambda_{ij,kl} = P_{ik} P_{jl} - \frac{1}{2} P_{ij} P_{kl}, \qquad P_{ij} = \delta_{ij} - n_i n_j. \tag{2.60}$$

It is convenient to introduce the traceless mass quadrupole moment,

$$\mathcal{I}_{ij}(t) = \int \rho(t, \mathbf{x}) \left(x_i x_j - \frac{1}{3} r^2 \delta_{ij} \right) d^3x, \tag{2.61}$$

which differs from I_{ij} only by a trace term. Since the TT projection automatically removes traces, one may equivalently replace I_{ij} by $\mathcal{I}_{ij}$ in the radiative formula.

Combining the far-zone expansion with the TT projection, the metric perturbation takes the form

$$h_{ij}^{\mathrm{TT}}(t, \mathbf{x}) = \frac{2G}{c^4 r} \Lambda_{ij,kl}\, \ddot{\mathcal{I}}_{kl}(t - r/c). \tag{2.62}$$

This is Einstein's quadrupole formula: the gravitational radiation measured far from an isolated source is proportional to the second time derivative of its traceless mass quadrupole moment, evaluated at the retarded time [34]. In particular, for compact binary systems the symmetry of the quadrupole tensor implies that the dominant gravitational radiation oscillates at twice the orbital frequency.

The quadrupole moment formalism provides the leading-order description of GW emission from isolated systems. Although originally derived by Einstein for non-gravitationally driven sources, it was later shown to hold also for self-gravitating systems in the weak-field, Newtonian regime. In this limit, the quadrupole moment is computed from the Newtonian mass density, and the resulting metric perturbation in the TT gauge gives the dominant radiative contribution, with higher-order multipoles and relativistic corrections appearing at post-Newtonian order [61].

2.1.5.2 Radiated energy

A system that radiates GWs necessarily undergoes an energy loss. It is therefore important to quantify the amount of energy carried away by the radiation. In other words, we aim to determine the energy flux transported by GWs away from the source.

The formula for the energy flux of GWs can be derived by considering the effective stress-energy tensor of the gravitational field in the wave zone. In the TT gauge, the energy flux per unit area is given by

$$\frac{dE}{dAdt} = \frac{c^3}{16\pi G}\langle \dot{h}_{ij}^{\mathrm{TT}} \dot{h}_{ij}^{\mathrm{TT}} \rangle, \tag{2.63}$$

where the angle brackets denote an average over several wavelengths or periods of the wave. This expression shows that the energy flux is proportional to the square of the time derivative of the metric perturbation, reflecting the fact that GWs carry energy away from their sources. Integrating the energy flux over a sphere of radius r centered on the source, one obtains the total power radiated in GWs,

$$P = \frac{c^3 r^2}{16\pi G}\int \langle \dot{h}_{ij}^{\mathrm{TT}} \dot{h}_{ij}^{\mathrm{TT}} \rangle d\Omega, \tag{2.64}$$

where the integral is taken over the solid angle Ω. Substituting the quadrupole formula for h_{ij}^{TT}, one finds the well-known expression for the power radiated by a source with mass quadrupole moment $\mathcal{I}_{ij}$,

$$P = \frac{G}{5c^5}\langle \dddot{\mathcal{I}}_{ij} \dddot{\mathcal{I}}_{ij} \rangle, \tag{2.65}$$

where the triple dot denotes the third time derivative. This formula quantifies the energy loss due to GW emission in terms of the third time derivative of the mass quadrupole moment of the source. For compact binary systems, this energy loss leads to a gradual inspiral of the orbit, which can be observed as a characteristic chirp signal in the GW waveform [55].

2.2 Astrophysical sources

GW astronomy identifies four broad categories of astrophysical signals, each characterized by distinct time-scales, amplitudes, and expected sources. These categories are commonly referred to as transient (compact binary coalescences (CBCs) and bursts), continuous waves (CWs) and stochastic gravitational wave background (SGWB) (see Fig. 2.4) [62].

2.2.1 Compact binary coalescences

CBCs are systems formed by two compact objects such as BHs, NSs, or a combination of both, which inspiral and merge due to the emission of GWs (see Fig. 2.5).

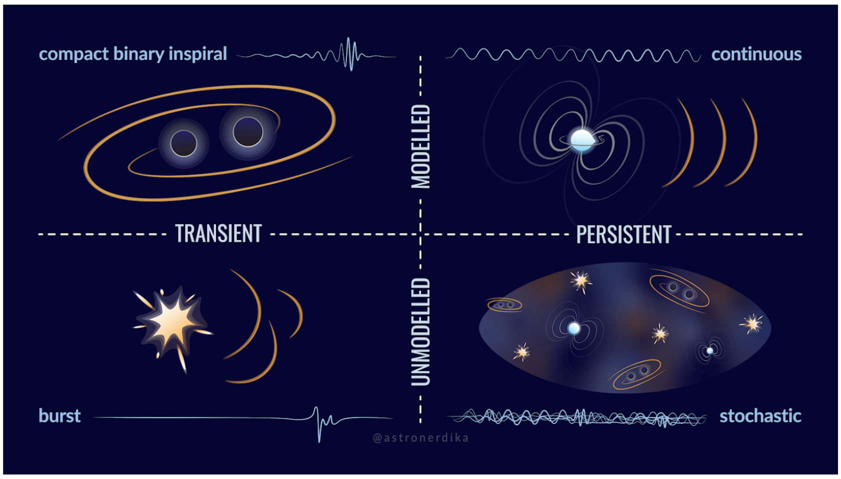


Figure 2.4: Classification of GW sources according to their temporal character (transient vs. persistent) and the availability of accurate waveform models (modelled vs. unmodelled), encompassing CBCs, burst events, continuous waves, and the stochastic background. Credit: Shanika Galaudage (@astronerdika).

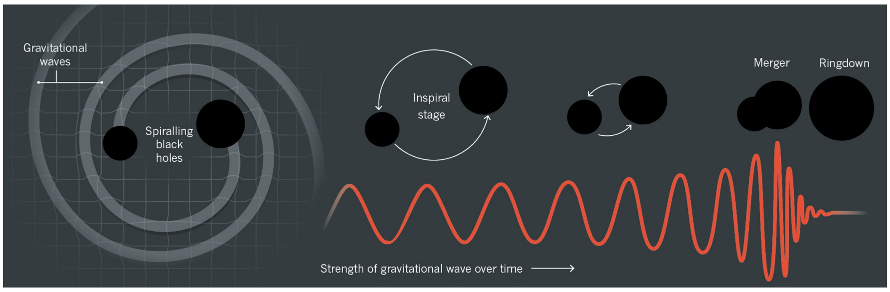


Figure 2.5: The three phases of a CBC: the inspiral, during which the two compact objects lose energy through GW emission and progressively approach each other; the merger, when the objects collide and coalesce; and the ringdown, in which the remnant settles into a stationary configuration by radiating the remaining perturbations. The corresponding gravitational waveform, showing the characteristic chirp of increasing frequency and amplitude, is displayed beneath each phase. Credit: Illustration by Lucy Reading-Ikkanda.

These systems are classified as BBHs, binary neutron stars (BNS), or NS and BH binaries (NSBH). The emitted signal is driven by the time varying quadrupole moment and exhibits a characteristic chirp pattern, with increasing frequency and amplitude as the orbit shrinks. The duration of the signal within the detector's sensitive frequency band (§2.3.3) depends on the total mass of the system: the rate at which the orbital frequency sweeps through this band scales with the chirp mass, so more massive systems evolve rapidly and merge at lower frequencies, leaving only fractions of a second of observable signal for BBH mergers, whereas the lower-mass BNS systems sweep slowly through the band and can remain visible for tens of seconds to minutes before merging [55].

To date, all GW detections made by ground based interferometers, including the LIGO Scientific Collaboration and the Virgo Collaboration, are consistent with CBC sources. During O1 and O2 [63], 11 events were identified, including the first BNS detection, GW170817 [64]. The number of detections increased significantly during O3 [65, 66] to around 90 events. During O4a [67], 128 new CBC candidates were identified, bringing the total to 218 when combined with previous catalogs. The most recent release, GWTC-5.0, extends the catalog through the end of O4b and adds 104 further observations, bringing the cumulative total to over 300 confirmed events [68], with binary BH mergers continuing to dominate the observed population.

The success in detecting CBCs is largely due to accurate waveform modelling, which enables matched filtering techniques to identify signals in noisy data [69]. Future space based detectors such as the Laser Interferometer Space Antenna (LISA) (see §2.3.5.2) will extend observations to other sources, including white dwarf binaries and extreme mass ratio inspirals (EMRIs), providing further insight into compact object populations and their formation [70].

2.2.2 Burst events

GW bursts are short duration signals produced by highly energetic and often poorly modelled astrophysical events. Typical sources include core collapse supernovae [71], gravitational collapses, highly eccentric compact binary mergers, and more speculative scenarios such as cosmic strings [72]. In addition, burst like emission can be associated with transient phenomena such as gamma ray bursts linked to NS or NS-BH mergers.

These signals typically last from a fraction of a second to a few tens of seconds and are characterized by their unpredictable morphology. Unlike CBCs or CWs, burst signals do not have accurate waveform models, which makes their detection more challenging. Instead of matched filtering techniques, searches rely on identifying excess power or coherent transients across multiple detectors [73].

2.2.3 Continuous waves

CWs are expected to be emitted by rapidly rotating compact objects, particularly NSs with small deviations from perfect axial symmetry. These asymmetries produce a time varying quadrupole moment, enabling the continuous emission of gravitational radiation. The resulting signal is nearly monochromatic, typically at a frequency related to the star's rotation, and evolves very slowly due to spindown effects [74].

In contrast to CBCs, CW signals are extremely long lived but much weaker in amplitude, as they originate from small deformations in the source. As a result, their detection requires long observation times, often spanning months or years, in order to accumulate sufficient signal-to-noise ratio (SNR). Despite their low amplitude, once detected, these signals can be tracked over extended periods, allowing for precise studies of the source properties.

The strength of the emission depends on factors such as the rotation rate and internal structure of the NS, including its equation of state and crust properties [74]. In addition to isolated NSs, other potential sources of CWs have been proposed, such as boson clouds [75] around rapidly spinning BHs or compact binaries in very early inspiral stages. Although no CW signal has been confirmed so far, current searches have placed increasingly stringent upper limits, improving the prospects for future detections [76].

2.2.4 Stochastic backgrounds

The SGWB is a diffuse signal produced by the superposition of many unresolved sources, which are either too weak or too numerous to be individually detected. This background can have both astrophysical and cosmological origins [77]. Astrophysical contributions arise from the cumulative effect of distant CBCs or populations of compact binaries such as WD systems, while cosmological contributions may be generated in the early Universe through processes such as inflation or phase transitions.

Although no definitive detection has yet been achieved by ground based detectors, current analyses have placed increasingly stringent upper limits on the energy density of the background [78]. Detecting the SGWB would provide unique information about the population of unresolved sources and could offer a direct probe of physical processes in the early Universe, complementing observations of the cosmic microwave background.

2.3 Detectors

The first experimental attempts to detect GWs began in the 1960s with resonant-mass detectors, which aimed to measure the mechanical response of massive bars to passing waves [79]. However, their limited sensitivity and narrow frequency range prevented successful detections. This motivated the development of interferometric detectors, capable of measuring extremely small changes in distance between test masses. After decades of technological progress, this approach led to the first direct detection of GWs in 2015 [54].

Today, GW observations are routinely performed by a global network of ground-based interferometers, marking the beginning of GW astronomy.

2.3.1 Interferometric detection principle

Current GW detectors are based on laser interferometry, using a modified Michelson configuration, as illustrated in Fig. 2.6. A coherent laser beam is split into two perpendicular arms by a beam splitter. The light travels along each arm, is reflected by mirrors, and recombines at the beam splitter, producing an interference pattern that is measured by a photodetector [80].

In the absence of a GW, the lengths of the two arms are adjusted so that the interference at the output is destructive, resulting in minimal detected light. When a GW passes through the detector, it induces a differential stretching and compression of the arms, analogous to the tidal deformation described in §2.1.4 by the geodesic deviation equation (2.38). This changes the optical path lengths and produces a phase difference between the two beams, leading to a measurable signal at the photodetector.

The quantity measured is the strain $h(t)$, defined as the relative change in the arm lengths, $h \sim \Delta L/L$ (see §2.3.2.1). This is precisely the quantity appearing in the TT metric perturbation (2.37): the two polarization amplitudes A_+ and $A_\times$ directly set the scale of the fractional length change experienced by the detector arms. For typical detectors with arm lengths of a few kilometers, the expected variations are extremely small, of order 10^{-21}, requiring highly sensitive

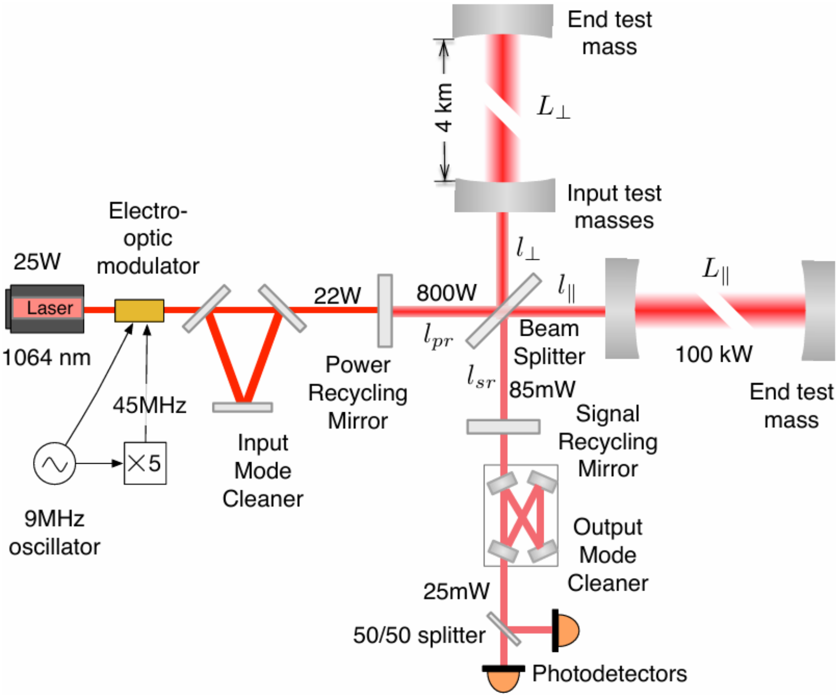


Figure 2.6: Optical layout of the advanced LIGO interferometer, illustrating the Fabry-Pérot arm cavities, the power recycling mirror (PRM) at the input port, and the signal recycling mirror (SRM) at the output port. Source: [80].

instrumentation.

Modern interferometers incorporate several enhancements to increase their sensitivity. These include Fabry-Pérot cavities in the arms, which effectively increase the optical path length, and power and signal recycling mirrors, which enhance the circulating power and the detector response. Despite these improvements, the measurement remains extremely challenging due to the tiny magnitude of the effect, set ultimately by the quadrupole formula discussed in §2.1.5.1 and the astrophysical distances involved.

2.3.2 Detector response

The response of an interferometric detector depends on the relative orientation between the incoming GW and the detector arms. This is described by antenna pattern functions, which determine the sensitivity to different sky locations and polarization states. As a result, a single detector has limited directional sensitivity, making a network of detectors essential for accurate sky localization and PE.

2.3.2.1 Antenna pattern of a ground-based interferometric detector

Here we derive the response of a ground-based Michelson interferometer to a plane GW in the long-wavelength approximation, where the detector arm length is much smaller than the reduced wavelength of the wave.

A complete description of the detection problem requires three distinct Cartesian frames [81]. The wave frame (x', y', z') has its z'-axis aligned with the propagation direction; in this frame the TT metric perturbation takes the standard form

$$H = \begin{pmatrix} h_+ & h_\times & 0 \\ h_\times & -h_+ & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{2.66}$$

where h_+ and $h_\times$ denote the plus and cross polarizations. The detector frame $(\bar{x}, \bar{y}, \bar{z})$ has its $\bar{z}$-axis pointing toward the local zenith and the two arms lying in the $(\bar{x}, \bar{y})$ plane along unit vectors $\hat{\mathbf{n}}_1$ and $\hat{\mathbf{n}}_2$, separated by an opening angle ζ. Finally, an ecliptic frame (x, y, z) centered at the Solar System Barycenter serves as a common inertial reference for source localization. The relative orientation of the wave and detector frames is parametrized by the polar angles $(\bar{\theta}_N, \bar{\phi}_N)$ specifying the source direction in the detector frame, and the polarization angle ψ (see Fig. 2.7).

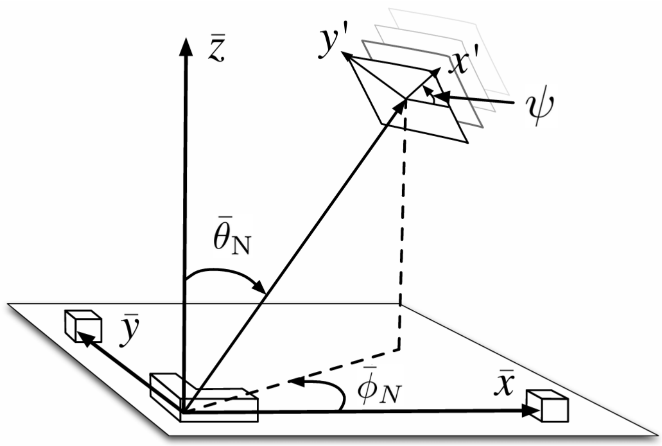


Figure 2.7: Geometric relationship between the detector frame $(\bar{x}, \bar{y}, \bar{z})$ and the wave frame (x', y', z'). The interferometer arms lie in the $(\bar{x}, \bar{y})$ plane, with $\bar{z}$ pointing toward the local zenith. The angles $\bar{\theta}_N$ and $\bar{\phi}_N$ give the polar and azimuthal coordinates of the incoming wave direction in the detector frame, and ψ denotes the rotation of the polarization axes about the propagation direction. Own production based on the source: [82].

The observable quantity is the differential arm-length variation induced by the passing wave. Working to first order in h and applying the geodesic deviation equation (2.45) in the detector frame, the fractional length change of a single arm oriented along $\hat{\mathbf{n}}$ is

$$\frac{\delta L(t)}{L} = \frac{1}{2}\,\hat{\mathbf{n}}^T \tilde{H}(t)\,\hat{\mathbf{n}}, \tag{2.67}$$

where $\tilde{H}(t) = MH(t)M^T$ is the wave-frame strain tensor (2.66) rotated into the detector frame by the orthogonal matrix M. The scalar strain measured by the interferometer is then

$$h(t) = \frac{\Delta L_{\mathrm{arms}}(t)}{L} = \frac{L_1(t) - L_2(t)}{L} = \frac{\delta L_1(t)}{L} - \frac{\delta L_2(t)}{L} = \frac{1}{2}\,\hat{\mathbf{n}}_1^T \cdot \tilde{H}(t)\,\hat{\mathbf{n}}_1 - \frac{1}{2}\,\hat{\mathbf{n}}_2^T \cdot \tilde{H}(t)\,\hat{\mathbf{n}}_2, \tag{2.68}$$

where the sign convention follows from choosing $\hat{\mathbf{n}}_1 \times \hat{\mathbf{n}}_2$ to point outward from the Earth's surface [83].

The frame transformation is given by the sequence of Euler rotations

$$M = R_z(-\bar{\phi}_N)\, R_y(-\bar{\theta}_N + \pi)\, R_z(\psi), \tag{2.69}$$

which encodes both the sky location of the source and the orientation of the polarization axes relative to the detector.

Substituting the explicit form of M and the arm vectors into (2.68), and simplifying with standard trigonometric identities, the detector response takes the form

$$h(t) = F_+(\psi, \zeta, \bar{\theta}_N, \bar{\phi}_N)\, h_+(t) + F_\times(\psi, \zeta, \bar{\theta}_N, \bar{\phi}_N)\, h_\times(t), \tag{2.70}$$

where the antenna pattern functions are [81, 83]

$$F_+ = \sin\zeta \left[\frac{1}{2}(1+\cos^2\bar{\theta}_N)\cos 2\bar{\phi}_N \cos 2\psi - \cos\bar{\theta}_N \sin 2\bar{\phi}_N \sin 2\psi\right], \tag{2.71}$$

$$F_\times = -\sin\zeta \left[\frac{1}{2}(1+\cos^2\bar{\theta}_N)\cos 2\bar{\phi}_N \sin 2\psi + \cos\bar{\theta}_N \sin 2\bar{\phi}_N \cos 2\psi\right]. \tag{2.72}$$

These expressions satisfy the symmetry relations

$$F_+(\psi + \tfrac{\pi}{4}) = F_\times(\psi), \qquad F_{+,\times}(\psi + \tfrac{\pi}{2}) = -F_{+,\times}(\psi), \tag{2.73}$$

reflecting the spin-2 character of GWs: a $\pi/4$ rotation of the polarization angle exchanges the two modes, while a $\pi/2$ rotation reverses their sign.

A polarization-independent measure of the directional sensitivity is provided by the quadrature sum

$$|F| = \sqrt{F_+^2 + F_\times^2}, \tag{2.74}$$

shown in Fig. 2.8 for $\zeta = \pi/2$. The characteristic quadrupolar structure reflects the tensorial nature of the GW strain and the geometric projection onto the detector arms. The individual patterns F_+ and $F_\times$ are displayed in Fig. 2.9 for several values of ψ; each successive column corresponds to a $\pi/8$ increment in polarization angle, illustrating the interchange symmetry above.

2.3.3 Noise and sensitivity

A GW with strain $h \sim 10^{-21}$ produces a differential arm-length change $\delta L = hL/2 \sim 10^{-18}\,\mathrm{m}$ for $L = 4\,\mathrm{km}$, three orders of magnitude smaller than the diameter of a proton. Achieving sensitivity at this level requires identifying and suppressing a broad set of noise sources that would otherwise overwhelm the signal. These are conveniently characterised by the one-sided power spectral density (PSD) $S_n(f)$ [84], defined as the Fourier transform of the autocorrelation function[4] of the detector output:

$$S_n(f) = \int C(\tau)\, e^{-i2\pi f\tau}\, d\tau, \tag{2.75}$$

[4] The autocorrelation function measures how strongly a signal is correlated with a delayed version of itself as a function of the time lag τ. In the context of detector noise, it quantifies the degree to which noise fluctuations at different times remain statistically related [85].

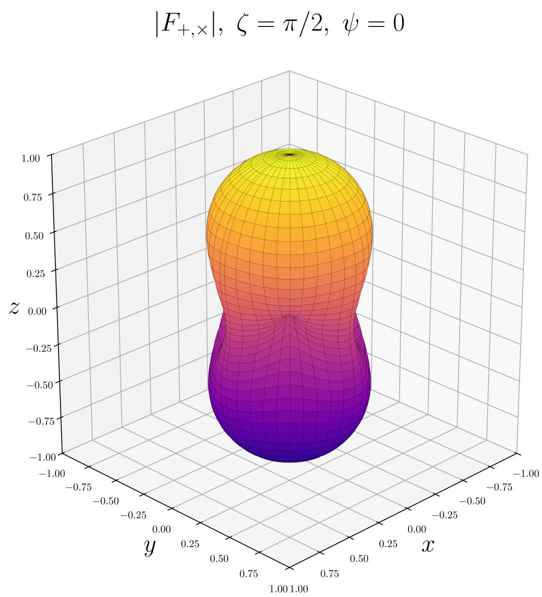


Figure 2.8: Polarization-averaged antenna pattern $|F|$ of a ground-based interferometer with orthogonal arms ($\zeta = \pi/2$), as a function of sky angles $(\bar{\theta}_N, \bar{\phi}_N)$. Original result.

where $C(\tau) = \langle n(t)n(t+\tau)\rangle$ is the autocorrelation function of the noise process $n(t)$, and $\langle\cdot\rangle$ denotes an ensemble average. Its square root $\sqrt{S_n(f)}$, expressed in units of $\mathrm{Hz}^{-1/2}$, defines the sensitivity curve of the instrument and sets the threshold below which signals are inaccessible.

Noise sources are broadly divided into two categories. Fundamental noise arises directly from the measurement principle and cannot be removed without redesigning the detector. Technical noise is not intrinsic to the design but still limits performance. The main contributions are the following [86, 87]:

- **Shot noise.** The Poissonian arrival statistics of photons at the output photodetector produce a phase uncertainty scaling as $1/\sqrt{N_\gamma}$, where N_γ is the number of detected photons per unit time. This contribution decreases with increasing circulating power and dominates the sensitivity above $\sim$200 Hz. It can be partially mitigated by injecting squeezed vacuum at the output port, a technique already incorporated in the Advanced LIGO and Advanced Virgo detectors.

- **Radiation-pressure noise.** Amplitude fluctuations of the intracavity field exert a stochastic force on the mirror surfaces, inducing position noise that grows with laser power and becomes significant at low frequencies. Shot noise and radiation-pressure noise are related by the Heisenberg uncertainty principle and together define the standard quantum limit for a free test mass. Squeezed-light injection redistributes the quantum uncertainty between the two quadratures, allowing simultaneous reduction of both contributions below the standard quantum limit.

- **Thermal noise.** Any component at finite temperature undergoes stochastic mechanical motion. Via the fluctuation-dissipation theorem[5], the amplitude of these fluctuations is

[5] The fluctuation-dissipation theorem states that any dissipative mechanism in a system in thermal equilibrium is necessarily accompanied by a fluctuating force of the same origin. In interferometric detectors, mechanical losses

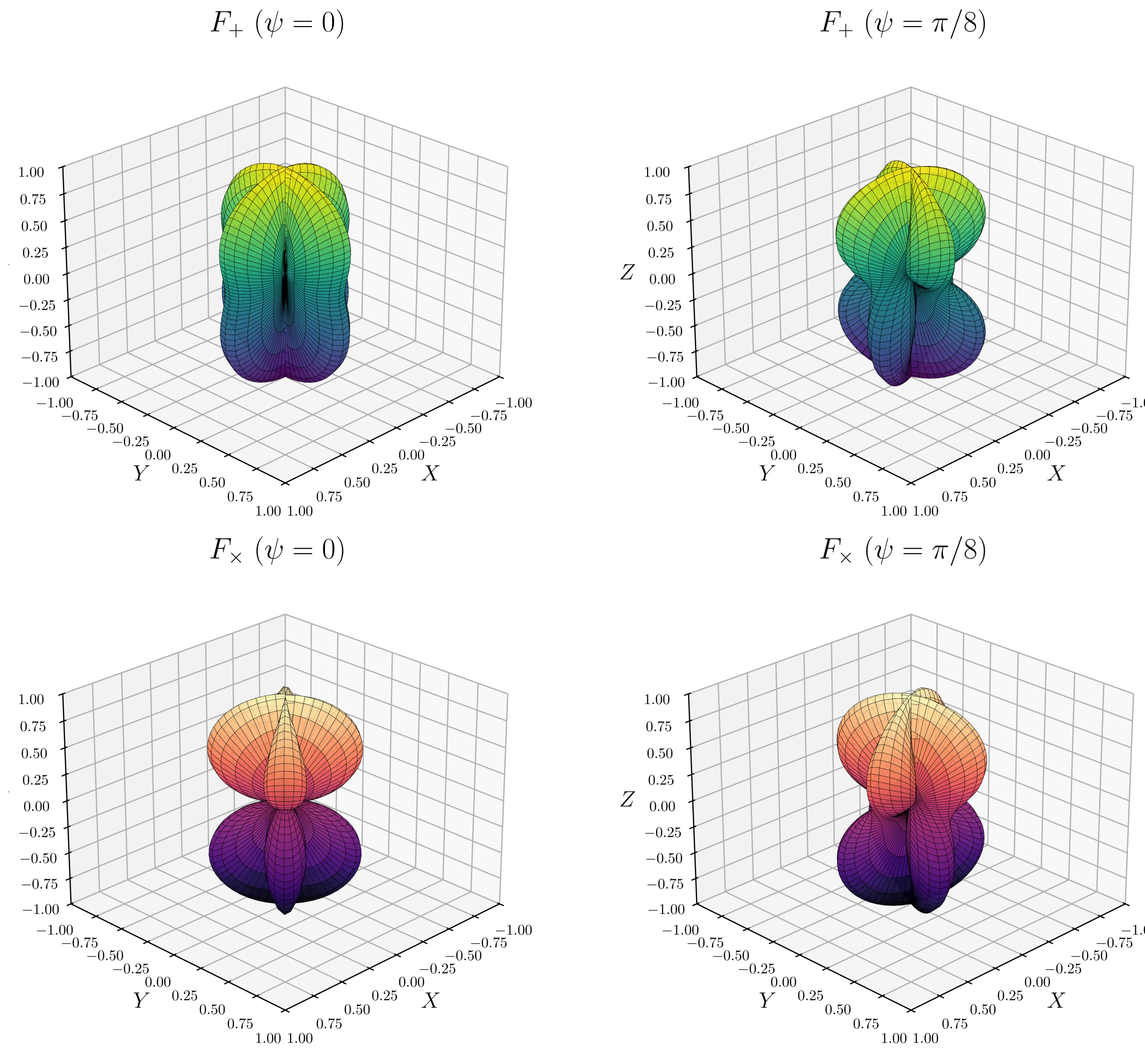


Figure 2.9: Antenna patterns for the plus (F_+, top) and cross ($F_\times$, bottom) polarizations for $\zeta = \pi/2$ and representative values of ψ. Original result.

proportional to the mechanical loss angle of the material. It affects the test masses, their dielectric mirror coatings, and the suspension fibers, and dominates the sensitivity in the intermediate band of roughly 40-200 Hz. Low-loss fused-silica substrates and fibers are employed to minimize this contribution; cryogenic operation, as adopted by KAGRA, provides an alternative route to its suppression.

- **Seismic noise.** Ground motion couples mechanically to the test masses through the suspension chain. The quadruple-pendulum isolation system attenuates seismic vibrations by more than ten orders of magnitude above $\sim$10 Hz, but residual motion from microseisms, wind, and anthropogenic activity still limits the sensitivity below $\sim$40 Hz.

- **Newtonian noise.** Density fluctuations in the surrounding ground and atmosphere produce direct gravitational forces on the test masses that cannot be shielded mechanically. This contribution, also known as gravity-gradient noise, sets a fundamental floor at low

in the test masses, coatings, and suspension fibers therefore set a fundamental limit on their positional stability [88].

frequencies for surface-based detectors.

- **Control noise.** Servo loops are required to maintain the arm cavities on resonance and keep the optical components aligned. The sensors and actuators in these feedback systems introduce noise that couples into the GW channel, representing the dominant technical limitation at low frequencies.

- **Residual gas noise.** Despite the ultra-high vacuum environment ($< 1\,\mu$Pa in the beam tubes), residual molecules scatter laser photons and cause fluctuations in the optical path length. This contribution is generally subdominant but must be controlled to avoid limiting the sensitivity in specific frequency bands.

- **Other technical noise sources.** Laser frequency and intensity fluctuations, electromagnetic coupling, scattered light, and photodetector dark noise contribute across various frequency ranges. These are monitored and subtracted using a network of $\gtrsim 2 \times 10^5$ auxiliary channels (including magnetometers, accelerometers, microphones and photodetectors) that record the state of the instrument and its environment continuously.

The noise process is approximately Gaussian but not stationary. Two classes of artefact further degrade the data quality. Lines are persistent narrow-band features visible as peaks above the noise floor in the PSD, arising from power-line harmonics (50 or 60 Hz and their multiples), mirror suspension violin modes, calibration injections, and environmental electromagnetic coupling (see Fig. 2.10) [89].

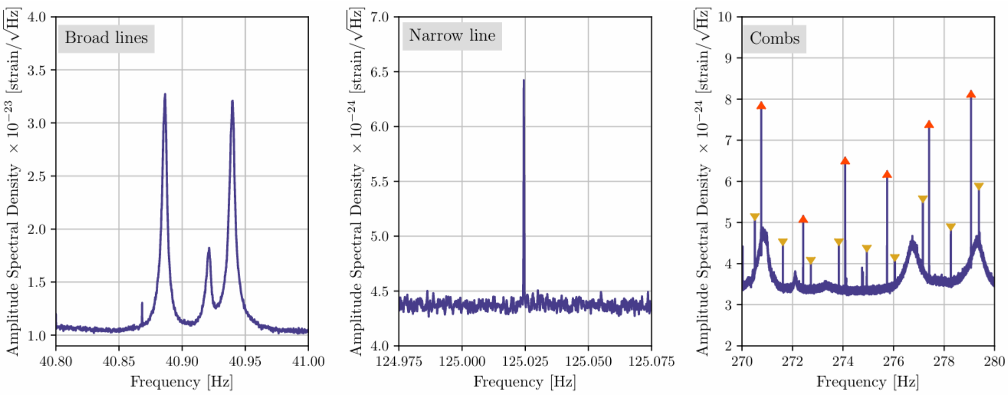


Figure 2.10: High-resolution amplitude spectral density ($\Delta f = 1/1800\,$Hz) of LIGO Hanford strain data illustrating three representative line artefacts. *Left:* a broad spectral feature associated with mirror suspension roll-mode resonances. *Centre:* an isolated narrow line of unknown origin. *Right:* two frequency combs with distinct spacings, marked by upright and inverted triangles, alongside broader noise structures. Source: [90].

Glitches are short-duration transient bursts of excess power whose morphology can mimic astrophysical signals, reducing the efficiency of detection pipelines (see Fig. 2.11) [91].

The interplay of all these contributions shapes the detector sensitivity curve shown in Fig. 2.12, where three distinct regimes are clearly visible: control and seismic noise below $\sim 50\,$Hz, a combination of thermal and quantum noise in the intermediate band (50-500 Hz), and shot noise above $\sim$500 Hz. Together they define the accessible frequency range of $\sim$10-5000 Hz for current ground-based detectors. The sharp spectral peaks superimposed on the broadband noise floor

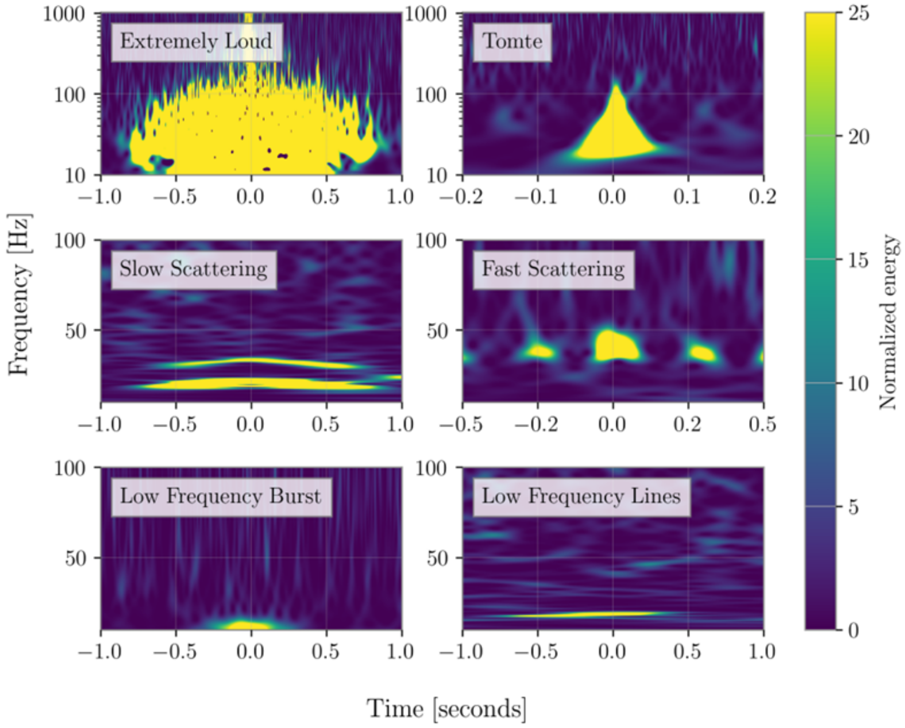


Figure 2.11: Time-frequency spectrograms of representative glitch classes recorded at LHO and LLO. Slow and Fast Scattering artefacts are attributed to light scattered by optics under increased ground motion, while the origin of Extremely Loud, Tomte, Low Frequency Burst, and Low Frequency Lines glitches remains unidentified. Source: [90].

correspond to the lines discussed above, most prominently the 60 Hz power-line harmonic and the mirror suspension violin modes.

2.3.4 Current detector network

A single interferometer provides limited directional information: the antenna pattern functions derived in §2.3.2.1 show that, at any given time, a ground-based detector is nearly blind along certain sky directions, an effect that continuosly shifts as the Earth's rotation changes the detector's orientation relative to the sky. A global network of detectors is therefore essential, both to confirm detections through coincident observations and to localise sources on the sky via time delay triangulation. Sky localisation precision improves with the number of detectors, their mutual separation, and their individual sensitivities. The current network consists of four facilities operating in the frequency band ~10-5000 Hz (see Fig. 2.13):

- **Advanced LIGO** [92] comprises two 4-km detectors: LIGO Hanford (H1) in Washington State and LIGO Livingston (L1) in Louisiana, separated by ~3000 km. Operated by the LIGO Scientific Collaboration (LSC), they have been the primary instruments for GW detections since the first observation of a BBH merger in September 2015 [54]. A third Advanced LIGO detector, LIGO-India, is currently under construction and expected to join the network in 2030.

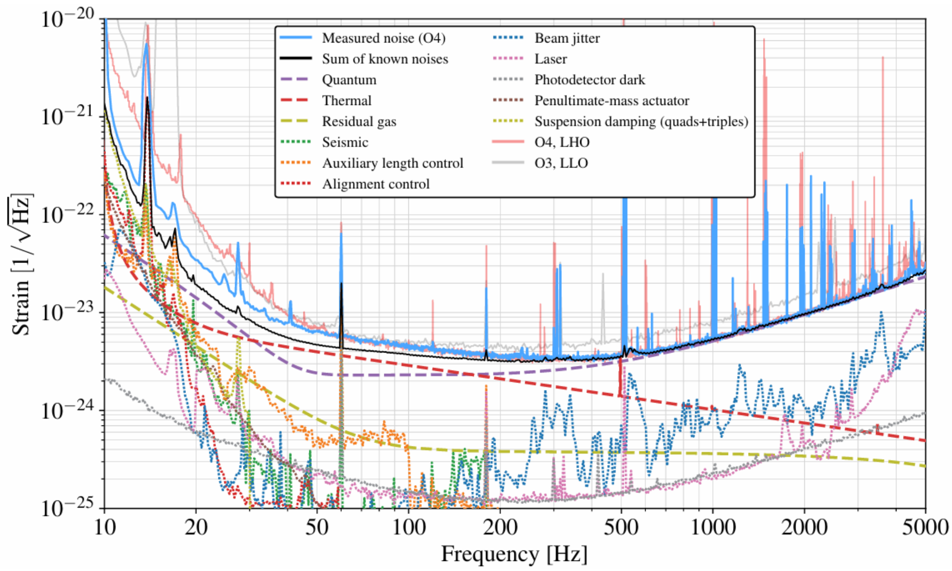


Figure 2.12: Strain noise amplitude spectral density of the LLO detector during O4, together with the O3 LLO and O4 LHO curves for comparison. Dashed lines indicate noise contributions estimated from analytical models or independent component measurements; dotted lines correspond to noise projected from auxiliary witness channels. The black curve gives the quadrature sum of all known noise sources. Three regimes are apparent: control noise dominates below ∼50 Hz, a combination of quantum and thermal noise governs the intermediate band, and shot noise sets the high-frequency floor above ∼500 Hz. Narrow spectral lines from power-line harmonics, violin modes, and calibration injections are visible throughout the spectrum. Source: [87].

- **Advanced Virgo (V1)** [93] is a 3-km interferometer located in Cascina, Italy, operated by the Virgo Collaboration. It joined the network in the final month of O2, contributing to the first three-detector observation of a BBH merger (GW170814 [94]) and to the precise sky localisation of the BNS merger GW170817 [64]. Although its sensitivity is somewhat below that of the LIGO instruments, its geographic separation from the US sites significantly improves source localisation.

- **KAGRA (K1)** [95] is a 3-km cryogenic underground detector situated 200 m beneath the Kamioka mountain in Gifu Prefecture, Japan. Its subterranean location provides natural seismic and atmospheric shielding, and its test masses are cooled to cryogenic temperatures (∼20 K) to suppress thermal noise, making it the first second-generation detector to adopt this approach. KAGRA participated in a short joint observing run with GEO600 at the end of O3, and has been part of the network since O4.

- **GEO600** [96] is a 600-m detector near Hannover, Germany, operated jointly by British and German institutions. Its shorter arms make it significantly less sensitive than the kilometre-scale instruments, so it does not contribute to standard astrophysical searches. Instead, it operates in astrowatch mode during maintenance periods of the other detectors and serves as a testbed for advanced instrumentation techniques (including squeezed-light injection) before their deployment in larger facilities. GEO600 is scheduled to conclude operations in 2026, marking the end of its role as the longest-running interferometric GW detector in Europe.

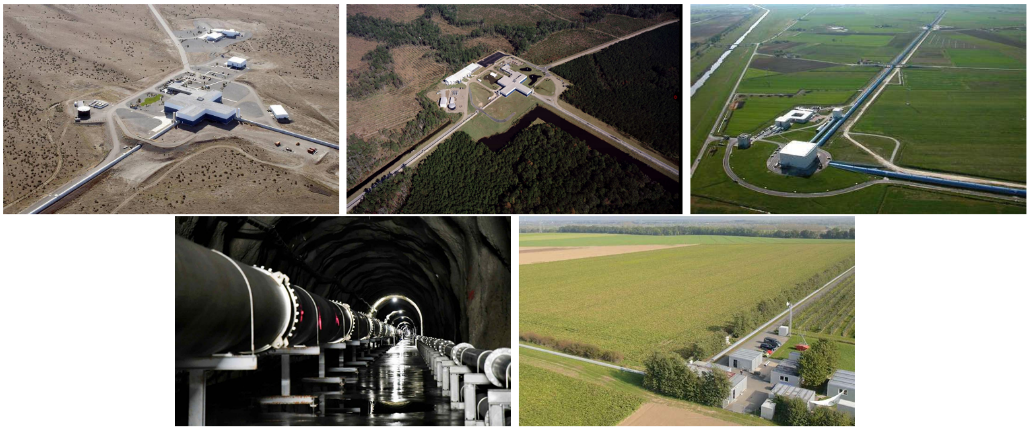


Figure 2.13: Aerial and underground views of the current LVK detector network. Top row, left to right: LIGO Hanford (Washington, USA), LIGO Livingston (Louisiana, USA), and Advanced Virgo (Cascina, Italy). Bottom row, left to right: KAGRA underground tunnel (Kamioka, Japan) and GEO600 (Hannover, Germany). Sources: [97, 98, 99, 100, 101].

The evolution of the network across observing runs is illustrated in Fig. 2.14. During O1 and the majority of O2, only H1 and L1 were operational; Virgo joined for the final weeks of O2 and participated throughout O3. A brief joint run (O3GK) at the close of O3 included GEO600 and KAGRA. Since O4, the network comprises all four facilities, with the BNS inspiral range (the standard figure of merit for detector sensitivity) improving progressively across runs as a result of the upgrades discussed in §2.3.3.

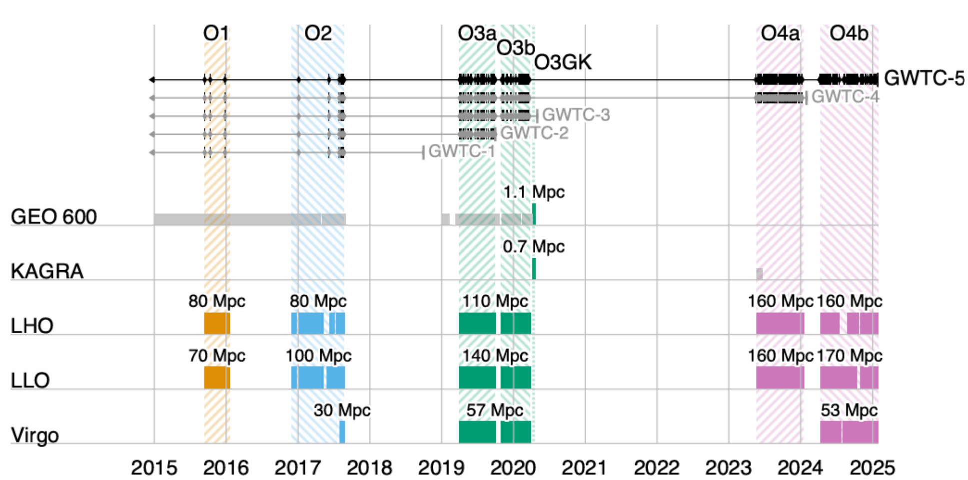


Figure 2.14: Timeline of LVK observing runs from the first detection in 2015 through the start of O4c in January 2025. Colored bars indicate the periods during which each detector was operating, together with the typical BNS inspiral range achieved in each run. GEO 600 astrowatch and KAGRA O4a periods (light gray) were not included in the main GW analyses. In O1 and O4a only LHO and LLO participated; Virgo joined for the final month of O2 and operated throughout O3a, O3b and O4b. A short joint O3GK run also included GEO 600 and KAGRA. Tick marks along the top indicate GW candidates from GWTC-1 through GWTC-5 with astrophysical probability $\geq 50\%$. Source: [68].

2.3.5 Future detectors

The current LVK network has demonstrated the transformative potential of GW astronomy, but its sensitivity and frequency coverage remain limited by fundamental and technical noise sources discussed in §2.3.3. The next generation of detectors, both ground-based and space-based, aims to extend the accessible parameter space by one to two orders of magnitude in strain sensitivity and to cover a much broader frequency band.

2.3.5.1 Ground-based third-generation detectors.

- **Einstein Telescope** (ET) [102] is the European third-generation concept. Its final geometry has not yet been decided and remains linked to the ongoing site-selection process, expected to conclude in 2027. The originally proposed reference design is an equilateral triangle with 10-km arms hosted underground at a single site to suppress seismic and Newtonian noise, where each side hosts a xylophone configuration: two nested interferometers, one optimised for high frequencies at room temperature and one for low frequencies operating cryogenically, jointly covering the band $\sim$2-10^4 Hz, providing full sky coverage and simultaneous measurement of both GW polarizations from a single facility. The alternative design, denoted 2L, instead places two separate 15-km L-shaped detectors at widely separated European sites; a detailed comparison of the scientific performance of both geometries is presented in [103].
- **Cosmic Explorer** (CE) [104] is the US counterpart, conceived as an L-shaped interferometer with 40-km arms (roughly ten times the length of Advanced LIGO) retaining the proven Michelson topology while achieving a strain sensitivity improvement of approximately one order of magnitude. Together, ET and CE would form a third-generation network capable of detecting CBC mergers throughout the majority of the observable Universe.
- **LIGO India** [105] is a planned Advanced LIGO-class detector to be built in Maharashtra, India. Although it will not exceed the sensitivity of the current LIGO instruments, its geographic location will substantially improve sky localisation and detection confidence for the global network.

2.3.5.2 Space-based detectors.

Ground-based detectors are fundamentally limited below ~ 10 Hz by seismic and Newtonian noise. Space-based missions overcome this barrier by operating in a quiet gravitational environment, opening the mHz frequency window inaccessible from the ground.

- **LISA** [106] is the flagship ESA mission, scheduled for launch in the mid-2030s. It consists of three spacecraft in heliocentric orbit forming a near-equilateral triangle with 2.5×10^6 km arms. Because the arm lengths are unequal and variable, standard Michelson cancellation of laser frequency noise is not possible; instead, Time Delay Interferometry (TDI) is employed to suppress this noise in post-processing. LISA will cover the band $\sim 10^{-4}$–10^{-1} Hz, giving access to sources including massive BH mergers, EMRIs, and the early inspiral of compact binaries centuries before they enter the ground-based band. The feasibility of the key technologies was demonstrated by the LISA Pathfinder mission in 2015-2017.

Further space-based concepts under study include the Japanese DECIGO [107], targeting the decihertz band (~ 0.1-$10\,\mathrm{Hz}$) that bridges LISA and ground-based detectors, and the Chinese missions TianQin and Taiji [108], which share a similar frequency range to LISA with different orbital configurations.

The combination of third-generation ground-based detectors and space-based observatories is expected to provide continuous frequency coverage from $\sim 10^{-4}\,\mathrm{Hz}$ to several kHz by the late 2030s, enabling a new era of multi-band GW astronomy.

2.4 Gravitational wave data analysis

The preceding sections established the theoretical framework for GW generation and propagation, and the experimental techniques employed to detect them. The remaining challenge is to extract astrophysical information from the noisy output of the detectors, a task that combines signal processing, physical modelling, and statistical inference. This relies on three crucial ingredients: matched filtering, which searches for candidate events buried in the noise by correlating the data against a bank of templates (§2.4.1); accurate waveform models, which provide these templates and are required both for the search and for the PE that follows (§2.4.2); and Bayesian PE, which builds directly on the matched-filtering framework introduced above to infer the full posterior distribution over the source parameters once a candidate has been identified (§2.4.3).

2.4.1 Matched filtering and signal-to-noise ratio

2.4.1.1 Detector output and noise model

The output of a GW detector can be written as the sum of a GW signal and instrumental noise,

$$s(t) = h(t;\boldsymbol{\theta}) + n(t), \tag{2.76}$$

where $h(t;\boldsymbol{\theta})$ is the GW strain parametrized by the source parameters $\boldsymbol{\theta}$, in practice provided by a theoretical waveform model (§2.4.2), and $n(t)$ is the noise contribution. With the sensitivity of current detectors, the noise amplitude is typically several orders of magnitude larger than the signal, $|h(t;\boldsymbol{\theta})| \ll |n(t)|$, so in most cases a detection cannot be claimed by direct inspection of $s(t)$ [55]. A systematic method is therefore required to reliably extract the vast majority of buried signals.

Although the true noise is non-stationary and non-Gaussian (exhibiting glitches and spectral lines as discussed in §2.3.3) a tractable analytic framework is obtained by modelling it as a zero-mean, stationary Gaussian random process fully characterized by its one-sided power spectral density (PSD) $S_n(f)$ (2.75),

$$\langle \tilde{n}^*(f)\,\tilde{n}(f')\rangle = \frac{1}{2}\,S_n(f)\,\delta(f-f'), \tag{2.77}$$

where tildes denote Fourier transforms, $\delta(f - f')$ is the Dirac delta encoding the statistical independence of different frequency components, and $\langle \cdot \rangle$ denotes an ensemble average, which for a single detector is identified with a time average via the ergodic theorem[6]. Under this Gaussian noise model, the probability density of a noise realization $n(t)$ is

$$p(n) \propto \exp\left(-\frac{1}{2}(n|n)\right), \tag{2.78}$$

where we have introduced the noise-weighted inner product (Wiener scalar product) between two real time series $a(t)$ and $b(t)$,

$$(a|b) \equiv 4\,\mathrm{Re}\int_0^\infty \frac{\tilde{a}^*(f)\,\tilde{b}(f)}{S_n(f)}\,df. \tag{2.79}$$

This inner product weights frequency bins inversely by the noise power, so that frequency bands where the detector is more sensitive contribute more to the overlap between two waveforms.

2.4.1.2 The matched filter and optimal SNR

Given a template waveform $h_m(t;\boldsymbol{\theta})$ representing our best theoretical model of the expected signal (see §2.4.2), the matched filter is the linear filter $K(t)$ that maximizes the SNR. The SNR is defined as the ratio of the expected filtered output when a signal is present to the root-mean-square of the filtered output when only noise is present. Concretely, one filters the detector output by computing the scalar product

$$\hat{s} = \int_{-\infty}^{\infty} s(t)\,K(t)\,dt, \tag{2.80}$$

and seeks the $K(t)$ that maximizes the ratio of the expected value of $\hat{s}$ when a signal is present to the root-mean-square of $\hat{s}$ when only noise is present. Applying the Cauchy-Schwarz inequality to this optimization in the frequency domain, one finds that the optimal filter is proportional to the signal template weighted by the inverse PSD,

$$\tilde{K}(f) \propto \frac{\tilde{h}_m(f)}{S_n(f)}, \tag{2.81}$$

which down-weights frequency bands where the detector is noisier. This is the key intuition behind matched filtering: the optimal filter is shaped by the signal itself, so that correlating the data against $K(t)$ effectively searches for a signal whose morphology matches the template $h_m(t;\boldsymbol{\theta})$. The additional weighting by $1/S_n(f)$ refines this correlation for realistic, coloured detector noise, suppressing frequency bands where the detector is noisier and amplifying those where it is most sensitive.

[6]The ergodic theorem states that, for a stationary random process, time averages computed from a single sufficiently long realization are equivalent to ensemble averages over all possible realizations. In practice, this means that the PSD $S_n(f)$ can be estimated directly from a single stretch of detector data, for example via the Welch method [109], without requiring multiple independent noise realizations.

With this optimal filter, the SNR is

$$\mathrm{SNR} \equiv \rho = \frac{(h_m|s)}{\sqrt{\langle (h_m|n)(n|h_m)\rangle}} = \frac{(h_m|s)}{\sqrt{(h_m|h_m)}} = (\hat{h}_m|s), \tag{2.82}$$

where we have used the property $\langle (a|n)(n|b)\rangle = (a|b)$ of the Wiener inner product, and defined the normalized template

$$\hat{h}_m \equiv \frac{h_m}{(h_m|h_m)^{1/2}}. \tag{2.83}$$

Substituting (2.76) and assuming the noise is uncorrelated with the template, $\langle (h_m|n)\rangle = 0$, the expectation value of ρ reduces to $\langle \rho \rangle = (\hat{h}_m|h)$, which is maximized when the template exactly matches the signal, $h_m = h$. In that ideal case one recovers the optimal SNR,

$$\rho^2_{\mathrm{opt}} = (h|h) = 4\int_0^\infty \frac{|\tilde{h}(f)|^2}{S_n(f)}\, df, \tag{2.84}$$

which quantifies the intrinsic detectability of a waveform given the detector noise [110]. In practice, ρ is computed as a time series by sliding the template across the data; a detection candidate is declared when ρ exceeds a predefined threshold, typically $\rho \gtrsim 8$ for a single detector and requiring coincident triggers across the network to suppress noise transients [69].

The matched filter is optimal only under the Gaussian noise assumption of (2.77). In practice, non-Gaussianities and glitches can produce large values of ρ in the absence of a real signal. To mitigate this, matched-filter searches are implemented through dedicated pipelines such as PyCBC [111] and GstLAL [112], which supplement the SNR with a χ^2 consistency test (verifying that the signal power across different frequency bands is consistent with the template) and require coincident triggers across multiple detectors to suppress noise transients. The data is further classified by data quality vetoes that exclude periods of poor detector performance. A candidate event is then ranked by a combined detection statistic and assigned a false alarm rate; events passing a significance threshold are passed to the PE stage described in §2.4.3. Finally, since ρ depends directly on the match between the template and the true signal, the effectiveness of the entire pipeline is tied to the accuracy of the waveform models discussed in the following section.

2.4.2 Waveform models for CBCs

The matched filter derived in §2.4.1 requires a template $h_m(t;\boldsymbol{\theta})$ that accurately represents the expected GW signal as a function of the source parameters. The quality of both the search and the subsequent PE depends directly on how faithfully these templates reproduce the true signal: a mismatch between template and signal reduces the recovered SNR (2.82) and biases the inferred parameters. In practice, matched-filter searches correlate the data against banks of $\sim 10^5$ templates covering the CBC parameter space, while PE requires evaluating $\sim 10^7$–10^9 waveforms [113]. Each template evaluation must therefore be both accurate and computationally cheap.

For a CBC, the GW signal evolves through the three phases described in §2.2 (inspiral, merger, and ringdown) and no single analytic method covers all three. During the inspiral, Post-Newtonian (PN) theory provides accurate closed-form waveforms by expanding the EFE in powers of v^2/c^2 [114], but breaks down near merger where velocities approach c. The merger phase requires full numerical relativity (NR) simulations of the EFE [115], which are highly accurate but computationally prohibitive and serve primarily to calibrate analytic models. The ringdown is described analytically by the quasi-normal modes (QNMs) of the remnant compact object [116].

To cover the full frequency evolution, these approaches are combined into inspiral-merger-ringdown (IMR) waveform families. Effective One Body (EOB) models reformulate the two-body problem as an effective single-body problem, resumming the PN series and matching to NR near merger; current implementations include `SEOBNRv5HM` and `SEOBNRv5PHM` [117, 118], which incorporate higher multipole modes and spin precession. Phenomenological models (IMRPhenom) fit closed-form frequency-domain expressions to hybrid PN-NR waveforms across the three phases; a widely used example, `IMRPhenomXPHM` [119], is calibrated to $\sim$400 NR simulations spanning mass ratios up to $q = 1000$ and dimensionless spins $|\chi_i| \lesssim 1$. NR surrogate models (NRSur) interpolate directly over NR simulations without an underlying analytic ansatz, achieving the highest fidelity at the cost of a restricted parameter space; `NRHybSur3dq8` [120] combines a PN-EOB inspiral with NR near merger, but is limited to non-precessing BBHs with $q \leq 8$. All three families have been progressively extended to include precessing spins, higher multipole modes, and tidal effects for NS components. Fast frequency-domain approximants such as `TaylorF2` or `IMRPhenomD` are favoured for matched-filter searches, while more sophisticated models are employed in PE, where waveform accuracy directly impacts posterior reliability.

2.4.3 Bayesian parameter estimation

Once a GW candidate has been identified, the goal is to determine what kind of source produced it and to measure its physical properties. Concretely, we seek to answer questions such as: what were the masses and spins of the merging objects? From which direction on the sky did the signal arrive? How far away was the source? The answers are encoded in the shape of the detected waveform, since the parameters of the binary uniquely determine the GW signal it emits. PE is the systematic procedure by which we compare the space of possible waveforms to the observed data and assign credible ranges to each source parameter [121].

The parameters of a GW signal fall into two categories. Intrinsic parameters describe the binary itself, and extrinsic parameters encode how the merger is observed from Earth. Together these constitute the parameter vector $\boldsymbol{\theta}$ over which inference is performed.

The intrinsic parameters of a quasi-circular CBC include the component masses $m_1 \geq m_2$ and the spin angular momenta $\mathbf{S}_1, \mathbf{S}_2$ of each object. The spins are conveniently decomposed into their dimensionless magnitudes $\chi_{1,2} = c|\mathbf{S}_{1,2}|/(Gm_{1,2}^2) \leq 1$, the tilt angles $\theta_{1,2}$ between each spin and the orbital angular momentum $\mathbf{L}$, and the azimuthal angle ϕ_{12} between the two spin components in the orbital plane. For BNS or NSBH systems, the tidal deformabilities $\Lambda_{1,2}$ are additional intrinsic parameters encoding the NS equation of state. In practice, several derived combinations of the intrinsic parameters appear naturally in the waveform and are therefore better constrained by the data: the chirp mass $\mathcal{M} = (m_1 m_2)^{3/5}/(m_1 + m_2)^{1/5}$, which governs the leading-order frequency evolution; the effective inspiral spin $\chi_{\text{eff}} = (m_1\chi_{1z} + m_2\chi_{2z})/(m_1 + m_2)$ [122], which captures the spin-orbit coupling along $\mathbf{L}$; and the effective precession spin χ_p [123], which parametrizes the in-plane spin components responsible for orbital precession.

The extrinsic parameters describe the location and orientation of the binary relative to the detector network: the sky position (α, δ) (right ascension and declination), the luminosity distance D_L, the inclination angle θ_{JN} between the total angular momentum $\mathbf{J}$ and the line of sight, the polarization angle ψ, the coalescence phase ϕ_c, and the geocentric coalescence time t_c. The angle ϕ_{JL} between $\mathbf{J}$ and $\mathbf{L}$ projected onto the plane perpendicular to the line of sight is also required when spins are precessing [124]. A complete description of all parameters, including their definitions and dimensions, is provided in Appendix §A.

Not all parameters are equally well constrained by a given observation. The chirp mass is typically the best-measured intrinsic quantity, as it directly controls the rate of frequency evolution visible in the detector band. Individual masses and mass ratio $q = m_2/m_1$ are less well constrained, and spin parameters are generally poorly measured unless the signal has high SNR or a long inspiral in band. Among the extrinsic parameters, the sky position and distance are generally poorly constrained by a single detector, improving significantly with a multi-detector network via time-delay triangulation [125].

In the present work, the injection parameters used to simulate the CBC signal are $\{m_1, m_2, D_L, \chi_1, \chi_2, \theta_1, \theta_2, \phi_{12}, \phi_{JL}, \theta_{JN}, \psi, \phi_c, t_c, \alpha, \delta\}$, corresponding to the full 15-dimensional parameter space of a precessing BBH system.

2.4.3.1 Bayes' theorem and posterior inference

The natural framework for GW PE is Bayesian inference, which provides a systematic way to update prior knowledge about the source parameters using the observed data. Unlike frequentist statistics, which interprets probability as the limiting frequency of repeated experiments, Bayesian inference treats probability as a degree of belief that is updated as new information becomes available. This distinction is particularly relevant in GW astronomy, where astrophysical events are unique and non-repeatable, making frequentist interpretations difficult to apply. The direct link between data and model provided by GR (where a BBH is fully described by just 15 parameters) makes Bayesian inference especially well-suited to this field [121].

Given detector strain data d and a waveform model $\mathcal{M}$, Bayes' theorem relates the posterior distribution $p(\boldsymbol{\theta}|d, \mathcal{M})$ to the likelihood $\mathcal{L}(d|\boldsymbol{\theta}, \mathcal{M})$ and the prior $p(\boldsymbol{\theta}|\mathcal{M})$ through

$$p(\boldsymbol{\theta}|d, \mathcal{M}) = \frac{\mathcal{L}(d|\boldsymbol{\theta}, \mathcal{M})\, p(\boldsymbol{\theta}|\mathcal{M})}{\mathcal{Z}(\mathcal{M})}, \tag{2.85}$$

where the evidence $\mathcal{Z}(\mathcal{M}) = \int \mathcal{L}(d|\boldsymbol{\theta}, \mathcal{M})\, p(\boldsymbol{\theta}|\mathcal{M})\, d\boldsymbol{\theta}$ normalizes the posterior. Each term has a distinct physical interpretation. The likelihood $\mathcal{L}(d|\boldsymbol{\theta}, \mathcal{M})$ quantifies how probable the observed data are given a particular set of source parameters (it encodes the noise model of the detector). The prior $p(\boldsymbol{\theta}|\mathcal{M})$ encodes our knowledge of the parameters before the measurement; for instance, an isotropic prior is assigned to the sky location, reflecting the absence of any preferred direction. The posterior is the central quantity of interest: it encodes the full probability distribution of the source parameters given the data, and its one-dimensional marginals,

$$p(\theta_i|d, \mathcal{M}) = \int \prod_{j \neq i} d\theta_j\; p(\boldsymbol{\theta}|d, \mathcal{M}), \tag{2.86}$$

yield credible intervals for individual parameters after integrating out all others.

For Gaussian, stationary noise characterized by the PSD $S_n(f)$ introduced in §2.3.3, the likelihood takes the form of a Gaussian in the residual between the data and the waveform template,

$$\mathcal{L}(d|\boldsymbol{\theta}) \propto \exp\left(-\frac{1}{2}(d - h(\boldsymbol{\theta}) \,|\, d - h(\boldsymbol{\theta}))\right), \tag{2.87}$$

where $(\cdot|\cdot)$ is the noise-weighted inner product defined in (2.79) and $h(\boldsymbol{\theta})$ is the waveform template evaluated at parameters $\boldsymbol{\theta}$. Taking the logarithm, one recovers (2.87) in the log form used in practice. This expression makes explicit the connection between PE and matched filtering: maximizing $\ln \mathcal{L}$ over $\boldsymbol{\theta}$ is equivalent to finding the template that best matches the data in the noise-weighted sense of §2.4.1, with the prior regularizing the solution across the full parameter space.

Because the posterior (2.85) is a 15-dimensional distribution for a precessing BBH, it cannot be evaluated analytically. Stochastic sampling methods are therefore required to explore it efficiently, as discussed in §2.4.3.3.

2.4.3.2 Hypothesis testing and the Bayes factor

Beyond PE within a fixed model, Bayesian inference also provides a rigorous framework for comparing competing hypotheses. The natural quantity for this is the odds ratio between two models $\mathcal{M}_1$ and $\mathcal{M}_2$,

$$\mathcal{O}_{12} = \frac{p(\mathcal{M}_1|d)}{p(\mathcal{M}_2|d)} = \frac{\mathcal{Z}(\mathcal{M}_1)}{\mathcal{Z}(\mathcal{M}_2)} \cdot \frac{p(\mathcal{M}_1)}{p(\mathcal{M}_2)}, \tag{2.88}$$

where the second factor is the prior odds, encoding our relative belief in the two models before observing the data. When both models are considered equally plausible a priori, the prior odds is unity and the odds ratio reduces to the Bayes factor,

$$\mathcal{B}_{12} = \frac{\mathcal{Z}(\mathcal{M}_1)}{\mathcal{Z}(\mathcal{M}_2)}, \tag{2.89}$$

which measures the factor by which the data update our relative belief between the two models. A value $\mathcal{B}_{12} \gg 1$ favours $\mathcal{M}_1$, while $\mathcal{B}_{12} \ll 1$ favours $\mathcal{M}_2$; by convention, $|\ln \mathcal{B}_{12}| \geq 8$ is often taken as the threshold for strong evidence in favour of one hypothesis over the other [126, 121].

The most fundamental application in GW astronomy is the signal-vs-noise Bayes factor, where $\mathcal{M}_1$ is the hypothesis that the data contain a GW signal and $\mathcal{M}_2$ is the noise-only hypothesis. In this case,

$$\mathcal{Z}_S = \int \mathcal{L}(d|\boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}, \qquad \mathcal{Z}_N = \mathcal{L}(d|\boldsymbol{\theta} = 0), \tag{2.90}$$

and $\mathcal{B}_{SN} = \mathcal{Z}_S/\mathcal{Z}_N$ quantifies how much more probable the data are under the signal hypothesis than under pure noise.

Beyond detection, Bayes factors are used throughout GW data analysis to address a variety of model selection questions: whether a signal requires spin precession or tidal deformability, which waveform approximant better describes the data, or whether the observations are consistent with GR or favour an alternative theory of gravity [124]. In each case the Bayes factor naturally penalizes more complex models through an Occam factor: a model with a large prior volume that is only marginally better supported by the data will yield a lower evidence than a simpler model with a tighter parameter space, reflecting the Bayesian preference for parsimony.

It is important to note that computing $\mathcal{Z}$ requires integrating the likelihood over the entire prior volume, which is precisely what nested sampling is designed to do efficiently, as discussed in §2.4.3.3.2.

2.4.3.3 Stochastic sampling methods

As we mentioned in this section, the posterior distribution (2.85) lives in a high-dimensional parameter space (15 dimensions for a precessing BBH) and cannot be evaluated analytically. A brute-force grid approach is equally intractable: with just 10 bins per dimension, evaluating the likelihood on the full grid would require 10^{15} waveform computations, far beyond any feasible computational budget. Stochastic sampling algorithms are therefore employed to explore the posterior efficiently, concentrating evaluations in the regions of parameter space where the posterior has significant support. The two dominant families used in GW astronomy are Markov Chain Monte Carlo (MCMC) [127, 128] and nested sampling [129], whose complementary strategies are illustrated in Fig. 2.15.

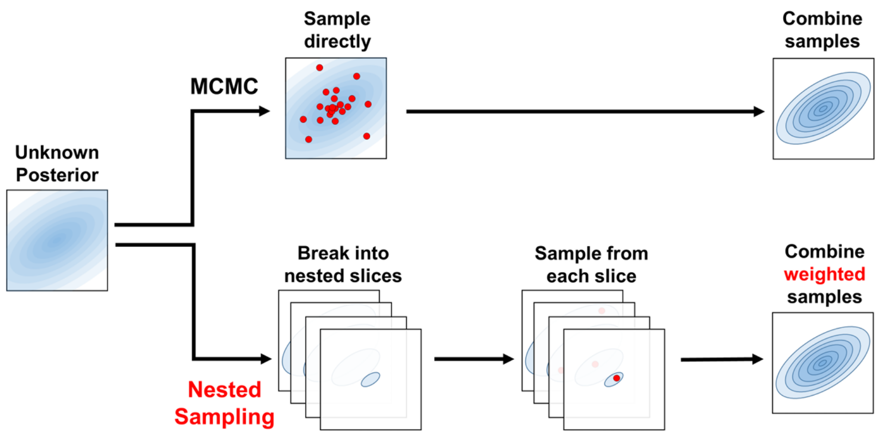


Figure 2.15: Schematic comparison of the two main stochastic sampling strategies. MCMC draws samples directly from the posterior via a random walk, combining them to reconstruct the distribution. Nested sampling instead decomposes the posterior into nested likelihood slices, samples from each slice independently, and combines the weighted samples to recover the posterior and compute the evidence $\mathcal{Z}$. Source: [130].

The output of both methods is a set of posterior samples $\{\boldsymbol{\theta}_k\}$ drawn from $p(\boldsymbol{\theta}|d, \mathcal{M})$, such that the density of samples in any region of parameter space is proportional to the posterior probability there. These samples can then be used to compute expectation values of any quantity of interest, to construct marginalized one- and two-dimensional posterior distributions for individual parameters, and to report credible intervals (the Bayesian analog of confidence intervals).

2.4.3.3.1 Markov Chain Monte Carlo.

MCMC methods construct a random walk through parameter space whose stationary distribution is the target posterior, as shown in Fig. 2.16. The most widely used variant is the Metropolis-Hastings algorithm, which proceeds as follows: starting from an initial point $\boldsymbol{\theta}_0$ (often drawn from the prior), at each iteration t a new point $\boldsymbol{\theta}'$ is proposed from a proposal distribution $q(\boldsymbol{\theta}'|\boldsymbol{\theta}_t)$ (a common choice for q is a Gaussian distribution with mean $\boldsymbol{\theta}_t$ and some fixed covariance matrix), and accepted with probability

$$A = \min\left(1, \frac{p(\boldsymbol{\theta}'|d)\, q(\boldsymbol{\theta}_t|\boldsymbol{\theta}')}{p(\boldsymbol{\theta}_t|d)\, q(\boldsymbol{\theta}'|\boldsymbol{\theta}_t)}\right). \tag{2.91}$$

If $A \geq 1$ the proposal is always accepted; otherwise it is accepted with probability A. When the proposal distribution q is symmetric, i.e. $q(\boldsymbol{\theta}'|\boldsymbol{\theta}_t) = q(\boldsymbol{\theta}_t|\boldsymbol{\theta}')$, the ratio of proposals cancels and A reduces to the simple likelihood ratio $p(\boldsymbol{\theta}'|d)/p(\boldsymbol{\theta}_t|d)$. This acceptance rule ensures detailed balance and guarantees that the chain converges to the correct stationary distribution. By recording the position of the chain at each accepted step, one accumulates draws from the posterior.

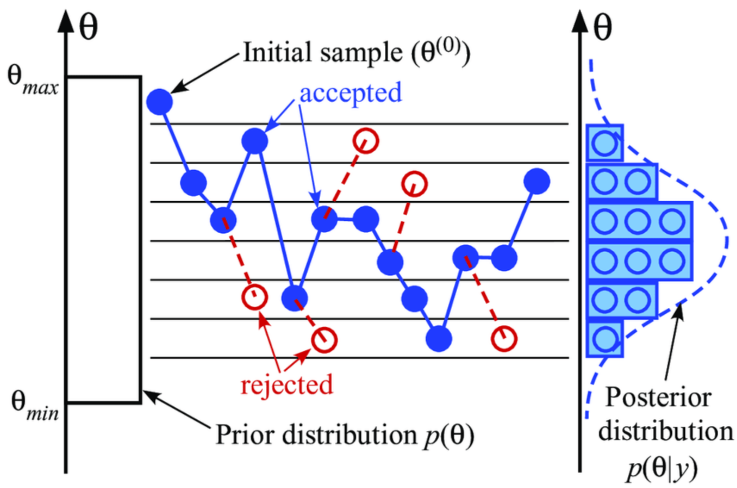


Figure 2.16: Illustration of the Metropolis-Hastings random walk in parameter space. Starting from initial samples drawn from the prior $\pi(\boldsymbol{\theta})$, walkers explore the parameter space following the acceptance criterion (2.91), progressively converging toward the high-probability region of the posterior $p(\boldsymbol{\theta}|y)$. Source: [121].

Several practical considerations arise. The early phase of the chain (the burn-in) is strongly dependent on the starting point and must be discarded before the chain has reached its stationary distribution. Successive samples within a chain are autocorrelated, so the chain must be thinned by retaining only samples separated by the autocorrelation length to obtain approximately independent draws. Finally, MCMC walkers can become trapped in a single mode of a multimodal posterior, a common situation in GW PE, where the sky position degeneracy produces two nearly symmetric modes. Parallel tempering mitigates this by running multiple chains at different temperatures (i.e., with artificially flattened posteriors) that periodically exchange states, allowing the cold chain to escape local maxima by borrowing proposals from hotter, more diffuse chains [124].

2.4.3.3.2 Nested sampling.

Nested sampling is an algorithm designed primarily to compute the evidence $\mathcal{Z}$ directly, with posterior samples produced as a byproduct. It is particularly well-suited to GW PE because it handles multimodal posteriors naturally and provides the evidence required for model comparison via the Bayes factor (2.89).

The algorithm proceeds as follows. A fixed number $N_{\rm live}$ of live points are drawn uniformly from the prior. At each iteration, the live point with the lowest likelihood $\mathcal{L}_{\rm min}$ is removed and replaced by a new point drawn from the prior subject to the constraint $\mathcal{L} > \mathcal{L}_{\rm min}$, progressively shrinking the prior volume explored. The discarded point contributes to the evidence integral via the trapezoid rule,

$$\mathcal{Z} \approx \sum_i \mathcal{L}_i\, w_i, \qquad w_i = X_{i-1} - X_i, \tag{2.92}$$

where X_i is the fraction of the prior volume enclosed by the i-th likelihood contour, estimated statistically from the number of live points. The algorithm terminates when the fractional contribution of the remaining prior volume to $\mathcal{Z}$ falls below a predefined threshold. Posterior samples are obtained by reweighting the discarded points by $\mathcal{L}_i w_i/\mathcal{Z}$. The geometric interpretation of the algorithm is shown in Fig. 2.17.

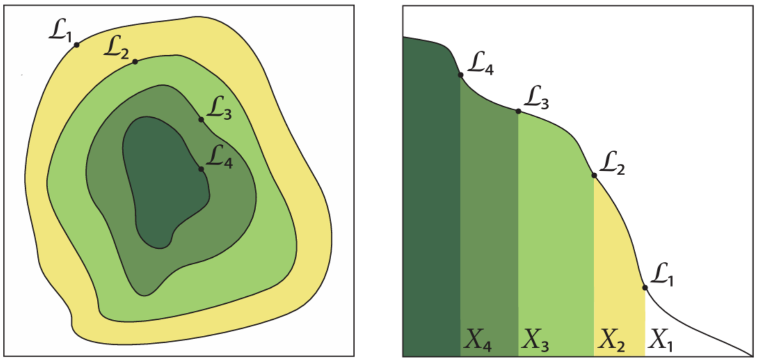


Figure 2.17: Geometric interpretation of nested sampling. *Left:* iso-likelihood contours in parameter space, with the discarded live points $\mathcal{L}_1 > \mathcal{L}_2 > \mathcal{L}_3 > \mathcal{L}_4$ marking successive likelihood shells. *Right:* the corresponding one-dimensional mapping $\mathcal{L}(X)$ as a function of the enclosed prior volume X, where the evidence $\mathcal{Z}$ is approximated as the area under the curve via the trapezoid rule. Source: [131].

The main algorithmic challenge is drawing new live points efficiently from the likelihood-constrained prior. The most widely used implementations in GW astronomy are MultiNest [132], which approximates the iso-likelihood contour with a set of ellipsoids that can be sampled exactly, and Dynesty [130], which uses dynamic live-point allocation to concentrate computational effort where the posterior has the most support. Both are accessible through the Bilby [133] inference library, which provides a modular Python interface for CBC PE and is used in this work.

In summary, this chapter has established the theoretical and experimental foundations of GW astronomy. Beginning from the linearized EFE, we derived the existence of GWs as propagating spacetime perturbations, characterized their two polarization states, and quantified their effect on test masses through the geodesic deviation equation. We then surveyed the main astrophysical source classes and described the detection principle underlying current interferometric observatories, from the antenna pattern functions and noise budget to the global detector network. The chapter closed with the Bayesian PE framework (matched filtering, waveform models, and stochastic sampling) that allows us to extract source properties from detected signals. These tools constitute the methodological backbone of the analysis presented in the following chapters.

Chapter 3

Gravitational lensing of gravitational waves

Having established the theoretical and observational foundations of GW astronomy in the preceding chapter, we now turn to a phenomenon that lies at the heart of this thesis: the gravitational lensing of GWs. We begin with the basics of gravitational lensing and its particular features in the GW context (§3.1), before developing the geometric optics formalism that governs strong lensing, including the lens equation, image formation, magnification, time delay, and the main lens models used in the literature (§3.2). We then characterize the signatures that strong lensing imprints on GW signals, including the amplification factor, image classification, the Morse phase, and their effect on the observed strain and inferred source parameters (§3.3). The chapter closes with a discussion of how the joint analysis of lensed image pairs can be exploited to improve sky localization (§3.4), motivating the methodology developed in the subsequent chapters.

3.1 Gravitational lensing

Gravitational lensing is a direct consequence of GR: massive objects curve spacetime, and any particle following a geodesic (including massless ones such as photons or GWs) will have its trajectory deflected [134, 135, 136]. The phenomenon was confirmed observationally by Eddington during the solar eclipse of 1919, when the apparent positions of stars near the Sun were displaced by exactly the amount predicted by GR [7]. Since then, gravitational lensing has become one of the most powerful tools in observational astrophysics, providing information about the mass distribution of the Universe independently of its luminous content [137].

When a GW propagates from its source to a detector, massive structures along the line of sight act as gravitational lenses (galaxies, galaxy clusters, and other compact objects). Depending on the geometry and lens mass, three regimes are distinguished [138]: strong lensing, caused by galaxies or galaxy clusters, produces multiple images of the same source arriving at different times and with different amplitudes; microlensing, caused by compact objects, produces overlapping images whose interference can modulate the observed strain; and weak lensing, due to large-scale structure, induces small coherent distortions. Figure 3.1 illustrates the general picture of GW lensing, including both strong-lensing and microlensing effects. This work focuses exclusively on the strong lensing regime.

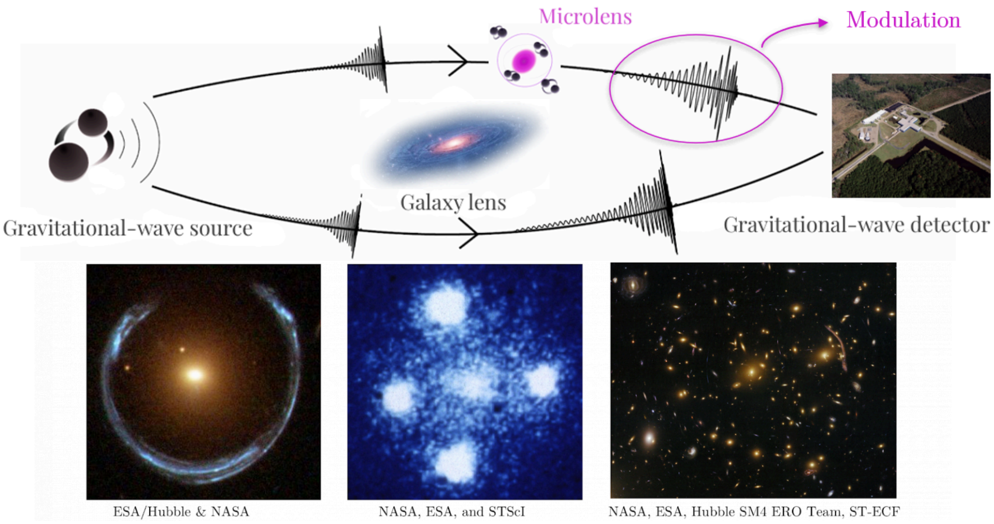


Figure 3.1: Schematic representation of GW lensing. A galaxy acting as a macrolens can produce multiple images of the same GW signal, arriving at the detector with different magnifications and time delays. Compact objects embedded within the lens galaxy (microlenses) can further perturb these images, introducing additional amplification, interference, and diffraction effects. The images shown at the bottom illustrate examples of strong gravitational lensing observed in electromagnetic astronomy. Original illustration based on sources: [139, 140].

3.1.1 Particularities of gravitational wave lensing

The deflection of GWs by a gravitational field follows the same geometric principles as for light, but several properties of GWs lead to qualitatively different lensing phenomenology.

The most fundamental difference is the wavelength regime. GWs detectable by ground-based interferometers have wavelengths ranging from roughly 10 km to 1 Mm, many orders of magnitude larger than EM wavelengths. The key parameter governing the lensing regime is the ratio $\lambda_{\rm GW}/r_E$, where $r_E = D_L\theta_E$ is the physical Einstein radius of the lens (see §3.2.3). When $\lambda_{\rm GW} \ll r_E$, each image travels along a distinct geometric path and the standard geometric optics formalism applies. When $\lambda_{\rm GW} \gtrsim r_E$, diffraction becomes important and wave optics must be used, producing frequency-dependent modulations of the waveform amplitude and phase with no EM analog [141]. For galaxy-scale lenses ($r_E \sim \mathcal{O}(\text{kpc})$), ground-based GW detectors always operate in the geometric optics regime. Wave optics effects become relevant for lower mass lenses such as stars or intermediate-mass black holes, which fall outside the scope of this work. Fig. 3.2 summarizes the lens-mass scales associated with each regime.

A second key difference is coherence. GWs are coherent waves with a well-defined amplitude and phase, so when two lensed images arrive within a time shorter than the signal duration they can interfere, producing observable beating patterns in the waveform. This effect has no counterpart in standard EM lensing, where sources are typically incoherent.

Finally, unlike EM radiation, GWs propagate through the intergalactic medium and through the lens galaxy itself without absorption or dispersion. The lensed GW signal therefore encodes clean information about the source and the lensing potential, uncontaminated by dust extinction or plasma effects that complicate EM observations.

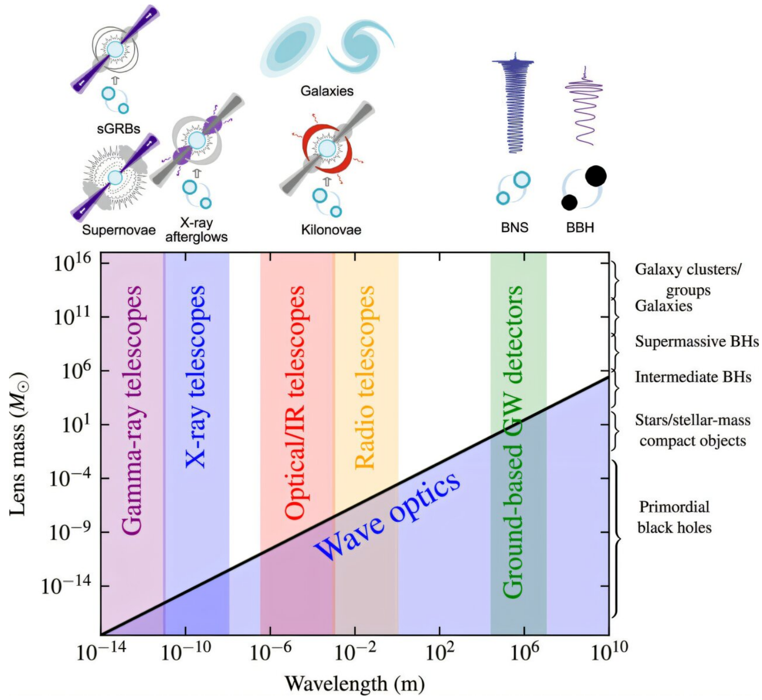


Figure 3.2: Illustration of the lens-mass scales relevant for gravitationally lensed signals. Depending on the relation between the radiation wavelength and the characteristic scale of the lensing potential, lensing occurs in either the geometric optics or wave optics regime. While geometric optics describes distinct propagation paths, wave optics accounts for diffraction and interference effects. Due to their long wavelengths, GW can exhibit wave-optics effects for relatively low lens masses. Source: [142].

3.1.2 Relevance for gravitational wave astronomy

No confirmed detection of gravitational lensing of GWs has been reported despite dedicated searches in GWTC-1 through GWTC-4 [143]. As detector sensitivities improve, detection is expected to become feasible within the next observing runs, with predictions suggesting more than one strongly lensed event per year at design sensitivity [144].

Such detections would open a rich set of applications: measuring the Hubble constant via lensing time delays combined with GW standard sirens [145, 146], probing dark matter distributions in lens galaxies [147], and testing GR by comparing the propagation speed of GWs and light through the same lensing potential [148]. Lensing also biases the apparent properties of detected sources (a magnified image appears closer and more massive than it truly is) making the identification of lensed events important for controlling systematics in population studies [149].

In addition, multiple images of the same source provide repeated observations that can be combined to improve PE and sky localization [146, 150]. This last application is the central motivation of the present work: by performing a joint Bayesian analysis of two strongly lensed images of the same BBH merger, we investigate how the sky localization of the source can be dramatically improved compared to the analysis of a single image.

3.2 Gravitational lensing formalism

3.2.1 Deflection of light by a gravitational field

The deflection of light by a gravitational field can be derived by studying null geodesics in a perturbed spacetime. Our Universe is well described by a spatially flat Friedmann-Robertson-Walker (FRW) metric, but for the scales relevant to lensing we can work locally with Minkowski spacetime as a starting point and incorporate cosmological corrections later through angular diameter distances.

The presence of a massive lens introduces inhomogeneities in the spacetime metric. Treating the lens in the weak-field limit (slow-moving, $v \ll c$, and small curvature), the energy-momentum tensor is dominated by the energy density component $T^{00} = \rho$, with all other components negligible. Linearizing the EFE to first order in the metric perturbation $\gamma_{\mu\nu}$, and neglecting time derivatives since the lens evolves slowly, the spatial components satisfy a Poisson equation whose solution yields the perturbed metric [151]

$$ds^2 = -(1 + 2U)\, dt^2 + (1 - 2U)\, \delta_{ij}\, dx^i\, dx^j, \tag{3.1}$$

where $U(\mathbf{x})$ is the Newtonian gravitational potential of the lens, satisfying $\nabla^2 U = 4\pi G\rho$. This is the same linearization procedure applied in §2.1 to derive the GW wave equation, now applied to a static mass distribution rather than a radiating source.

A photon travelling along a null geodesic of (3.1) at closest approach distance ξ from a mass M (the impact parameter) accumulates a total deflection angle [152]

$$\hat{\alpha} = \frac{4GM}{c^2 \xi}, \tag{3.2}$$

twice the value predicted by Newtonian gravity, and confirmed by Eddington's 1919 measurement [7]. Since the deflection is linear in the perturbation, the contributions of multiple masses add linearly, so any mass distribution can be treated as a superposition of point-mass deflectors.

3.2.2 The lens equation

The standard framework for gravitational lensing adopts the thin lens approximation: all deflection is assumed to occur at a single plane perpendicular to the line of sight, the lens plane, while the source lies on a parallel source plane [153]. This approximation is valid when the physical extent of the lens is much smaller than the distances between observer, lens, and source. The geometry of the problem is illustrated in Fig. 3.3, which defines the angular positions $\boldsymbol{\theta}$ (image) and $\boldsymbol{\beta}$ (source), the angular diameter distances D_L, D_S, and D_{LS}, and the impact parameter $\boldsymbol{\xi} = D_L \boldsymbol{\theta}$.

Projecting the geometry onto the source plane and applying the small-angle approximation, one obtains the lens equation

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}), \tag{3.3}$$

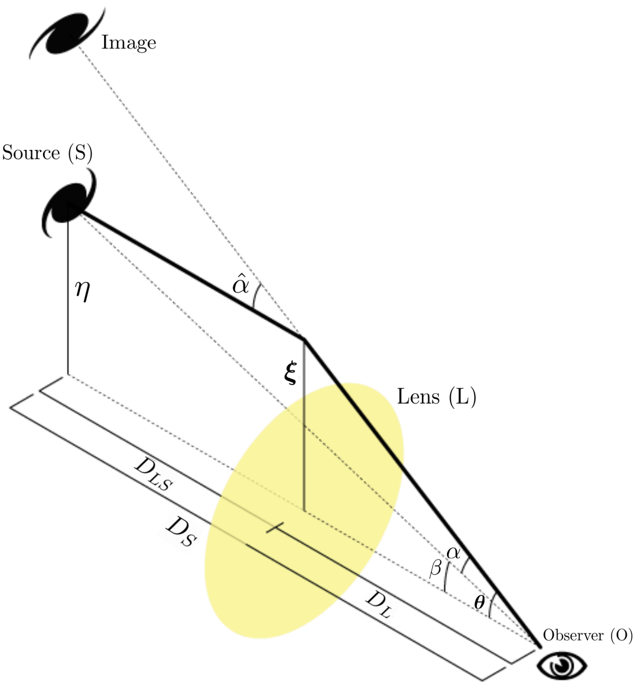


Figure 3.3: Geometry of gravitational lensing in the thin lens approximation. The observer (O), lens (L), and source (S) define the angular diameter distances D_L, D_S, and D_{LS}. The true source position $\boldsymbol{\beta}$ and the apparent image position $\boldsymbol{\theta}$ are related by the lens equation through the reduced deflection angle $\boldsymbol{\alpha}$. Adapted from the source: [154].

where the reduced deflection angle is defined as

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) \equiv \frac{D_{LS}}{D_S}\,\hat{\boldsymbol{\alpha}}(\boldsymbol{\theta}). \tag{3.4}$$

The lens equation maps each image position $\boldsymbol{\theta}$ to a unique source position $\boldsymbol{\beta}$, but the inverse mapping can be multi-valued: a single source can produce multiple images. Equivalently, by Fermat's principle, images form at the stationary points of the Fermat potential (or scaled time delay surface)

$$\tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = \frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta}), \tag{3.5}$$

where $\psi(\boldsymbol{\theta})$ is the projected gravitational potential of the lens, obtained from the Newtonian potential $U(\boldsymbol{\xi}, Z)$ by integration along the line of sight,

$$\psi(\boldsymbol{\theta}) = \frac{2D_{LS}}{D_S D_L}\int U(\boldsymbol{\xi}, Z)\,dZ. \tag{3.6}$$

The deflection angle is then $\boldsymbol{\alpha} = \nabla_{\boldsymbol{\theta}}\psi$, and the lens equation (3.3) is recovered from $\nabla_{\boldsymbol{\theta}}\tau = 0$.

Since stationary points of τ always appear and disappear in pairs (a minimum together with a saddle point, or a maximum together with a saddle point) as the source position varies, the total number of images is always odd [155]: there is always at least one image corresponding to the global minimum of τ, and any additional images appear in pairs of two. The wavefront interpretation of this result is illustrated in Fig. 3.4.

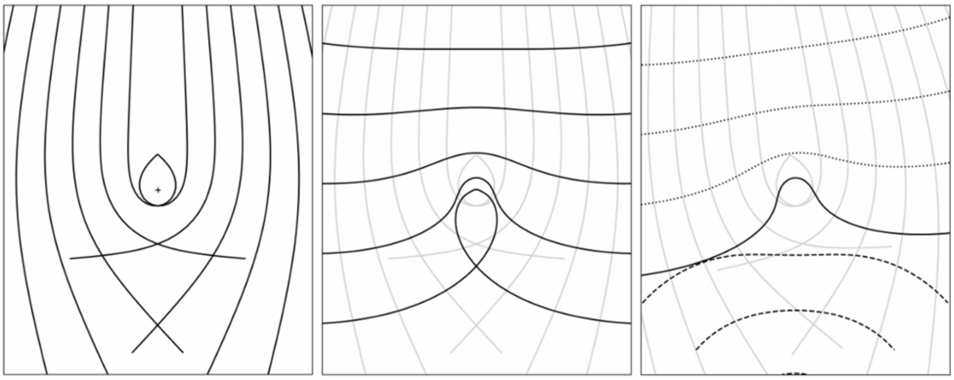


Figure 3.4: Wavefront picture of gravitational lensing. Left: light rays from a distant source are deflected near the centre, where the gravitational field is strongest. Centre: the corresponding wavefronts at a fixed time. The small closed loop arises from rays undergoing very large deflections. Right: a second wavefront traced backwards from the observer (dashed) is superimposed on the frozen source wavefront (solid). Lensed images form where the two wavefronts share the same local direction of propagation, which corresponds to the stationary points of the Fermat potential (3.5). Source: [153].

3.2.3 Einstein radius and multiple image formation

For a point mass M located at the origin, the lens equation (3.3) admits an analytic solution. When the source lies exactly behind the lens ($\boldsymbol{\beta} = 0$), perfect circular symmetry produces a ring image at the Einstein radius

$$\theta_E = \sqrt{\frac{4GM}{c^2}\frac{D_{LS}}{D_L D_S}}, \tag{3.7}$$

which sets the characteristic angular scale of lensing [153]. For a general source position, the lens equation reduces to a quadratic in $|\boldsymbol{\theta}|$, always producing exactly two images at positions

$$\theta_{1,2} = \frac{1}{2}\left(\beta \pm \sqrt{\beta^2 + 4\theta_E^2}\right), \tag{3.8}$$

where $\beta = |\boldsymbol{\beta}|$. The image at $\theta_1 > \theta_E$ lies outside the Einstein ring, while $\theta_2 < \theta_E$ lies inside. More generally, multiple images form when the projected surface mass density of the lens,

$$\Sigma(\boldsymbol{\xi}) = \int \rho(\boldsymbol{\xi}, Z)\, dZ, \tag{3.9}$$

obtained by integrating the three-dimensional mass density ρ of the lens along the line of sight coordinate Z, exceeds the critical surface density

$$\Sigma_{\rm cr} = \frac{c^2}{4\pi G}\frac{D_S}{D_L D_{LS}}, \tag{3.10}$$

a combination of distances that sets the lensing strength for a given geometry. It is convenient to introduce the dimensionless convergence

$$\kappa(\boldsymbol{\xi}) \equiv \frac{\Sigma(\boldsymbol{\xi})}{\Sigma_{\rm cr}}, \tag{3.11}$$

which measures the projected mass density in units of the critical value. When $\kappa > 1$ in some region of the lens plane, the lensing effect is strong enough to produce multiple images. Equivalently, κ satisfies the two-dimensional Poisson equation $\nabla^2\psi = 2\kappa$, linking the lensing potential to the projected mass distribution of the lens.

3.2.4 Magnification and time delay

Since gravitational lensing preserves surface brightness but distorts the solid angle subtended by a source, it produces a net magnification. The magnification of each image is given by the inverse determinant of the Jacobian matrix of the lens mapping,

$$\mu = \frac{1}{\det \mathbf{A}}, \qquad A_{ij} = \delta_{ij} - \frac{\partial^2\psi}{\partial\theta_i\,\partial\theta_j}. \tag{3.12}$$

The sign of μ encodes the parity of the image: images at minima and maxima of τ have positive magnification, while images at saddle points have negative magnification (i.e. they are mirror-inverted). Explicit expressions for the magnifications of each image type are given in §3.2.5 for the specific lens models considered in this work.

The arrival time of each image is delayed with respect to an unlensed ray by two contributions. The geometric delay arises from the longer path length of the deflected ray, while the Shapiro delay [156, 157] results from the slowing of the signal in the gravitational potential of the lens. Together, they yield the time delay between two images i and j

$$\Delta t_{ij} = \frac{(1+z_L)}{c}\frac{D_L D_S}{D_{LS}}\left[\tau(\boldsymbol{\theta}_i, \boldsymbol{\beta}) - \tau(\boldsymbol{\theta}_j, \boldsymbol{\beta})\right], \tag{3.13}$$

where the $(1+z_L)$ factor accounts for the cosmological redshifting of the delay at the lens. For galaxy-scale lenses, Δt_{ij} ranges from minutes to months [158], and constitutes one of the key observables in GW lensing searches.

3.2.5 Lens models

The choice of lens model determines the deflection angle $\hat{\boldsymbol{\alpha}}$ and hence all lensing observables. While real lenses are complex, extended, and often non-circular, a small set of idealized models captures the essential physics and is widely used in both analytical work and population studies [153].

3.2.5.1 Point mass lens

The simplest model treats the entire lens mass M as concentrated at a single point, so that the projected surface mass density is $\Sigma(\boldsymbol{\xi}) = M\delta^{(2)}(\boldsymbol{\xi})$ [159]. The deflection angle is then exactly (3.2), and the lens equation reduces to a quadratic whose two solutions are the image positions (3.8). The corresponding magnifications are

$$\mu_{\pm} = \frac{1}{2} \pm \frac{u^2+2}{2u\sqrt{u^2+4}}, \qquad u \equiv \frac{\beta}{\theta_E}, \tag{3.14}$$

where u is the normalized source-lens separation [153]. The positive image (μ_+) is always magnified, while the negative image (μ_-) can be magnified or demagnified. The total magnification diverges as $u \to 0$ (Einstein ring) and approaches unity for $u \gg 1$ (no lensing). The time delay between the two images is

$$\Delta t = 4M_{Lz}\left[\frac{u\sqrt{u^2+4}}{2} + \ln\frac{\sqrt{u^2+4}+u}{\sqrt{u^2+4}-u}\right], \tag{3.15}$$

where $M_{Lz} = M_L(1+z_L)$ is the redshifted lens mass [160].

3.2.5.2 Singular isothermal sphere

A more realistic description of galaxy-scale lenses is the singular isothermal sphere (SIS), which models the lens as a self-gravitating gas of stars with a Maxwellian velocity distribution characterized by a constant velocity dispersion σ_v. The three-dimensional density profile is

$$\rho(r) = \frac{\sigma_v^2}{2\pi G r^2}, \tag{3.16}$$

which yields a flat rotation curve $v_c = \sqrt{2}\,\sigma_v$, consistent with observations of early-type galaxies [138]. Projecting onto the lens plane gives the surface mass density

$$\Sigma(\xi) = \frac{\sigma_v^2}{2G\xi}, \tag{3.17}$$

which depends only on the radial distance $\xi = |\boldsymbol{\xi}|$, confirming that the SIS is an axisymmetric lens [161]. Expressing this in terms of the convergence gives

$$\kappa(\theta) = \frac{\theta_E}{2\theta}, \qquad \theta_E = 4\pi\frac{\sigma_v^2}{c^2}\frac{D_{LS}}{D_S}, \tag{3.18}$$

and the deflection angle has the particularly simple form $\boldsymbol{\alpha} = \theta_E\,\hat{\boldsymbol{\theta}}$, independent of the image position. When $u \leq 1$ (source inside the Einstein radius) two images form with magnifications $\mu_\pm = \pm 1 + 1/u$ and time delay $\Delta t = 8M_{Lz}u$; for $u > 1$ only a single image is produced [159].

The SIS is the simplest axisymmetric model for strong lensing by individual galaxies. Its elliptical generalization, the singular isothermal ellipsoid (SIE), extends the convergence to elliptical isocontours parametrized by the projected axis ratio q_l, and has traditionally been the model of choice in electromagnetic surveys of galaxy-galaxy lensing [162]. External shear can be added to either model to account for the tidal field of surrounding mass. For galaxy cluster lenses, more complex profiles are required, typically a Navarro-Frenk-White (NFW) dark matter halo combined with a SIS profile for the brightest central galaxy [138].

3.2.5.3 Extended power-law lens models

Observations of early-type galaxies show that their density slopes are close to isothermal but not exactly so, with a distribution of slopes $\gamma \sim \mathcal{N}(2.0, 0.2)$ [163]. A more flexible family is the elliptical power law (EPL), which generalizes the SIE by allowing the density slope to vary freely.

The three-dimensional density profile is

$$\rho(r) \propto r^{-\gamma}, \tag{3.19}$$

where $\gamma = 2$ recovers the isothermal case [164]. The projected convergence for an elliptical mass distribution with axis ratio q_l takes the form

$$\kappa(\boldsymbol{\theta}) = \frac{3-\gamma}{2}\left(\frac{\theta_E}{\sqrt{q_l\theta_1^2 + \theta_2^2/q_l}}\right)^{\gamma-1}, \tag{3.20}$$

where θ_1 and θ_2 are coordinates aligned with the major and minor axes of the lens. For $\gamma = 2$ and $q_l = 1$ this reduces to the SIS convergence (3.18).

The EPL model has become the preferred choice for population studies of galaxy-scale GW lensing, as its variable slope and ellipticity naturally produce the range of image configurations observed in strongly lensed systems [165]. In practice, external shear is added to account for the tidal contribution of the lens environment, yielding the EPL + shear model that constitutes the current standard in strong lensing analyses [166] and the model adopted in this work for the generation of the lensed GW population.

3.3 Strong lensing of gravitational waves

Strong gravitational lensing occurs when a GW propagates through the gravitational potential of a sufficiently massive foreground object, producing multiple images of the same source. In GW astronomy these images manifest as repeated signals arriving at the detector with identical intrinsic parameters but different amplitudes, arrival times, and phases [136]. This section develops the theoretical framework that describes these effects within the geometric optics limit relevant for ground-based detectors and galaxy-scale lenses.

3.3.1 From geometric optics to the amplification factor

The effect of lensing on a GW signal can be modeled by an amplification factor $F(f;\boldsymbol{\phi})$ modifying the unlensed waveform in the frequency domain [159]

$$\tilde{h}_L(f;\, \boldsymbol{\theta}, \boldsymbol{\phi}) = F(f;\, \boldsymbol{\phi})\, \tilde{h}_U(f;\, \boldsymbol{\theta}), \tag{3.21}$$

where $\tilde{h}_U(f;\boldsymbol{\theta})$ is the frequency-domain unlensed strain with binary parameters $\boldsymbol{\theta}$, $\tilde{h}_L(f;\boldsymbol{\theta},\boldsymbol{\phi})$ is the corresponding lensed strain, and $\boldsymbol{\phi}$ collects the lensing parameters. The simplicity of this relation arises because F depends solely on the mass density profile of the lens and the source position relative to it, decoupling cleanly from the intrinsic source parameters.

In the general wave optics case, $F(f;\boldsymbol{\phi})$ is given by a Fresnel-Kirchhoff diffraction integral over the lens plane [167]

$$F(w) = \frac{w}{2\pi i}\int d^2x\, e^{iw\,\phi(x,y)}, \tag{3.22}$$

where x and y are dimensionless coordinates in the lens and source planes, $\phi(x,y)$ is the Fermat potential (3.5), and

$$w \equiv 8\pi G M_{Lz} f = 4\pi R_S f = \frac{2R_S}{\lambda}, \tag{3.23}$$

is a dimensionless frequency proportional to the product of the lens mass and the GW frequency, with $M_{Lz} = M_L(1+z_L)$ the redshifted effective lens mass, $R_S = 2GM_{Lz}/c^2$ the corresponding Schwarzschild radius, and $\lambda = c/f$ the GW wavelength. The value of w controls the lensing regime: for $w \ll 1$ (equivalently $\lambda \gg R_S$) lensing effects are negligible (perturbative regime); for $w \sim 1$ diffraction and interference are important (wave optics regime); and for $w \gg 1$ (equivalently $\lambda \ll R_S$) the geometric optics limit applies. For a fixed GW frequency f, the transition between regimes is set by the lens mass, as illustrated in Fig. 3.5.

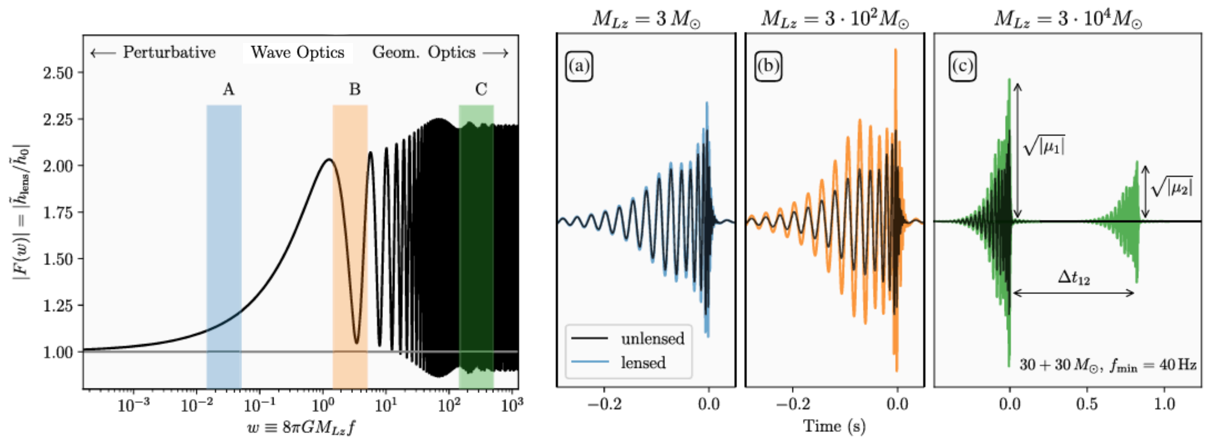


Figure 3.5: *Left:* modulus of the dimensionless amplification factor $|F(w)|$ as a function of dimensionless frequency $w = 8\pi G M_{Lz} f$ for a SIS lens with impact parameter $y = 0.7$. Three regimes are distinguished: perturbative (low w, negligible lensing), wave optics (intermediate w, diffraction and interference), and geometric optics (high w, $|F|$ converges to the sum of image magnifications). *Right:* time-domain waveforms of a $30+30\,M_\odot$ BBH system (unlensed in black, lensed in colour) for three effective lens masses representative of each regime (panels a, b, c). In the geometric optics limit (panel c), the two images are clearly separated in time by Δt_{12} and have distinct amplitudes $\sqrt{|\mu_1|}$ and $\sqrt{|\mu_2|}$. Source: [168].

In the geometric optics limit $w \to \infty$, the integral is dominated by the stationary points of ϕ, and each image j contributes one term, which we write following the convention of [169]:

$$F(f; \boldsymbol{\phi}) = \sum_j \sqrt{\mu_j}\, e^{-2\pi i f t_j + i\pi n_j \,\mathrm{sign}(f)}, \tag{3.24}$$

where μ_j is the magnification (3.12) evaluated at image j, t_j is the arrival time of image j relative to a reference, and n_j is the Morse index introduced in the following section.

3.3.2 Image classification and the Morse phase

The Morse index n_j classifies each image according to the topology of the Fermat potential at its location [170]

$$n_j = \begin{cases} 0 & \text{Type I (minimum of } \phi) \\ 1/2 & \text{Type II (saddle point of } \phi) \\ 1 & \text{Type III (maximum of } \phi) \end{cases} \tag{3.25}$$

This classification is directly related to the parity of the image: Type I and III images have $\det \mathbf{A} > 0$ (positive parity), while Type II images have $\det \mathbf{A} < 0$ (negative parity, mirror-inverted), consistent with the magnification matrix defined in (3.12). The structure of the arrival-time surface and the resulting image positions are illustrated in Fig. 3.6 for three different source positions with the same lens potential.

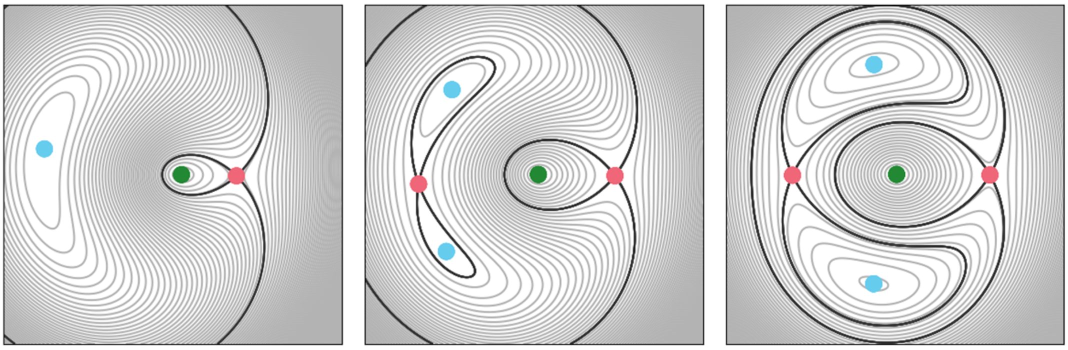


Figure 3.6: Contour maps of the Fermat potential $\tau(\boldsymbol{\theta})$ for three different source positions with the same lens potential, shown qualitatively in the image plane $\boldsymbol{\theta}$. Type I images (minima) are marked in blue, Type II (saddle points) in red, and Type III (maxima) in green. The thick black contours pass through the saddle points. The figure is intended to illustrate the topological structure of the arrival-time surface rather than a specific lens model or parameter space. Source: [153].

Galaxy-scale lenses typically produce two bright images of Type I and II. A faint central image of Type III can in principle exist, but is usually demagnified below detectability by the central mass concentration of the lens.

The factor $e^{-i\pi n_j}$ in (3.24) is the Morse phase. A Type I image ($n = 0$) leaves the waveform morphology unchanged. A Type III image ($n = 1$) introduces an overall sign flip. A Type II image ($n = 1/2$) applies a $\pi/2$ phase shift equivalent to a Hilbert transform of the time-domain signal. For non-precessing binaries with only the dominant quadrupole mode, this phase shift is degenerate with the coalescence phase and cannot be detected. For signals with spin precession or higher-order multipole moments, such as those generated with IMRPhenomXPHM in this work, the Morse phase produces measurable imprints on the waveform [136].

3.3.3 Lensed gravitational wave strain and apparent source parameters

Combining the geometric optics amplification factor (3.24) with the unlensed waveform, the frequency-domain strain of the j-th lensed image is

$$\tilde{h}_L^j(f;\boldsymbol{\theta},\boldsymbol{\phi}_j) = \sqrt{\mu_j}\, e^{-2\pi i f t_j + i\pi n_j\,\mathrm{sign}(f)}\, \tilde{h}_U(f;\boldsymbol{\theta}), \tag{3.26}$$

where $\boldsymbol{\phi}_j = \{\mu_j, t_j, n_j\}$ collects the lensing parameters of image j, consistent with the general form (3.21) [169]. The three multiplicative factors encode amplitude rescaling, time delay, and Morse phase respectively. All images share the same intrinsic parameters $\boldsymbol{\theta}$, which is the fundamental property exploited in joint PE analyses of lensed pairs.

Because (3.26) enters the detector identically to an unlensed signal, a PE pipeline that does not account for lensing will infer biased source parameters. The factor $\sqrt{\mu_j}$ is degenerate with the luminosity distance, since $h \propto D_L^{-1}$, yielding an apparent distance

$$D_L^{\mathrm{obs},j} = \frac{D_L}{\sqrt{\mu_j}}, \tag{3.27}$$

so a magnified image appears closer than the true source. Since redshift is inferred from D_L, the source-frame masses are also biased: the inferred chirp mass $\mathcal{M}_c^{\mathrm{obs}} = (1+\tilde{z})\mathcal{M}_c^{\mathrm{src}}$ corresponds to an incorrect redshift $\tilde{z}$, and lensed events at high redshift may be misidentified as nearby events with anomalously massive black holes [171]. Similarly, the apparent coalescence time is

$$t_c^{\mathrm{obs},j} = t_c + t_j, \tag{3.28}$$

so the two images of the same event appear to arrive at different times.

3.3.4 Lensing observables

In a joint analysis of two lensed images the absolute lensing parameters $\boldsymbol{\phi}_j$ are not individually measurable; only their relative values between the two images are observable. The primary lensing observables for an image pair are the relative magnification

$$\mu_{\mathrm{rel}} \equiv \frac{\mu_2}{\mu_1}, \tag{3.29}$$

the time delay between images $\Delta t = t_2 - t_1$ (3.13), and the difference in Morse indices $\Delta n = n_2 - n_1$ [172]. The relative magnification can span several orders of magnitude depending on the source-lens alignment, while Δn is only detectable when the signal carries precession or higher-mode content. Together, $\{\mu_{\mathrm{rel}}, \Delta t, \Delta n\}$ constitute the complete set of lensing parameters sampled by Bilby in the joint PE analysis carried out in this work.

3.4 Sky localization with lensed image pairs

Strong gravitational lensing offers a natural opportunity to improve the sky localization of GW sources beyond what is achievable from a single image [173]. This section describes the limitations of single-image sky localization, explains how the joint analysis of lensed image pairs overcomes them, and introduces the statistical framework used to quantify the evidence for the lensed hypothesis.

3.4.1 Limitations of single-image sky localization

The sky localization of a GW source relies primarily on triangulation: the difference in signal arrival times across the detector network constrains the direction of propagation, while the antenna pattern functions and strain amplitudes provide complementary information [138]. The achievable localization depends critically on the number and geometry of the detectors: a single detector provides no directional information for short-duration transients; two detectors constrain the source to a ring on the celestial sphere, centered on the axis connecting them; a third non-collinear detector reduces this to a bimodal region; and a fourth non-coplanar detector resolves the position to a single sky area [68].

With only two detectors, a single GW image therefore produces an annular sky region of typically hundreds of square degrees, containing millions of potential host galaxies and tens of thousands of galaxies within the 90% credible volume [146], making electromagnetic follow-up or lens identification essentially impossible in the absence of an independent counterpart. The analyses presented in this work use only H1 and L1, reflecting the most realistic and conservative scenario for strong lensing searches: this is the most common observing configuration given current network uptime and sensitivity, and ensures that the sky localization improvement demonstrated here is achievable under realistic observing conditions.

3.4.2 Joint parameter estimation of lensed pairs

When a GW source is strongly lensed, the multiple images arrive at the detector network at different times. Because the Earth rotates during the delay between arrivals (typically minutes to months for galaxy-scale lenses, see §3.2.4), each image is observed with a different detector orientation, introducing varying antenna pattern functions and effectively enlarging the triangulation baseline [173]. Each lensed image therefore contributes independent constraints on the source position through different noise realizations and detector responses, and their combination reduces the overall sky localization uncertainty compared to a single image [146].

This is exploited through a coherent joint PE of the two images: under the lensed hypothesis $\mathcal{H}_L$ (formally defined in §3.4.3), both data streams d_1 and d_2 share the same intrinsic binary parameters $\boldsymbol{\theta}$, as established in (3.26), so the joint likelihood is a product of the individual likelihoods evaluated at the same $\boldsymbol{\theta}$, with separate lensing parameters $\boldsymbol{\phi}_j = \{\mu_j, t_j, n_j\}$ for each image:

$$p(d_1, d_2 \mid \mathcal{H}_L) \equiv \mathcal{Z}_L = \int p(d_1 \mid \boldsymbol{\theta}, \boldsymbol{\phi}_1)\, p(d_2 \mid \boldsymbol{\theta}, \boldsymbol{\phi}_2)\, p(\boldsymbol{\theta}, \boldsymbol{\phi}_1, \boldsymbol{\phi}_2)\, d\boldsymbol{\theta}\, d\boldsymbol{\phi}_1\, d\boldsymbol{\phi}_2. \tag{3.30}$$

The shared parameters include the masses, spins, and crucially the sky location (α, δ), while the lensing parameters per image, namely the relative magnification μ_{rel} and the image type n_j sampled from a discrete uniform prior over $\{1, 2, 3\}$, remain free independently for each image. Under the unlensed hypothesis $\mathcal{H}_U$, the two signals are independent and the joint evidence reduces to the product of the individual evidences,

$$p(d_1, d_2 \mid \mathcal{H}_U) = \mathcal{Z}_1 \cdot \mathcal{Z}_2, \tag{3.31}$$

obtained from the single-image analyses. The improvement in sky localization is quantified through the 90% credible area of the posterior on (α, δ), defined as the minimum sky region $\mathcal{R}_{90}$ satisfying

$$\int_{\mathcal{R}_{90}} p(\alpha, \delta \mid d)\, d\Omega = 0.90, \tag{3.32}$$

where $d\Omega = \cos\delta\, d\alpha\, d\delta$ is the solid angle element and $p(\alpha, \delta \mid d)$ is obtained by marginalizing the full PE posterior over all remaining parameters.

3.4.3 Identifying lensed events: the lensing Bayes factor

The Bayesian hypothesis testing framework introduced in §2.4.3.2 applies directly to the identification of lensed GW events. Given two detected signals, the competing hypotheses are

$$\mathcal{H}_{\mathbf{L}} : \text{the two signals are lensed images of the same source,} \tag{3.33}$$

$$\mathcal{H}_{\mathbf{U}} : \text{the two signals originate from two independent sources.} \tag{3.34}$$

The coherence ratio

$$\mathcal{C}^{\mathrm{L}}_{\mathrm{U}} \equiv \frac{\mathcal{Z}_L}{\mathcal{Z}_1 \cdot \mathcal{Z}_2}, \tag{3.35}$$

quantifies how much more probable the data are under $\mathcal{H}_L$ than under $\mathcal{H}_U$, without accounting for selection effects or population priors [138]. In practice, since each PE run outputs a signal-vs-noise Bayes factor $\mathcal{B}_{SN}$ and the noise evidences cancel in the ratio, it can be computed as

$$\ln \mathcal{C}^{\mathrm{L}}_{\mathrm{U}} = \ln \mathcal{B}^{\mathrm{joint}}_{SN} - \left(\ln \mathcal{B}^{(1)}_{SN} + \ln \mathcal{B}^{(2)}_{SN}\right). \tag{3.36}$$

A more complete statistic is the full lensing Bayes factor $\mathcal{B}^{\mathrm{L}}_{\mathrm{U}}$, which additionally incorporates selection effects and astrophysical population priors [172]. The primary focus of this work is the improvement in sky localization achieved by the joint analysis; the coherence ratio provides complementary validation that the joint PE correctly identifies the two images as originating from the same source.

Part II

Original results

With the theoretical framework for gravitational lensing of GWs now established, including the geometric optics formalism, the signatures of strong lensing on GW signals, and the joint PE approach that exploits lensed image pairs to improve sky localization, we are in a position to present the original results of this thesis. Chapter 4 describes the simulation setup and the strongly lensed BBH population generated with LeR. Chapter 5 presents the PE pipeline. Each image is first analyzed individually with BILBY, and the two images of each event are then analyzed jointly with HANABI, which coordinates the coherent analysis of the image pair, sharing the intrinsic binary parameters between images while sampling the lensing parameters, on top of the BILBY inference backend. The chapter also describes the automation infrastructure deployed on the Picasso HPC cluster. Finally, Chapter 6 presents the sky localization results, comparing the 90% credible sky area obtained from single-image and joint analyses across the final set of analyzed events.

Chapter 4

Simulation setup and implementation

This chapter describes the construction of the lensed BBH population used as input to the PE pipeline. The simulation proceeds in two stages: first, a large Monte Carlo (MC) population of strongly lensed BBH events is generated with LeR v0.4.2 [174], and then a realistic detectability filter is applied to select the subset of events that would be observed by the H1 and L1 detectors under O5 sensitivity conditions.

4.1 Lensed BBH population with LeR

LeR (LVK Event Rate calculator) is a Python framework for the joint simulation and rate computation of unlensed and strongly lensed GW populations [174]. It integrates source and lens population sampling, lens equation solving via lenstronomy [175], and detector selection into a single reproducible pipeline. All lensing calculations are performed in the geometric optics regime and under the thin lens approximation, both of which are valid for galaxy-scale lenses in the LIGO frequency band (see §3.1).

The simulation is initialized with the following configuration, specifying the detector network, noise curves, and waveform approximant:

```
ler = LeR(
    npool=6,
    event_type="BBH",
    ifos=['L1', 'H1'],
    psds={'L1': 'T2500310-v2_O5a_strain_psd.txt',
          'H1': 'T2500310-v2_O5a_strain_psd.txt'},
    waveform_approximant="IMRPhenomXPHM",
    spin_zero=False,
    spin_precession=True,
)
```

Here, `npool` sets the number of parallel worker processes used for the MC sampling and SNR calculations. The simulation was run in a Jupyter notebook on the CIT JupyterHub, using the `igwn` conda environment (Python 3.11.14) distributed via CVMFS, with 48 CPU cores available. The `event_type="BBH"` argument selects the BBH source population model described in Table 4.1, as opposed to the BNS or NS-BH options also supported by LeR. The `ifos` argument specifies the detector network over which the network SNR $\rho_{\rm net}$ is computed, here restricted to the two LIGO detectors, H1 and L1.

The `psds` argument sets the noise curve assigned to each detector, here the O5a design sensitivity PSD [176]. This choice is motivated by the goal of simulating a realistic future observing scenario.

The default PSD in LeR approximates O4 sensitivity, but no strongly lensed events were detected in O4. Simulating at O5a sensitivity instead, the first epoch of the O5 run with a BNS range of 225 Mpc [176], provides a more informative and forward-looking setting for this study. Figure 4.1 shows the projected O5a strain sensitivity curve used in this work, compared against the L1 O4 sensitivity achieved in September 2024 and the later O5b and O5c projections.

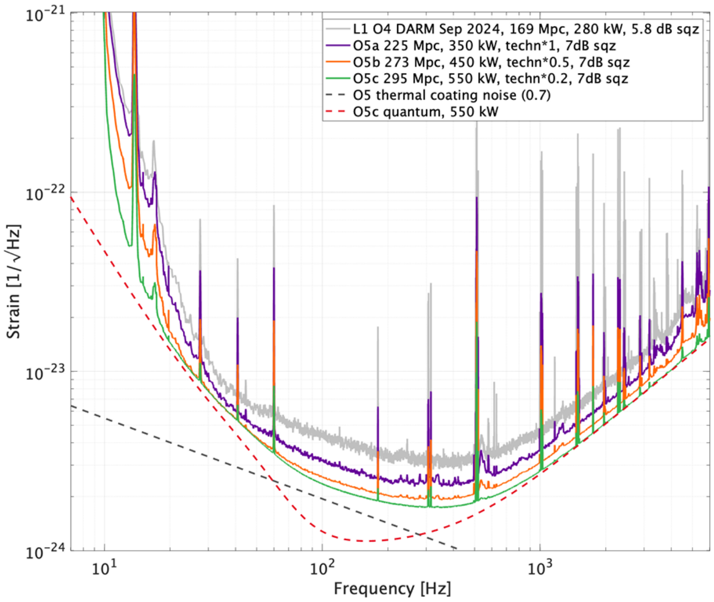


Figure 4.1: Projected strain sensitivity curves for the O5 observing run (O5a, O5b, and O5c), compared against the L1 O4 DARM sensitivity achieved in September 2024 (grey). The O5a curve (purple), adopted in this work, corresponds to a binary neutron star range of 225 Mpc. Dashed lines show the O5 thermal coating noise and O5c quantum noise budgets. Source: [176].

The `waveform_approximant` argument selects `IMRPhenomXPHM` [119] for the SNR computation, which incorporates spin precession and higher-order multipole moments. Finally, `spin_zero=False` enables non-zero component spins in the source population, and `spin_precession=True` allows for spin tilts misaligned with the orbital angular momentum, so that the resulting waveforms include precession effects. Both are necessary for the spin parameters listed in Table 4.1 to be sampled rather than fixed to zero.

The population is generated by drawing events in batches of 20 000 until 500 strongly lensed detectable events are collected, each requiring that at least two images independently satisfy $\rho_{\rm net} \geq 10$:

```
lensed_params = ler.selecting_n_lensed_detectable_events(
    size=500,
    batch_size=20000,
    snr_threshold=10,
    num_img=2,
    resume=False,
    output_jsonfile="n_lensed_params_bbh.txt",
)
```

The SNR threshold of $\rho_{\rm net} \geq 10$ and the requirement of at least two detectable images follow standard detection criteria adopted in GW searches, so that the resulting population reflects

Parameter	Distribution / value
Cosmology	
H_0, Ω_m, Ω_Λ	$70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, 0.3, 0.7 (flat ΛCDM) [174]
BBH source parameters	
Component masses m_1, m_2	Power Law + Gaussian (LIGO O3) [177]
Spin magnitudes a_1, a_2	Uniform [0, 1]
Spin tilt angles θ_1, θ_2	Sine prior on $[0,\, \pi]$
Azimuthal spin angles ϕ_{12}, ϕ_{JL}	Uniform $[0,\, 2\pi]$
Inclination θ_{JN}	Sine prior on $[0,\, \pi]$
Polarization ψ	Uniform $[0,\, \pi]$
Coalescence phase ϕ_c	Uniform $[0,\, 2\pi]$
Sky location $(\alpha,\, \delta)$	Isotropic
Source redshift z_s	Merger rate density [178]
Lens galaxy parameters	
Optical depth $\tau(z_s)$	$(D_c[\mathrm{Gpc}]/62.2)^3$ (SIS analytic) [179]
Velocity dispersion σ_v	$161 \cdot \mathrm{GenGamma}(0.869,\, 2.67)\ \mathrm{km\,s^{-1}}$ [180]
Lens redshift z_L	Conditioned on z_s (SDSS catalogue)
Density slope γ	$\mathcal{N}(2.0,\, 0.2)$ [181, 182]
Axis ratio q	Rayleigh$(s(\sigma_v))$, $q \in [0.2,\, 1]$ [183]
External shear γ_1, γ_2	$\mathcal{N}(0,\, 0.05^2)$ [174]
Lens model	EPL + external shear [184]
Detectability	
Network SNR threshold	$\rho_{\mathrm{net}} \geq 10$
Minimum detectable images	2

Table 4.1: Source and lens population parameters used by LeR v0.4.2 [174] to generate the strongly lensed BBH population. All models correspond to its default configuration.

events that a real detector network could plausibly identify and confidently claim as lensed, rather than a theoretical population extending down to marginal or sub-threshold significance.

A total of approximately 152 000 candidate events were drawn to collect the 500 detectable ones, yielding a total lensed event rate of $\mathcal{R}_L \approx 0.75\,\mathrm{yr}^{-1}$ for this detector configuration and sensitivity. This value is consistent with early forecasts for second-generation detector networks at design-like sensitivity, which span roughly 0.1-10 yr^{-1} depending on the assumed high-redshift merger rate and detector network [183], and is somewhat lower than forecasts for the full three- or four-detector network, reflecting the more limited H1+L1 configuration adopted here.

4.2 Injected parameter distributions

The 500 events produced by LeR represent the strongly lensed population detectable under ideal observing conditions. To obtain a more realistic injection set, a duty cycle filter is applied: for each image of each event, both H1 and L1 are independently activated with probability $p_{\mathrm{det}} = 0.70$, reflecting typical O4/O5 single-detector duty cycles [68]. An image is retained only if both detectors are simultaneously active and $\rho_{\mathrm{net}} \geq 10$. Events with fewer than two surviving images are discarded; when more than two images survive, only the two with the highest network SNR are kept. After applying this filter once to the fixed 500-event population, 157 events are selected as the final injection set analyzed with bilby.

The 500 events produced by LeR represent the strongly lensed population detectable under ideal observing conditions. To obtain a more realistic injection set, a duty cycle filter is applied: for each image of each event, both H1 and L1 are independently activated with probability $p_{\mathrm{det}} = 0.70$, reflecting typical O4/O5 single-detector duty cycles [68]. An image is retained only if both detectors are simultaneously active and $\rho_{\mathrm{net}} \geq 10$. Events with fewer than two surviving images are discarded; when more than two images survive, only the two with the highest network SNR are kept.

After applying this filter once to the fixed 500 event population, 157 events are selected as the final injection set analyzed with bilby. Table 4.2 summarizes the successive stages of this selection process, from the initial MC draws performed by LeR to this final injection set.

Selection stage	Number of events	Retained (%)
Candidate events drawn by LeR	$\sim 152\,000$	—
Detectable lensed events (≥ 2 images with $\rho_{\mathrm{net}} \geq 10$)	500	100
Duty cycle filter ($p_{\mathrm{det}} = 0.70$ per detector)	309	61.8
Duty cycle + SNR cut (≥ 2 images with $\rho_{\mathrm{net}} \geq 10$)	157	31.4

Table 4.2: Number of events surviving each stage of the selection process. Events drawn by LeR are first required to have at least two images with $\rho_{\mathrm{net}} \geq 10$ under ideal observing conditions, yielding the 500 detectable events used as the baseline for the percentages below. A duty cycle filter is then applied, independently activating each detector per image with probability $p_{\mathrm{det}} = 0.70$ (309 events with at least two images surviving, regardless of SNR). The SNR threshold is finally re-applied, since some surviving images may fall below $\rho_{\mathrm{net}} = 10$ once only active detectors are considered, leaving 157 events. All percentages are relative to the 500 detectable events.

Figures 4.2–4.4 show the distributions of the key parameters for the full detectable population ($N = 500$, dashed) and the final injection set ($N = 157$, solid). Figure 4.2 compares the optimal SNR distributions for L1, H1, and the network. The network SNR distribution for the filtered population is truncated at $\rho_{\mathrm{net}} = 10$ by construction, and its peak shifts to higher SNR relative to the full population, reflecting the removal of images that fall below the threshold once only the active detectors are considered; the individual L1 and H1 distributions can extend below $\rho = 10$ even after filtering, since the selection criterion applies to the network SNR rather than to each detector separately.

Figure 4.3 shows the primary source-frame mass m_1^{src} (left) and the number of lensed images per

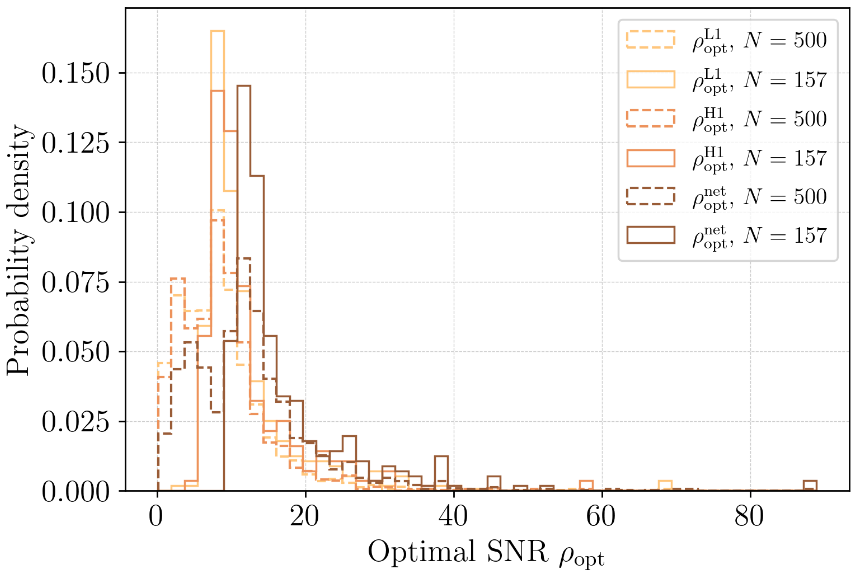


Figure 4.2: Optimal SNR for L1 ($\rho_{\rm opt}^{\rm L1}$, light orange), H1 ($\rho_{\rm opt}^{\rm H1}$, orange), and the network ($\rho_{\rm opt}^{\rm net}$, brown), for the full detectable population ($N = 500$, dashed) and the final injection set ($N = 157$, solid).

event (right). The mass distribution reflects the underlying LIGO O3 Power Law + Gaussian population model adopted by LeR; the duty cycle and SNR selection preferentially retains events with $m_1^{\rm src}$ in the 25-35 $M_\odot$ range, consistent with higher-mass systems producing intrinsically louder signals and thus being more robust to the loss of detector uptime, while the high-mass tail above $\sim 40\,M_\odot$ is largely absent from the filtered population. While the underlying population is dominated by four-image (quad) systems ($\sim 75\%$), the joint requirement of both detectors active and $\rho_{\rm net} \geq 10$ typically leaves only two images per event above the threshold, and almost never leaves all four. This reflects the fact that quad systems produce two brightly magnified images and two much fainter ones, the latter rarely reaching the detection threshold even when both detectors are active.

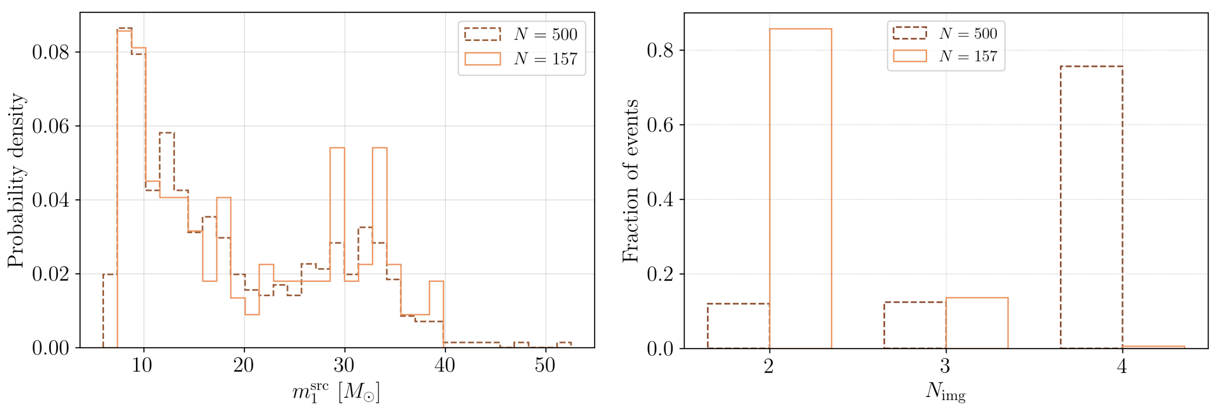


Figure 4.3: Left: primary source-frame mass $m_1^{\rm src}$. Right: number of lensed images per event, comparing the true image multiplicity ($N = 500$, dashed) against the number surviving the duty cycle and SNR filters before the top-2 cut ($N = 157$, solid).

Figure 4.4 compares the true luminosity distance d_L against the effective luminosity distance $d_L^{\rm eff}$ (left), and the image magnification μ (right). Because magnification boosts the observed signal amplitude, $d_L^{\rm eff} = d_L/\sqrt{|\mu|}$ is systematically smaller than the true distance, shifting the effective-distance distributions toward lower values relative to d_L. The magnification distribution, shown prior to the duty cycle filter for the full population and after it for the injection set, is strongly peaked near $\mu \sim 0$ with extended tails; negative values correspond to Type II images (saddle points of the Fermat potential), which have mirror-inverted parity and acquire a $\pi/2$ Morse phase shift (see §3.3). The horizontal axis of the magnification panel is clipped to the 0.5-99.5 percentile range of the combined data to make the shape of the peak and tails visible, excluding a small number of extreme outliers reaching $|\mu| \sim 10^3$.

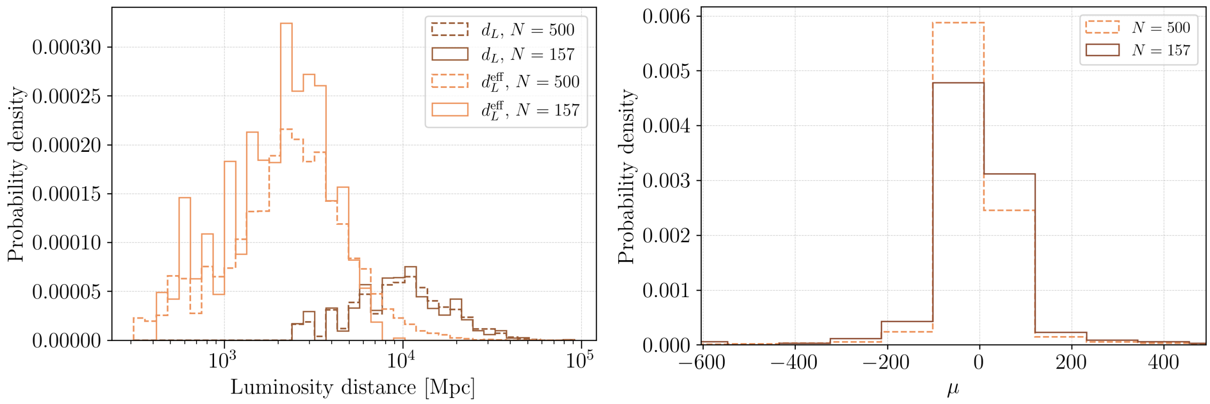


Figure 4.4: *Left*: true luminosity distance d_L (brown) versus effective luminosity distance $d_L^{\rm eff}$ (orange). *Right*: image magnification μ.

Figure 4.5 shows the joint $m_1^{\rm src}$-$m_2^{\rm src}$ distribution for the 500 detectable events, and Figure 4.6 shows the time delay distribution between the two lensed images for the same population.

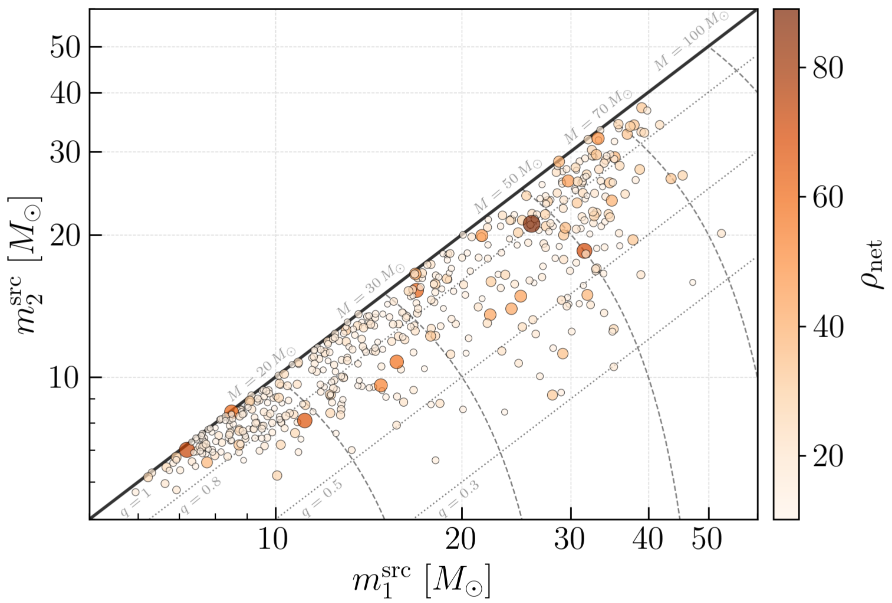


Figure 4.5: Source-frame component masses $m_1^{\rm src}$ and $m_2^{\rm src}$ for the 500 detectable lensed BBH events selected by LeR, colour-coded by network SNR $\rho_{\rm net}$. Diagonal dashed lines mark constant total mass M and dotted lines mark constant mass ratio q. The population traces the underlying LIGO O3 Power Law + Gaussian model (§4.1), with most events clustering near $q = 1$ and the highest-SNR events concentrated around $M \sim 50$-$70\,M_\odot$.

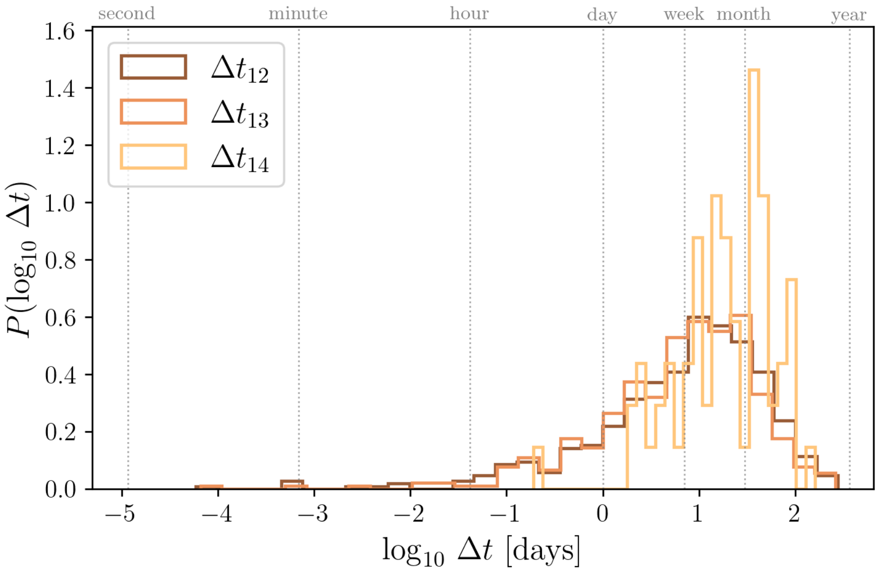


Figure 4.6: Distribution of time delays Δt between the two lensed images for the 500 detectable lensed BBH events, shown on a logarithmic scale. The distribution spans from minutes to months, consistent with galaxy-scale lenses in the geometric optics regime. The Earth's rotation during this interval is the primary mechanism enabling improved sky localization through joint analysis (§3.4.2).

Chapter 5

Parameter estimation pipeline

5.1 Pipeline overview

The PE pipeline processes each of the 157 injected events through the same sequence of automated stages, illustrated in Figure 5.1. Starting from the filtered population `n_lensed_params_bbh_filtered.txt` produced in §4.2, a single script, `Generator_inis_lenskyloc.py`, generates for every event the two single-image `.ini` configurations, the joint `.ini` configuration, and the companion SLURM scripts needed for the downstream steps described below, populating priors and injection parameters directly from the corresponding LeR output (§5.4).

Each single-image `.ini` file is analyzed independently with BILBY_PIPE, using the lensed waveform source model provided by HANABI (§5.2). This yields, for every event, two independent posterior distributions, one per image. Once both single-image jobs have completed, the joint `.ini` file is submitted to `hanabi_joint_pipe` (§5.3), which coordinates a coherent analysis of the image pair, sharing the intrinsic source parameters and sky location between images while sampling the lensing parameters, on top of the same BILBY inference backend used for the single-image runs.

From this point the pipeline branches into two independent post-processing tracks. In the first, the joint posterior is merged into a lightweight result file and passed, together with the single-image data realizations, to `hanabi_postprocess_result`, which recomputes the per-image and network signal-to-noise ratios consistently across the single and joint analyses; a dedicated script then extracts these recovered SNRs into a per-event summary table (§5.4), later combined across all events for the SNR consistency checks of §5.5. In the second track, the single-image and joint posteriors are combined with PESummary [**TODO:pesummary**] into a single metafile, from which sky maps are generated for each image and for the joint analysis using `ligo-skymap-from-samples`, `ligo-skymap-stats`, and `ligo-skymap-constellations`, providing the 90% credible sky areas compared between single-image and joint analyses in Chapter 6.

The entire pipeline, from `.ini` generation to job submission and the SNR post-processing track, is automated on the Picasso HPC cluster; the sky-map track is run separately, as described in §5.4.

5.2 Single-image analysis with bilby

Each of the two images of every event is first analyzed independently with BILBY_PIPE [133], treating it as an isolated signal characterized by its own data segment, noise realization, and set

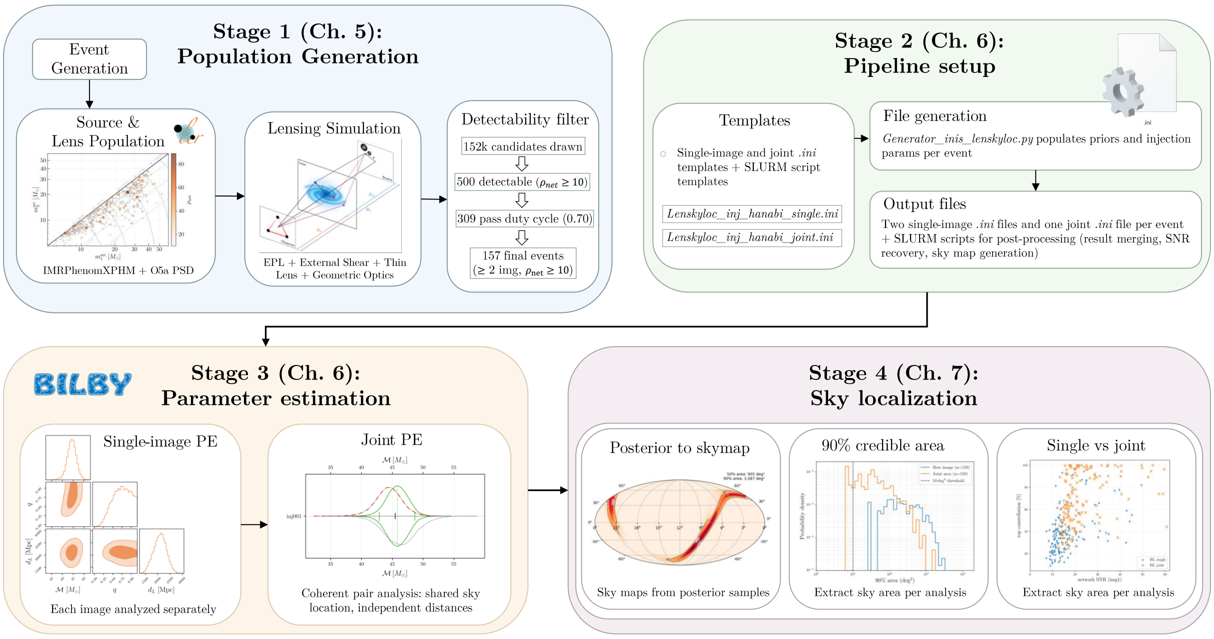


Figure 5.1: Overview of the parameter estimation pipeline. Each event is analyzed with two independent single-image BILBY_PIPE runs and one joint HANABI_JOINT_PIPE run, sharing the intrinsic binary parameters and sky location between images while sampling the lensing parameters. Post-processing splits into an SNR-recovery track and a sky-localization track.

of extrinsic parameters. The `.ini` configuration is generated automatically for each image from a shared template (§5.4); the key data-conditioning and sampler settings are as follows:

```
detectors=[H1, L1]
sampling-frequency=2048
reference-frequency=20
minimum-frequency={'H1': 20, 'L1': 20, 'waveform': 13.33}
maximum-frequency={'H1': 896, 'L1': 896}
psd-dict={"H1": "T2500310-v2_O5a_strain_psd.txt",
          "L1": "T2500310-v2_O5a_strain_psd.txt"}
```

The same O5a PSD used to generate the population with LeR (§4.1) is adopted for the analysis, so that the noise assumptions are consistent between simulation and recovery. Unlike the data duration, which is fixed for the whole population in LeR, the segment `duration` used by BILBY_PIPE is computed individually for every image from its injected component masses and spins, using LALSimulation bounds on the merger and ringdown time together with the expected waveform length at $f_{\rm low} = 20\,$Hz, rounded up to the next power of two. This keeps each analysis as short as the signal allows without truncating the inspiral, rather than using a single conservative duration for the whole population.

The injected and recovered waveforms both use the `IMRPhenomXPNR` approximant [**TODO:IMRPhenomXPNR** rather than the `IMRPhenomXPHM` approximant used by LeR to generate the population and compute detection SNRs (§4.1). This is a deliberate choice rather than an oversight: `IMRPhenomXPHM` is computationally cheap and sufficiently accurate for the population-level detectability calculation performed by LeR, whereas `IMRPhenomXPNR` incorporates numerical-relativity-tuned precession effects and is the more accurate, PE-oriented model of the two. Using it for parameter estimation follows standard practice in recent LVK analyses, at the cost of a waveform mismatch between the population-generation and inference stages that should be kept in mind when interpreting parameter recovery (§5.5).

The waveform is generated through HANABI's lensed source model, `hanabi.lensing.waveform.strongly_lense` [**TODO:hanabi**], which implements the lensed strain expression of §3.3, combining the intrinsic `IMRPhenomXPNR` waveform with an image-dependent amplitude and Morse phase factor. In the single-image configuration this factor is fixed to the injected image type rather than treated as a free parameter,

```
prior-dict={..., image_type = 1.0}
```

since a single lensed image is fully degenerate with an unlensed signal of appropriately rescaled amplitude and shifted phase: the magnification and Morse phase cannot be separated from the ordinary distance and coalescence-phase parameters using one image alone. Breaking this degeneracy is precisely the role of the joint analysis (§5.3).

The remaining priors follow standard choices for BBH parameter estimation: a chirp mass prior adapted individually for each event to a ±30% range around its injected value (rather than a single wide range for the whole population, which would waste live points on regions of parameter space far from every event's true chirp mass), mass ratio and component-mass constraints, uniform spin magnitudes and precession angles, an isotropic sky location, and a 50-20 000 Mpc uniform luminosity distance prior. Sampling is performed with `dynesty` [130] using the acceptance-walk method,

```
n-parallel=3
sampler-kwargs={'nlive': 1000, 'naccept': 60,
                 'sample': 'acceptance-walk', 'npool': 128}
```

with three independent `dynesty` runs combined per image (`n-parallel=3`) to mitigate sampler-to-sampler variance, and 128 parallel workers per run.

5.3 Joint analysis with hanabi

Once both single-image configurations exist, the joint analysis is performed with `hanabi_joint_pipe` [**TODO:hanabi**], which coherently analyzes the two images of an event as two triggers of the same underlying source rather than as independent signals:

```
n-triggers=2
trigger-ini-files=[single_event_xxx_image1.ini,
                   single_event_xxx_image2.ini]

common-parameters=[chirp_mass, mass_ratio, mass_1, mass_2,
                   a_1, a_2, tilt_1, tilt_2, phi_12, phi_jl,
                   cos_theta_jn, ra, dec, psi, phase]
```

Each trigger inherits its own data segment, noise realization, and PSD from its single-image `.ini` file, but the parameters listed in `common-parameters`, namely the intrinsic binary parameters, sky location, polarization, and coalescence phase, are constrained to be identical between the two images, reflecting the fact that both are gravitationally lensed copies of the same source. Notably, `luminosity_distance` is deliberately left out of this shared list: each image retains its own independent effective distance parameter, so that the relative magnification between images is recovered through the ratio of the two independently fitted distances rather than sampled as a separate parameter. This is why the lensing-specific parameters reduce to the image type alone,

```
lensing-prior-dict={relative-magnification^(1): 1,
    image_type^(1) = hanabi.lensing.prior.DiscreteUniform(
        name='image_type^(1)', minimum=1, N=3),
    relative_magnification^(2) = 1,
    image_type^(2) = hanabi.lensing.prior.DiscreteUniform(
        name='image_type^(2)', minimum=1, N=3)}
```

with the relative magnification fixed to unity (its physical content already captured by the two independent distance posteriors) and each image's Morse-index class sampled independently from the three discrete types introduced in §3.3. Sharing the sky location across two images observed at different times, typically separated by minutes to months (Figure 4.6), is the key mechanism behind the improvement in sky localization: the Earth's rotation between the two epochs changes the detectors' antenna pattern for the same fixed (α, δ), effectively lengthening the interferometric baseline available for triangulation (§3.4.2).

The joint likelihood is evaluated over both images simultaneously, using the same sampler settings as the single-image runs (`dynesty`, `nlive=1000`, `naccept=60`, acceptance-walk, `n-parallel=3`), but with substantially higher resource requirements, reflecting the added cost of a two-detector, two-image coherent likelihood:

```
request-memory=180.0
scheduler-analysis-time=71:30:00
```

compared to `request-memory=110.0` and `scheduler-analysis-time=48:00:00` for each single-image run.

5.4 Automation and infrastructure on Picasso

The full pipeline, from the filtered injection set to the final SNR tables, is automated for all 157 events on the Picasso HPC cluster (SLURM scheduler), with jobs executed directly inside a dedicated `conda`/`miniforge` environment (`hanabi_bilby260`), without containerization.

Configuration and script generation

A single script, `Generator_inis_lenskyloc.py`, converts the filtered injection catalogue `n_lensed_params_bbh` into a complete, ready-to-submit run directory for every event. For each event it: (i) computes the true mass ratio, chirp mass, and a per-event data duration from the injected masses and spins (§5.2); (ii) adapts the chirp-mass prior to a ±30% range around the injected value; (iii) populates the `injection-dict` of both single-image templates directly from the corresponding LeR parameters, mapping, e.g., `effective_luminosity_distance` to `luminosity_distance` and `effective_geocent_time` to `geocent_time`; and (iv) writes out the two single-image `.ini` files, the joint `.ini` file, and the companion SLURM scripts for the downstream post-processing chain (final-result merging, joint SNR post-processing, sky map generation), substituting the event-specific paths into a set of shared templates. This guarantees that the 157 event directories are internally consistent and reproducible, and that no configuration is edited by hand.

Submission and job dependencies

Job submission is split into two stages, reflecting the pipeline's dependency structure. The first stage, `submit_all_part1.sh`, loops over all event directories and, for each one, submits the two single-image `bilby_pipe` jobs and then the joint `hanabi_joint_pipe` job with an explicit SLURM dependency on both single-image job IDs,

```
sbatch --dependency=afterok:$jid_s1:$jid_s2 $joint_script
```

so that the joint analysis for an event is only launched once both of its single-image analyses have completed successfully. The second stage, `submit_all_part2.sh`, runs after the first stage completes and chains, again via `afterok` dependencies, the merging of the joint posterior into a lightweight result file (`bilby_result`, capped at 20 000 samples), the HANABI post-processing step (`hanabi_postprocess_result`, which recomputes per-image SNRs from the joint posterior using the noise realizations generated for the single-image analyses), and the final SNR extraction (`compute_snr.py`).

Post-processing splits into the two tracks introduced in §5.1. The SNR-recovery track, final-result merging, `hanabi_postprocess_result`, and SNR extraction, is fully automated on Picasso as part of `submit_all_part2.sh`. The sky-map track, generating PESUMMARY pages and sky maps for every event, is instead run on the CIT cluster using HTCondor, within the `igwn` CONDA environment distributed via CVMFS: `Pesummaries_automated_condor_all.py` and `Skymap_condor.py` are each launched as an HTCondor job array of 157 processes, indexed automatically by `$(Process)`, generating the summary pages and sky maps for all events in parallel. The parameter estimation itself, both single-image and joint, was always run on Picasso and never on CIT; only this post-processing track uses HTCondor, since the version of SUMMARYPAGES available in the Picasso environment produced errors, while the `igwn` environment on CIT ran it correctly.

Compute losses

The joint analysis is the most expensive step of the pipeline, requiring up to 71.5 h of wall time per event (§5.3). Partway through the production run, the Picasso allocation available for this project was reduced to a 24 h wall-time limit, below what the remaining joint jobs required. The last seven joint analyses were interrupted mid-run when the allocation was exhausted; rather than repeatedly resubmit and monitor these jobs against an uncertain future allocation, the analysis was finalized with the 150 events that had completed successfully. This loss of 7 out of 157 events (157 → 150) is a consequence of the compute allocation available at the time, not a scientific selection criterion, and does not bias the final sample in any systematic way tied to source or lens parameters.

5.5 Parameter recovery and validation

Chapter 6

Sky localization

6.1 From posterior samples to sky maps

6.2 Sky localization: single-image versus joint analysis

6.2.1 90% credible area

6.2.2 Constellation localization

To summarize, in this chapter we have...

Chapter 7

Conclusions

Future work

In this thesis work we have obtained two original results: 1. ... 2. ... These results are shown in Chapter ? and in Chapter ?, respectively. In particular, in Section ? we have shown... . Instead, in Sections ? and ? we have studied...

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Appendix A

Parameters of Compact Binary Coalescences

This appendix provides a reference summary of the parameters used to describe a CBC system with quasi-circular orbits, following the conventions of [185, 143]. The parameters are grouped into mass, spin, tidal, and extrinsic categories; those employed in the injections and parameter estimation analyses of this work are indicated with $\star$.

Table A.1: Parameters describing a CBC system with quasi-circular orbits [185, 143]. Parameters used in this work are indicated with $\star$.

Parameter name	Symbol	Definition [Dimensions]
Mass parameters		
Primary/secondary mass	m_1, $m_2^\star$	Mass of the more (less) massive component, $m_1 \geq m_2$ [$M_\odot$]
Chirp mass	$\mathcal{M}$	$\mathcal{M} = \eta^{3/5} M$ [$M_\odot$]
Total mass	M	$M = m_1 + m_2$ [$M_\odot$]
Final mass	M_f	Mass of the remnant [$M_\odot$]
Mass ratio	q	$q = m_2/m_1 \leq 1$ [dimensionless]
Symmetric mass ratio	η	$\eta = m_1 m_2 / M^2 \leq 1/4$ [dimensionless]
Radiated energy	$E_{\rm rad}$	$E_{\rm rad} = (M - M_f)c^2$ [energy]
Spin parameters		
Primary/secondary spin vectors	$\mathbf{S}_1$, $\mathbf{S}_2$	Spin angular momentum of the primary/secondary [ang. mom.]
Dimensionless spin magnitudes	χ_1, $\chi_2^\star$	$\chi_{1,2} = c\lvert\mathbf{S}_{1,2}\rvert/(Gm_{1,2}^2) \leq 1$ [dimensionless]
Remnant spin magnitude	χ_f	$\chi_f = cS_f/(GM_f^2) \leq 1$ where S_f is the magnitude of the remnant's spin angular momentum [dimensionless]
Orbital ang. momentum	$\mathbf{L}$	Instantaneous orbital angular momentum about the center of mass [ang. mom.]
Total ang. momentum	$\mathbf{J}$	$\mathbf{J} = \mathbf{L} + \mathbf{S}_1 + \mathbf{S}_2$ [ang. mom.]
Primary/secondary tilt angles	θ_1, $\theta_2^\star$	Angle between $\mathbf{S}_{1,2}$ and $\mathbf{L}$ [angle]
Spin azimuthal difference	$\phi_{12}^\star$	Angle between $\mathbf{L} \times (\mathbf{S}_1 \times \mathbf{L})$ and $\mathbf{L} \times (\mathbf{S}_2 \times \mathbf{L})$ [angle]

Continued on next page

Table A.1 *(continued)*

Parameter name	Symbol	Definition [Dimensions]
Effective inspiral spin	χ_{eff}	$(m_1\chi_{1z} + m_2\chi_{2z})/M$ [dimensionless]
Effective precession spin	χ_p	Avg. over precession cycles (see Eq. (16) of [68]) [dimensionless]
Tidal parameters		
Tidal deformabilities	Λ_1, Λ_2	Dimensionless tidal deformability; $\Lambda_{1,2} = 0$ for BHs [dimensionless]
Effective tidal deformability	$\tilde{\Lambda}$	Dominant tidal combination; $\tilde{\Lambda} = 0$ for BBH [dimensionless]
Extrinsic parameters		
Luminosity distance	$D_L^\star$	Distance to the source [Mpc]
Redshift	z	Computable from D_L assuming a cosmological model [dimensionless]
Source inclination	$\theta_{JN}^\star$	Angle between **J** and line of sight **N** [angle]
Orbital inclination	ι	Angle between **L** and **N** [angle]
Coalescence time	$t_c^\star$	GPS time at the geocenter [time]
Coalescence phase	$\phi_c^\star$	Orbital phase at coalescence [angle]
Right ascension	$\alpha^\star$	Celestial longitude [angle]
Declination	$\delta^\star$	Celestial latitude [angle]
Polarization angle	$\psi^\star$	Rotates polarizations in the plane orthogonal to propagation [angle]
Azimuthal spin angle	$\phi_{JL}^\star$	Angle between **J** and **L** projected on the sky plane [angle]

Appendix B

Reproducibility tutorial. Linux environment and supercomputing tasks.

The motivation for including this section comes from the desire to make life easier for future students who will use the same tools, especially since many parts of this tutorial came with their fair share of headaches.

This appendix provides a detailed, step-by-step overview of the workflow followed to run my experiments on a Linux-based high-performance computing (HPC) system. The goal is to ensure full reproducibility and transparency. The system used was Picasso, which uses a Linux environment and the SLURM workload manager.

Accessing the Supercomputer

Access to the HPC cluster was provided via secure shell (SSH). The connection was established using:

```
ssh username@cluster.domain.edu
```

For convenience, an SSH configuration file was set up:

```
# ~/.ssh/config
Host mycluster
    HostName cluster.domain.edu
    User myusername
    IdentityFile ~/.ssh/id_rsa
```

3. File and Directory Organization

The working directory was structured as follows:

```
/home/username/
 project/
  data/
   scripts/
    results/
   logs/
```

This structure helped maintain organization and simplified automation.

4. Working Environment

The HPC system used a module system to load required software. The Python environment was prepared as follows:

```
module load python/3.10
python -m venv env
source env/bin/activate
```

Additional Python packages were installed using `pip` or `requirements.txt`.

5. Key Commands and Utilities

Here is a summary of the most commonly used commands:

Command	**Description**
`ssh`	Secure connection to the HPC server.
`scp`	Copy files between local and remote systems.
`sbatch`	Submit job scripts to SLURM.
`squeue`	View active jobs in the queue.
`ls`	List files and directories.
`ll`	Detailed list of files (usually alias for ls -l).
`cd`	Change directory.
`pwd`	Show current working directory.
`mkdir`	Create a new directory.
`rm`	Remove files or directories.
`cp`	Copy files or directories.
`mv`	Move or rename files or directories.
`less`	View file contents page by page.
`tail`	Display the last lines of a file (useful for logs).
`head`	Display the first lines of a file.
`grep`	Search for patterns in files.
`nano`	Simple terminal text editor.
`df`	Show disk space usage.

6. Job Submission and SLURM Scripts

Jobs were submitted to the scheduler using SLURM. Below is a sample job script:

```
#!/bin/bash
#SBATCH --job-name=my_job
#SBATCH --output=logs/output_%j.txt
#SBATCH --error=logs/error_%j.txt
#SBATCH --time=01:00:00
#SBATCH --mem=4G
#SBATCH --cpus-per-task=2

module load python/3.10
source env/bin/activate
python scripts/my_script.py
```

To submit this job:

```
sbatch job.sh
```

To check job status:

```
squeue -u myusername
```

7. Running the Experiments

Experiments were launched using prepared scripts with configurable arguments. For example:

```
python train_model.py --dataset data/input.csv --epochs 50 --lr 0.001
```

Each run was logged using ‘tee‘ and redirected output:

```
python script.py > logs/run1.out 2>&1
```

8. Troubleshooting and Common Issues

Typical problems encountered included:

- **Module conflicts:** Solved by purging modules with `module purge`.
- **Permission errors:** Solved by adjusting file permissions via `chmod`.
- **Job hangs:** Resolved by checking SLURM node load and memory limits.

9. Optimization and Performance Tuning (Optional)

Where relevant, job parameters were optimized to use resources efficiently. For example:

- Multi-threading enabled via `-cpus-per-task`.
- Memory profiling using `/usr/bin/time -v`.
- Parallel jobs launched via SLURM job arrays.

10. Additional Resources

- SLURM documentation: https://slurm.schedmd.com/documentation.html
- SSH key setup: https://www.ssh.com/academy/ssh/keygen
- GitHub repository :

Appendix C

Detailed implementation of the pipeline